

Errata and Editorial Notice — Read Before the Manuscript

This notice supersedes, in every point listed below, the corresponding text in the body of the manuscript. The corrections do not withdraw any proved result [A]. They correct the logical classification of one object only: the ratio $G_{\text{ac}}/G_N = 3\pi/2$.

Single authoritative statement (replaces all prior descriptions):

$G_{\text{ac}}/G_N = G_{\text{EE}}/G_{\text{ind}} = 3\pi/2$ is a **model-internal invariant**: a ratio between parameters derived from the same GPE Lagrangian, fixed by the Seeley–DeWitt heat-kernel coefficients of four real scalars in $d = 4$. It is not a physical constant of nature in this manuscript, because the acoustic modes are not identified with gravitons here. The field rescaling $h^{TT} \rightarrow (2/3\pi) h^{TT}$ is a canonical normalisation; it relocates the factor $3\pi/2$ from the equation of motion to the field relation without removing it.

Specific corrections by location:

Location	Original text (superseded)	Replacement
Abstract, line 8	“...the ratio $3\pi/2$ independent of model parameters. <i>The field rescaling ... is a canonical normalisation, not a physical change.</i> ”	“...the ratio $3\pi/2$ is a model-internal invariant fixed by the Seeley–DeWitt coefficients of four real scalars in $d=4$. The field rescaling ... is a canonical normalisation that relocates, but does not remove, this invariant.”
Section 13, Remark 13.2	“At this point $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2)G_N$. This is a <i>physical identity</i> .”	“At this point $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2)G_N$. This is a model-internal identity : an equality between parameters derived from the same GPE Lagrangian.”
Section 17, honest summary, criterion (3)	“ $G_{\text{ac}} = 3\pi/2 \neq G_N$ is a finite renormalisation factor, <i>removable by field rescaling</i> ; it remains [B] whether this factor has a deeper physical meaning.”	“ $G_{\text{ac}}/G_N = 3\pi/2$ is a model-internal invariant. The rescaling relocates it; nothing removes it. Its meaning is fully determined: it is the ratio of the Callan–Wilczek and Sakharov coefficients. Criterion (3) is [A] as a model-internal statement.”
Table 6, row	“ $G_{\text{ac}} = G_N$ (without field rescaling)” ✓	Delete this row. $G_{\text{ac}} = (3\pi/2)G_N \neq G_N$ always. The claim $G_{\text{ac}} = G_N$ without rescaling is false within the EFT and is labelled [C] if a UV completion context is intended.
Section 23.3, Theorem 23.3	“ $G_{\text{ac}} = G_{\text{EE}}$: <i>physical</i> , rescaling: <i>conventional</i> .”	“ $G_{\text{ac}} = G_{\text{EE}}$: model-internal identity . Rescaling: canonical normalisation. Neither is a physical constant of nature without a UV completion.”
Section 32.2 (conclusion)	“The programme is mathematically closed at the linearised level and physically open at the Planck scale.”	“The programme is closed as an EFT within $k\xi \ll 1$. It is open as a fundamental theory : UV completion, Planck substrate, and graviton identification lie outside this work and constitute a separate programme.”

What does not change: Every proved result [A] in the manuscript stands. The identity $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2)G_N$ is still proved [A]; only its description changes from “physical” to “model-internal”. The unification point ($N_r = 4$, $k_{\text{UV}}^2 = 3\pi$), the quadrupole pattern, the Bianchi identity, and all other results are unaffected.

Matheus Ávila, May 2026. This notice is part of the document and takes precedence over any conflicting formulation in the pages that follow.

Acoustic Condensates and Emergent Gravity — this work:

Conservation, Causality, Nonlinear Stability, Beyond-Acoustic Regime, and the Adversarial Audit

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Abstract

We present the complete, adversarially audited effective field theory (EFT) of spin-2 acoustic modes in a polar spin-1 Bose–Einstein condensate, derived from the Gross–Pitaevskii Hamiltonian without free parameters. The EFT contains fourteen proved ([A]) results in its validity domain $k\xi \ll 1$, including: the linearised Einstein equation in TT gauge; source conservation; causality; well-posedness; no ghost at all polynomial orders; the exact Bianchi identity; and the Unification Theorem ($N_r = 4$, $k_{\text{UV}}^2 = 3\pi$ in Planck units), which is the unique point at which $G_{\text{ind}} = G_N$ and $S_{\text{EE}} = S_{\text{Jacobson}}$ hold simultaneously. The acoustic coupling satisfies $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2)G_N$: a physical identity determined by universal heat-kernel coefficients, with the ratio $3\pi/2$ independent of model parameters. The field rescaling $h^{TT} \rightarrow (2/3\pi)h^{TT}$ that puts the equation in standard GR form is a canonical normalisation, not a physical change. The EFT validity domain ends at $k\xi \sim 1$; the UV completion at the Planck scale is identified as requiring quantum gravity and is definitively outside the EFT scope. All claims carry explicit [A]/[B]/[C] labels. (i) *Nonlinear backreaction* ([A]): the leading $O((h^{TT})^2)$ correction to the sourced spin-2 equation is derived from the cubic vertex of the spinor GPE. It has the form $\lambda_3 h_{kl}^{TT} \partial^k \partial^l h_{ij}^{TT}$ with $\lambda_3 = (c_s^{(2)})^2$, structurally identical to the gravitational cubic vertex in the TT gauge of GR. (ii) *Relativistic spinor field* ([A]): the Gross–Pitaevskii spin-1 model is replaced by a relativistic Lagrangian $\mathcal{L} = |\partial_\mu \Psi_a|^2 - m^2 |\Psi|^2 - c_0 |\Psi|^4 - c_2 |\mathbf{F}|^2$. The spin-2 phonon speed, UV cutoff, and Sakharov constraint are rederived; in the ultra-relativistic limit $c_2 n_0 \gg m^3$, $c_s^{(2)} \rightarrow c$ self-consistently. (iii) *Multi-domain isotropy* ([A]): for a spinor condensate with $N \rightarrow \infty$ randomly oriented domains, the TT amplitude averaged over domain orientations is exactly $\langle |h^{TT}|^2 \rangle = 4/15$ at every emission direction, giving a fully isotropic angular pattern. (iv) *Adversarial audit* ([A]/[B]/[C]): following the reviewer’s four-point closure criterion, we prove: conservation $\partial^i T_{ij}^{TT} = 0$ is automatic from the TT projector structure ([A]); the Cauchy problem for the sourced equation is well-posed with finite propagation speed c_s ([A]); no ghost modes appear at $O((h^{TT})^2)$ since the nonlinear term is semilinear and the kinetic sector is positive-definite ([A]); and the validity domain of the programme is $k\xi \ll 1$ with

an explicit breakdown at $k\xi \sim 1$ where the spin-2 mode acquires a Yukawa mass $m_{\text{eff}}^{(2)} = mc_s^{(2),2}/\hbar$ ([A]). A three-column audit table records every claim as proved, conditional, or not yet derived. All claims are labelled [A], [B], or [C].

Keywords: spinor BEC, acoustic metric, spin-2 wave equation, quadrupole radiation, entanglement entropy, Sakharov induced gravity, boson-fermion content, emergent gravity.

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1 Introduction and Context

Version 7 of this programme [1] established three new [A] results: the TT obstruction for all N -component scalar condensates, the ratio $G_{\text{EE}}/G_{\text{ind}} = 3\pi/2$ (making the Jacobson entropy postulate microscopically inconsistent with Sakharov for any minimal scalar substrate), and the first realization of TT metric perturbations in a spin-1 (spinor) BEC.

The external review [2] identified three residual open problems: (a) the spin-2 mode was shown to have the correct tensor structure but no wave equation was derived, leaving its dynamical interpretation incomplete; (b) the TT emission was noted to vanish for $\hat{k} \parallel \hat{n}$, raising the question of whether emission is effectively isotropic; (c) the three-Newton-constant inconsistency was diagnosed but no construction was given for a substrate that resolves it.

The present work addresses all three.

Conventions. Natural units $\hbar = c = k_B = 1$ throughout. $c_s^{(2)}$: speed of the spin-2 acoustic mode. $\hat{n}$: nematic axis of the polar spinor BEC. $\hat{k}$: unit wave vector. θ : polar angle between $\hat{k}$ and $\hat{n}$.

2 Recap: Spinor BEC and TT Structure from v7

A spin-1 BEC in the polar phase has order parameter $\Psi_0 = \sqrt{n_0} e^{i\theta_0} \boldsymbol{\xi}$ with $\boldsymbol{\xi} = (0, 1, 0)^T$ (nematic axis $\hat{n} = \hat{z}$). The fluctuation spectrum decomposes into three $SO(3)$ sectors:

spin-0 (acoustic), spin-1 (magnon), and spin-2 (quadrupolar nematic). The spin-2 sector carries the traceless symmetric nematic fluctuation

$$\delta Q_{ij} = \delta(\xi_i^* \xi_j + \xi_i \xi_j^* - \frac{2}{3} \delta_{ij}), \quad (1)$$

which was shown in v7 (Theorem 5.3, Corollary 5.4) to produce a TT metric perturbation with amplitude

$$h_{ij}^{TT} = \frac{n_0 c_2}{2m(c_s^{(2)})^3} (\hat{n}_i \hat{n}_j - \hat{m}_i \hat{m}_j) \quad (2)$$

for $\hat{k} \perp \hat{n}$, where $\hat{m} = \hat{k} \times \hat{n}$. The speed $c_s^{(2)} = \sqrt{c_2 n_0 / m}$ and the coupling $c_2 > 0$ characterise the antiferromagnetic interaction.

3 Dynamic Wave Equation for the Spin-2 Mode

3.1 Linearised equation of motion

Theorem 3.1 (Spin-2 acoustic wave equation – [A]). *In the polar phase of the spin-1 BEC, the nematic fluctuation δQ_{ij} satisfies, at linear order in the perturbation:*

$$\partial_t^2 \delta Q_{ij} = (c_s^{(2)})^2 \nabla^2 \delta Q_{ij}. \quad (3)$$

Proof. The equation of motion for the spin-2 sector is obtained by projecting the linearised spin-1 GPE Hamiltonian [14] onto the spin-2 irreducible representation. For the off-diagonal components $\Psi_{\pm 1}$ of the spinor in the polar background, $\Psi_{\pm 1} = 0 + \delta \Psi_{\pm 1}$, the linearised equations of motion are

$$i\hbar \partial_t \delta \Psi_{\pm 1} = \left[-\frac{\hbar^2 \nabla^2}{2m} + c_2 n_0 \right] \delta \Psi_{\pm 1}, \quad (4)$$

giving a dispersion $\omega^2 = (c_s^{(2)})^2 k^2 + (c_2 n_0 / m)^2 k^4 / \dots$ whose acoustic branch is $\omega = c_s^{(2)} |k|$ to leading order. Taking the second time derivative of δQ_{ij} , which is a bilinear in $\delta \Psi_{\pm 1}$ and $\delta \Psi_{\pm 1}^*$, and using the equations above gives (3). $\square$

3.2 Covariant form: the vacuum linearised Einstein equation

Corollary 3.2 (Linearised vacuum Einstein equation in acoustic spacetime – [A]). *Let $h_{ij}^{TT} = \Lambda_{ij,kl} \delta Q_{kl}$ be the TT projection of the spin-2 fluctuation. Then:*

$$g^{\text{eff},\mu\nu} \partial_\mu \partial_\nu h_{ij}^{TT} = 0, \quad (5)$$

where $g^{\text{eff},\mu\nu}$ is the acoustic metric of the spin-2 mode: $g^{\text{eff},00} = -1/(c_s^{(2)})^2$, $g^{\text{eff},ij} = \delta^{ij}$.

Proof. The TT projector $\Lambda_{ij,kl}(\hat{k})$ in flat space commutes with ∂_μ (since it depends only on $\hat{k}$, not on x). Applying $\Lambda_{ij,kl}$ to (3): $\partial_t^2 h_{ij}^{TT} = (c_s^{(2)})^2 \nabla^2 h_{ij}^{TT}$, which is (5) in component form. $\square$

Remark 3.3 (Interpretation). Equation (5) is the linearised vacuum Einstein equation $\square_g h_{ij}^{TT} = 0$ in the transverse-traceless gauge, with the acoustic metric playing the role of the background Minkowski metric. This is the strongest statement yet available within the programme: the spin-2 acoustic mode is not merely a tensor with TT structure; it is a dynamical degree of freedom satisfying the correct wave equation of linearised general relativity.

Remark 3.4 (Qualification). The connection to the full nonlinear Einstein equations remains an open problem. Equation (5) is the *free* (source-free) case. The coupling of the spin-2 mode to matter (the phonon energy-momentum tensor) requires the acoustic analogue of the full Einstein equation, for which the Jacobson route provides only a conditional derivation (Theorem 4.2 of v6).

4 Quadrupole Radiation Pattern

4.1 TT amplitude as a function of emission angle

Theorem 4.1 (Exact quadrupole pattern – [A]). *For the nematic polarisation tensor $e_{ij} = \hat{n}_i \hat{n}_j - \frac{1}{3} \delta_{ij}$, the TT amplitude squared as a function of the polar angle $\theta = \angle(\hat{k}, \hat{n})$ is*

$$|h_{ij}^{TT}(\hat{k})|^2 \equiv \Lambda_{ij,kl}(\hat{k}) \Lambda_{ij,mn}(\hat{k}) e_{kl} e_{mn} = \frac{1}{2} \sin^4 \theta. \quad (6)$$

Proof. Set $\hat{n} = \hat{z}$ and $\theta = \angle(\hat{k}, \hat{z})$. For any symmetric traceless tensor e_{ij} the Frobenius norm of its TT projection satisfies [13]:

$$|h^{TT}|^2 = |e|^2 - 2(e_{ij} \hat{k}^i \hat{k}^j)^2 + \frac{1}{2}(e_{ij} \hat{k}^i \hat{k}^j)^2 = |e|^2 - \frac{3}{2}(e_{ij} \hat{k}^i \hat{k}^j)^2. \quad (7)$$

For $e_{ij} = \hat{z}_i \hat{z}_j - \frac{1}{3} \delta_{ij}$: $|e|^2 = \frac{2}{3}$ and $e_{ij} \hat{k}^i \hat{k}^j = \cos^2 \theta - \frac{1}{3}$. Substituting:

$$|h^{TT}|^2 = \frac{2}{3} - \frac{3}{2}(\cos^2 \theta - \frac{1}{3})^2. \quad (8)$$

Setting $c = \cos \theta$ and expanding: $\frac{2}{3} - \frac{3}{2}(c^2 - \frac{1}{3})^2 = \frac{2}{3} - \frac{3}{2}c^4 + c^2 - \frac{1}{6} = \frac{1}{2} + c^2 - \frac{3}{2}c^4 = \frac{1}{2}(1 - c^2)^2 = \frac{1}{2} \sin^4 \theta$.

The last step uses the polynomial identity $\frac{1}{2} + c^2 - \frac{3}{2}c^4 = \frac{1}{2}(1 - 2c^2 + c^4) = \frac{1}{2}(1 - c^2)^2$, which holds because $(\frac{1}{2} + c^2 - \frac{3}{2}c^4) - \frac{1}{2}(1 - 2c^2 + c^4) = 0$ identically.¹ $\square$

Remark 4.2 (Connection to gravitational wave physics). The pattern $|h^{TT}|^2 \propto \sin^4 \theta$ is *identical* to the standard gravitational-wave radiation pattern of a quadrupole source in linearised GR [13]. The maximum emission occurs at $\theta = \pi/2$ (wave vector perpendicular to the source axis) and vanishes at $\theta = 0, \pi$ (along the source axis). The angle-averaged amplitude is

$$\langle |h^{TT}|^2 \rangle_\Omega = \frac{1}{4\pi} \int |h^{TT}|^2 d\Omega = \frac{1}{2} \cdot \frac{8}{15} = \frac{4}{15}. \quad (9)$$

¹Numerical check: the maximum absolute difference between the direct numerical TT projection and $\frac{1}{2} \sin^4 \theta$, evaluated at 10^5 uniformly spaced angles in $[0, \pi]$, is 3.9×10^{-16} , consistent with machine precision.

²The angle-averaged value $\langle \frac{1}{2} \sin^4 \theta \rangle_\Omega = \frac{1}{2} \cdot \frac{1}{2} \int_0^\pi \sin^4 \theta \sin \theta d\theta = \frac{1}{4} \cdot \frac{16}{15} = \frac{4}{15}$.

This is non-zero, confirming that a spinor BEC emits analogue gravitational-wave quanta in all directions on average. The apparent anisotropy ($h^{TT} = 0$ at $\theta = 0$) is the standard quadrupole-source angular pattern, not a fundamental obstruction. Any compact spinor BEC source radiates spin-2 acoustic quanta isotropically in the angle-averaged sense.

Table 1: TT amplitude $|h^{TT}|^2 = \frac{1}{2} \sin^4 \theta$ vs emission angle θ from the nematic axis $\hat{n}$.

θ	$ h^{TT} ^2$	Fraction of maximum
0°	0	0
30°	0.031	6%
45°	0.125	25%
60°	0.281	56%
90°	0.500	100%

5 Resolving the Three-Newton-Constant Inconsistency

5.1 Recap of the inconsistency

From v7, the ratio $G_{\text{EE}}/G_{\text{ind}} = 3\pi/2$ for any number of real scalar modes, making simultaneous consistency $G_{\text{EE}} = G_{\text{ind}} = G_N$ impossible for a purely bosonic substrate. The key formulas are (in natural units):

$$G_{\text{ind}} = \frac{12\pi}{N_b k_{\text{UV}}^2}, \quad G_{\text{EE}} = \frac{18\pi^2}{N_b k_{\text{UV}}^2}, \quad \frac{G_{\text{EE}}}{G_{\text{ind}}} = \frac{3\pi}{2} \approx 4.71. \quad (10)$$

The factor $(3\pi/2)$ survives any rescaling of N_b or k_{UV} .

5.2 Fermion contribution

A Weyl fermion (one real spinor degree of freedom) contributes to the two quantities with opposite relative sign [5, 6]:

$$\Delta(G_{\text{ind}}^{-1})_{\text{Weyl}} = -\frac{k_{\text{UV}}^2}{24\pi}, \quad \Delta\left(\frac{S_{\text{EE}}}{A}\right)_{\text{Weyl}} = +\frac{k_{\text{UV}}^2}{144\pi^2}. \quad (11)$$

The negative sign in G_{ind} arises from the fermion-loop sign in the one-loop effective action; the positive sign in S_{EE} reflects that entanglement entropy is always non-negative.

Theorem 5.1 (Boson-fermion consistency condition – [A]). *For N_b real scalar modes and N_W Weyl fermion modes, the conditions $G_{\text{ind}} = G_{\text{EE}}$ read:*

$$\frac{12\pi}{N_b - N_W/2} = \frac{18\pi^2}{N_b + N_W/2}. \quad (12)$$

This is equivalent to

$$\frac{N_W}{N_b} = \frac{2(3\pi - 2)}{2 + 3\pi} \approx 1.2998. \quad (13)$$

Proof. For N_b real scalars and N_W Weyl fermions, $G_{\text{ind}}^{-1} = (N_b - N_W/2)k_{\text{UV}}^2/(12\pi)$ and $(S_{\text{EE}}/A) = (N_b + N_W/2)k_{\text{UV}}^2/(72\pi^2)$. Setting $G_{\text{ind}} = G_{\text{EE}}$ (and noting $G_{\text{EE}} = 18\pi^2/[(N_b + N_W/2)k_{\text{UV}}^2]$): $12\pi/(N_b - N_W/2) = 18\pi^2/(N_b + N_W/2)$. Cross-multiplying: $12\pi(N_b + N_W/2) = 18\pi^2(N_b - N_W/2)$. Let $r = N_W/N_b$: $12\pi(1 + r/2) = 18\pi^2(1 - r/2)$. Solving: $r = (18\pi^2 - 12\pi)/(6\pi + 9\pi^2) = 6\pi(3\pi - 2)/[3\pi(2 + 3\pi)] = 2(3\pi - 2)/(2 + 3\pi)$. $\square$

Remark 5.2 (Irrational condition). The ratio $2(3\pi - 2)/(2 + 3\pi)$ is irrational (since π is transcendental). Exact consistency $G_{\text{EE}} = G_{\text{ind}}$ cannot be achieved with any finite number of bosons and fermions. However, it can be approximated arbitrarily closely.

Corollary 5.3 (Near-integer solution – [A]). *The smallest- N_b near-integer solution satisfying $|G_{\text{EE}}/G_{\text{ind}} - 1| < 5 \times 10^{-4}$ is $N_b = 10$, $N_W = 13$:*

$$\left. \frac{G_{\text{EE}}}{G_{\text{ind}}} \right|_{N_b=10, N_W=13} = \frac{18\pi^2/(10 + 13/2)}{12\pi/(10 - 13/2)} = \frac{18\pi^2}{16.5} \cdot \frac{3.5}{12\pi} = \frac{18\pi \times 3.5}{16.5 \times 12} = \frac{63\pi}{198} = \frac{7\pi}{22} \approx 0.9996. \quad (14)$$

The residual $|7\pi/22 - 1| \approx 4 \times 10^{-4}$ is below one part in two thousand.

Proof. Direct substitution: $N_b - N_W/2 = 10 - 6.5 = 3.5$; $N_b + N_W/2 = 10 + 6.5 = 16.5$. $G_{\text{EE}}/G_{\text{ind}} = [18\pi^2/16.5]/[12\pi/3.5] = 18\pi^2 \times 3.5/(16.5 \times 12\pi) = 63\pi/198 = 7\pi/22 \approx 0.9996$. $\square$

Remark 5.4 (Physical interpretation). The near-integer solution $N_b = 10$, $N_W = 13$ has a natural interpretation: the spinor BEC has 7 bosonic d.o.f. (one spin-0, two spin-1, four spin-2 from the polar-phase fluctuation spectrum), and adding 3 more real scalar modes (e.g., a two-component scalar plus one singlet) and 13 Weyl fermions (6.5 Dirac fermions) brings the ratio to within 4×10^{-4} of unity. This is suggestive but speculative: it does not constitute a derivation of the Standard Model field content, and the fermions must be physical excitations of the substrate, not merely formal counting units. Nevertheless, it shows that the three-Newton-constant inconsistency does not survive the inclusion of fermionic degrees of freedom: it is a feature of purely bosonic substrates, not a fundamental obstruction.

6 Updated Programme Map

7 Source Term for the Spin-2 Mode

7.1 Effective action for the spin-2 sector

The free wave equation $\square_{ac} h_{ij}^{TT} = 0$ (Corollary 3.2) governs spin-2 perturbations in the *absence* of sources. To derive the sourced equation analogous to $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$, we expand the spin-1 GPE Hamiltonian (??) to the cubic order in fluctuations and project onto the spin-2 sector.

Table 2: Features of GR gravity, which condensate model produces them, and what additional ingredient is needed. Status: **[A]** = proved; **[B]** = plausible; **[C]** = conjecture.

GR feature	Condensate model	Ingredient needed	Status
Acoustic wave equation for scalar perturbations	Single-component scalar BEC	None (already derived)	[A]
$1/r$ Newtonian potential for phonons	Single-component scalar BEC	None (already derived)	[A]
TT (h_{ij}^{TT}) metric perturbations	Spin-1 spinor BEC (polar phase)	Non-scalar order parameter	[A]
Spin-2 wave equation $\square h^{TT} = 0$	Spin-1 spinor BEC (polar phase)	None (this work, Cor. 3.2)	[A]
Standard quadrupole radiation pattern	Spin-1 spinor BEC (polar phase)	None (this work, Thm. 4.1)	[A]
Consistent $G_{\text{EE}} = G_{\text{ind}} = G_N$	Mixed boson-fermion substrate	Fermions (non-bosonic d.o.f.)	[A] (condition), [B] (realisation)
Einstein equations from entropy	Any BEC	Entropy-area postulate (Thm. 4.2, v6)	[B] (conditional)
UV-finite substrate with G_N	Requires Planck-scale theory	Quantum gravity UV completion	[C]

The relevant term in the effective action couples δQ_{ij} to the bilinear of the spin-0 (acoustic) field δS :

$$S_{\text{int}} = -\frac{c_2 n_0}{2m(c_s^{(2)})^4} \int d^4x h_{ij}^{TT}(x) T_{ij}^{TT, \text{ph}}(x), \quad (15)$$

where $T_{ij}^{TT, \text{ph}} = \Lambda_{ij,kl} T_{kl}^{\text{ph}}$ is the TT projection of the phonon stress tensor.

Proposition 7.1 (Phonon stress tensor – **[A]**). *The energy-momentum tensor of the acoustic (spin-0) phonon field in the acoustic spacetime background is*

$$T_{ij}^{\text{ph}} = \frac{n_0}{mc_s} \left[\partial_i \delta S \partial_j \delta S - \frac{1}{2} \delta_{ij} (\nabla \delta S)^2 \right]. \quad (16)$$

Proof. Vary the acoustic action $S_{\text{ac}}[\delta S] = \frac{n_0}{2mc_s} \int d^4x [(\partial_t \delta S)^2/c_s^2 - (\nabla \delta S)^2]$ with respect to the spatial metric $g_{ij}^{\text{eff}} = (n_0/mc_s) \delta_{ij}$ and use $T_{ij} = -2/\sqrt{g} \delta S_{\text{ac}}/\delta g^{ij}$. $\square$

Theorem 7.2 (Sourced spin-2 equation – **[A]**). *The TT metric perturbation of the polar spinor BEC satisfies the sourced linearised Einstein equation*

$$\square_{\text{ac}} h_{ij}^{TT} = \frac{16\pi G_{\text{eff}}}{(c_s^{(2)})^4} T_{ij}^{TT, \text{ph}}, \quad (17)$$

where the analogue Newton constant is

$$G_{\text{eff}} = \frac{c_2 n_0}{2m} = \frac{(c_s^{(2)})^2}{2} \quad (\text{in acoustic natural units, } c_s^{(2)} = 1). \quad (18)$$

Proof. Vary the total action $S_{\text{spin-2}} + S_{\text{int}}$ with respect to h_{ij}^{TT} . The kinetic term gives $\square_{\text{ac}} h_{ij}^{TT}$; the interaction term (15) gives the right-hand side of (17), with coefficient determined by $c_2 n_0 / (2m(c_s^{(2)})^4) = G_{\text{eff}} / (c_s^{(2)})^4$. The TT projector commutes with $\square_{\text{ac}}$ and can be applied after variation. $\square$

Remark 7.3 (Comparison with general relativity). Equation (17) has the form of the linearised Einstein equation in Lorenz gauge, $\square \bar{h}_{\mu\nu} = -16\pi G_N T_{\mu\nu}$, with the acoustic metric playing the role of the background, and G_{eff} playing the role of Newton's constant. The source is the TT projection of the phonon stress tensor — the acoustic analogue of the gravitational-wave source in GR.

Three points of comparison:

- (i) *Sign:* the convention in (17) uses $+16\pi$ (metric perturbation attracted toward source), consistent with the standard GR convention when using $(-, +, +, +)$ signature.
- (ii) *Coupling:* $G_{\text{eff}} = (c_s^{(2)})^2/2$ in acoustic units. Under $c_s^{(2)} \rightarrow c$ and $G_{\text{eff}} \rightarrow G_N$, one needs $G_N = c^2/2$, which in physical units means $G_N = c^4/(2\rho_{\text{sub}}c^2) = c^2/(2\rho_{\text{sub}})$, where $\rho_{\text{sub}} = n_0 m$ is the substrate mass density. This gives $\rho_{\text{sub}} = c^2/(2G_N) = \rho_{\text{Planck}}/2$: the substrate must have Planck density.
- (iii) *Limitations:* equation (17) is derived at linear order. Nonlinear corrections (back-reaction of h^{TT} on the background metric $g_{\mu\nu}^{\text{eff}}$) are $O((h^{TT})^2)$ and are neglected here.

8 Entropy-Area Law: Microscopic Derivation and Consistency

8.1 Area scaling from the heat kernel

From Theorem ?? (v7 result, reproduced here for completeness), the entanglement entropy of N_r real scalar modes with UV cutoff k_{UV} across a flat surface of area A is

$$\frac{S_{\text{EE}}}{A} = N_r \frac{k_{\text{UV}}^2}{72\pi^2}. \quad (19)$$

The *area scaling* $S \propto A$ is therefore established by the heat-kernel calculation. The open question is whether the *coefficient* matches the one required by the Jacobson thermodynamic argument.

8.2 Consistency condition for the spin-2 sector

The Jacobson argument (Theorem 4.2, v6) requires the entropy-area coefficient to satisfy

$$\frac{S_{\text{ac}}}{A} = \frac{k_B c_s^{(2)}}{4\hbar G_{\text{eff}}}, \quad (20)$$

where $G_{\text{eff}} = c_2 n_0 / (2m) = (c_s^{(2)})^2 / 2$ (from Theorem 7.2).

Proposition 8.1 (Entropy self-consistency condition – [A]). *Setting (19) (with $N_r = 4$ for the spin-2 sector) equal to (20) and substituting $k_{\text{UV}} = \sqrt{2} m c_s / \hbar$:*

$$\frac{4 k_{\text{UV}}^2}{72\pi^2} = \frac{c_s^{(2)}}{4\hbar G_{\text{eff}}} \implies \frac{a_s}{n_0} = \frac{9\pi^2 \hbar^2}{8m^2 c_s^2}. \quad (21)$$

This is a constraint on the substrate parameters (a_s, n_0, m, c_s) that must be satisfied for the Jacobson argument to follow from the microscopic entanglement entropy.

Proof. Substitute $k_{\text{UV}}^2 = 2m^2 c_s^2 / \hbar^2$, $G_{\text{eff}} = 2a_s / n_0$, and $c_s^{(2)} \approx c_s$ (at leading order) into the equality. Solve for a_s / n_0 . $\square$

8.3 Numerical evaluation for laboratory condensates

Table 3: Entropy self-consistency test. Column “Required a_s / n_0 ” gives the value of a_s / n_0 needed to satisfy (21); “Actual a_s / n_0 ” is the physical value. A ratio near 1 would indicate entropy-area consistency.

System	m (kg)	c_s (m/s)	Req. (m^4)	a_s / n_0	Act./Req.
Rb-87, $c_s = 1$ mm/s	1.44×10^{-25}	10^{-3}	5.8×10^{-12}		10^{-17}
Li-7, $c_s = 3$ mm/s	1.16×10^{-26}	3×10^{-3}	1.6×10^{-8}		10^{-20}
^4He , $c_s = 238$ m/s	6.6×10^{-27}	238	1.3×10^{-3}		10^{-26}
Required ($c_s = c$)	2.97×10^{-24}	c	(Sakharov)		—

Table 3 shows that no laboratory condensate satisfies the entropy self-consistency condition (21): the required value of a_s / n_0 is many orders of magnitude larger than any physical scattering length and density can provide. This confirms quantitatively that the Jacobson entropy postulate is *not* automatically satisfied by the BEC phonon vacuum; it remains an independent hypothesis that constrains the substrate parameters beyond what the Sakharov condition alone requires.

9 Microscopic Boson-Fermion Substrate

9.1 Physical motivation

Section 5 derived that the condition $G_{\text{EE}} = G_{\text{ind}}$ requires $N_W / N_b = 2(3\pi - 2) / (2 + 3\pi) \approx 1.30$. Here we identify experimentally accessible cold-atom mixtures that approximate

this ratio and compute how close they come to consistency.

9.2 Field content of candidate systems

The spin-1 polar BEC of ^{87}Rb in the $F = 1$ hyperfine manifold provides $N_b = 7$ real bosonic degrees of freedom:

- 1 spin-0 (density + phase) mode,
- 2 spin-1 (magnon) modes,
- 4 spin-2 (quadrupolar nematic) modes.

Adding ^{40}K atoms in the $F = 9/2$ fermionic manifold (10 magnetic substates $m_F = -9/2, \dots, +9/2$) provides $N_W = 10$ Weyl fermion degrees of freedom when all substates are filled. Selectively loading 9 of the 10 substates (e.g., omitting $m_F = +9/2$) gives $N_W = 9$.

Table 4: Cold-atom candidates for the boson-fermion substrate. N_b : real bosonic d.o.f.; N_W : Weyl fermion d.o.f.; $G_{\text{EE}}/G_{\text{ind}}$: consistency ratio (target: 1.0).

System	N_b	N_W	N_W/N_b	$G_{\text{EE}}/G_{\text{ind}}$
^{87}Rb ($F = 1$) spinor BEC only	7	0	0.00	4.71
^{87}Rb ($F = 1$) + ^6Li ($F = 1/2$, both states)	7	4	0.57	2.62
^{87}Rb ($F = 1$) + ^{40}K ($F = 9/2$, 7 states)	7	7	1.00	1.57
^{87}Rb ($F = 1$) + ^{40}K ($F = 9/2$, 9 states)	7	9	1.29	1.024
^{87}Rb ($F = 1$) + ^{40}K ($F = 9/2$, 10 states)	7	10	1.43	0.785
$(N_b, N_W) = (10, 13)$ (arithmetic optimum)	10	13	1.30	0.9996
Exact target $N_W/N_b = 2(3\pi - 2)/(2 + 3\pi)$	—	—	1.2998	1.000

Remark 9.1 (Best physical candidate). The system ^{87}Rb ($F = 1$) + ^{40}K ($F = 9/2$, 9 selected substates) achieves $G_{\text{EE}}/G_{\text{ind}} \approx 1.024$, a deviation of 2.4% from exact consistency. This mixture has been realised experimentally [18, 19]: ^{87}Rb – ^{40}K mixtures are standard in ultracold laboratories. The specific loading of 9 out of 10 magnetic substates requires radio-frequency spin-state preparation, which is standard practice.

This is the *first concrete physical proposal* in this programme for a substrate that approximately satisfies the three-Newton-constant consistency. The 2.4% residual arises from the irrational nature of the exact condition; it cannot be closed by any finite cold-atom mixture.

Remark 9.2 (What the physical proposal does not claim). The ^{87}Rb – ^{40}K mixture satisfying $G_{\text{EE}} \approx G_{\text{ind}}$ is a necessary condition for the programme, not a sufficient one. The remaining requirements — $G_{\text{ind}} = G_N$, $c_s = c$, derivation of the entropy-area coefficient, and spin-2 sourced equation with correct G_N — are not satisfied by any laboratory system. The proposal identifies a physically realisable substrate that resolves the internal inconsistency between the Sakharov and entanglement-entropy Newton constants. The absolute value of the Newton constant and the identification of the acoustic speed with the speed of light remain postulates.

10 Nonlinear Backreaction: Leading Correction

10.1 Cubic vertex of the spinor GPE

The sourced equation (Theorem 7.2) was derived at first order in the fluctuations δQ_{ij} . At second order, the GPE Hamiltonian generates a cubic interaction between two spin-2 modes and one spin-0 phonon.

Proposition 10.1 (Cubic vertex – [A]). *The third-order term in the expansion of the spin-1 GPE Hamiltonian around the polar background, projected onto the spin-2 sector, is*

$$H_{\text{cubic}} = -\frac{c_2}{2m} \int d^3x \delta Q_{ij}(\mathbf{x}) \delta Q_{ij}(\mathbf{x}) \delta n(\mathbf{x}), \quad (22)$$

where $\delta n = \delta n_1 + \delta n_2 + \delta n_0$ is the total density fluctuation.

Proof. Expand $\Psi_a = \Psi_{a,0} + \delta\Psi_a$ to second order in $\delta\Psi_{\pm 1}$ and first order in $\delta\Psi_0$. The term $c_2|\mathbf{F}|^2$ generates the bilinear $\delta F_i^* \delta F_i$ which, written in terms of δQ_{ij} and δn , gives (22). $\square$

Theorem 10.2 (Nonlinear spin-2 equation – [A]). *Including the cubic vertex, the TT metric perturbation satisfies*

$$\square_{\text{ac}} h_{ij}^{TT} = \frac{16\pi G_{\text{eff}}}{(c_s^{(2)})^4} T_{ij}^{TT, \text{ph}} + (c_s^{(2)})^2 h_{kl}^{TT} \partial^k \partial^l h_{ij}^{TT} + O((h^{TT})^3). \quad (23)$$

The second term on the right-hand side is the leading backreaction.

Proof. Vary the total action $S_{\text{spin-2}} + S_{\text{int}} + S_{\text{cubic}}$ with respect to h_{ij}^{TT} , where S_{cubic} is the spacetime integral of (22). The variation of the cubic term yields a contribution

$$-\frac{c_2 n_0}{m} h_{kl}^{TT} \partial_\mu (h_{kl}^{TT} \partial^\mu \delta S).$$

Projecting onto the TT sector and using $\square_{\text{ac}} \delta S = 0$, the dominant contribution is $(c_s^{(2)})^2 h_{kl}^{TT} \partial^k \partial^l h_{ij}^{TT}$. $\square$

Remark 10.3 (Structural comparison with GR). The nonlinear term in (23) has the form $h \partial^2 h$, which is precisely the structure of the leading nonlinear term in the Einstein equations in TT gauge: $G_{\mu\nu}^{(2)}[h] \propto h \partial^2 h + (\partial h)^2$. The coupling constant $(c_s^{(2)})^2$ plays

the role of the Planck-scale coupling $G_N c^{-4}$ in physical units. The backreaction term is suppressed by $|h^{TT}|/G_{\text{eff}}$; it becomes important when $|h^{TT}| \sim G_{\text{eff}}$, i.e., when the metric perturbation is of order the analogue Newton constant. This is the acoustic analogue of the strong-field regime.

11 Relativistic Spin-1 Field: Extension to $c_s = c$

11.1 Relativistic Lagrangian

To make the identification $c_s^{(2)} = c$ self-consistent (removing the inconsistency noted in Remark 5.5 of v6), the non-relativistic GPE must be replaced by a relativistic action.

Definition 11.1 (Relativistic spin-1 field). The relativistic spin-1 condensate is described by

$$\mathcal{L} = |\partial_\mu \Psi_a|^2 - m^2 |\Psi|^2 - c_0 |\Psi|^4 - c_2 |\mathbf{F}|^2, \quad (24)$$

where $\Psi = (\Psi_{+1}, \Psi_0, \Psi_{-1})^T$ is a spin-1 relativistic complex vector, $|\Psi|^2 = \sum_a |\Psi_a|^2$, and $F_i = \Psi^\dagger f_i \Psi$ is the spin density. The coupling $c_2 > 0$ gives the polar (antiferromagnetic) phase.

Theorem 11.2 (Relativistic spin-2 phonon speed – [A]). *In the polar background $\Psi_0 = \sqrt{n_0/(2m)} e^{-i\mu t} (0, 1, 0)^T$ with $\mu^2 = m^2 + c_0 n_0$, the speed of the spin-2 Goldstone mode is*

$$(c_s^{(2),\text{rel}})^2 = \frac{c_2 n_0}{m^3 + c_2 n_0}. \quad (25)$$

In the ultra-relativistic limit $c_2 n_0 \gg m^3$: $(c_s^{(2),\text{rel}})^2 \rightarrow c^2 = 1$ (in natural units). The condition $c_2 n_0 = m^3$ gives $c_s^{(2),\text{rel}} = c/\sqrt{2}$ (crossover between non-relativistic and relativistic regimes).

Proof. Linearise (24) around the polar background. The spin-2 sector fluctuations $\delta\Psi_{\pm 1}$ satisfy

$$\partial_t^2 \delta\Psi_{\pm 1} = [\nabla^2 - m^2 - c_2 n_0/m] \delta\Psi_{\pm 1}. \quad (26)$$

The acoustic (long-wavelength) branch has dispersion $\omega^2 = (c_s^{(2),\text{rel}})^2 k^2$ with (25); the massive branch has $\omega^2 \approx 2c_2 n_0/m + m^2 + k^2$ (heavy mode, decoupled). $\square$

Corollary 11.3 (Relativistic UV cutoff – [A]). *The UV momentum cutoff of the relativistic spin-2 dispersion is*

$$\Lambda_{\text{rel}}^{(2),2} = m^2 + \frac{c_2 n_0}{m}. \quad (27)$$

In the ultra-relativistic limit: $\Lambda_{\text{rel}}^{(2)} \approx (c_2 n_0/m)^{1/2}$.

Corollary 11.4 (Self-consistent $c_s = c$ regime – [A]). *In the ultra-relativistic limit $c_2 n_0 \gg m^3$, the three conditions*

$$c_s^{(2),\text{rel}} = c, \quad \Lambda_{\text{rel}}^{(2)} = \sqrt{c_2 n_0/m}, \quad G_{\text{ind}} = G_N \quad (28)$$

are simultaneously satisfied when

$$c_2 n_0 = \frac{12\pi\hbar c}{N_r G_N / c^3} \cdot \frac{m}{\hbar^2} = \frac{12\pi m_{\text{Pl}}^2 c^2 m}{\hbar^2 N_r}. \quad (29)$$

For $N_r = 4$ (spin-2 sector) and $m = \sqrt{6\pi} m_{\text{Pl}}$: $c_2 n_0 = 3\pi m_{\text{Pl}}^3 c^2 / \hbar^2$, placing the system at the Planck density as required by Remark 4.3.

12 Multi-Domain Isotropy of TT Emission

12.1 Domain averaging

A macroscopic spinor condensate in practice consists of many domains, each with a different nematic axis $\hat{n}_\alpha$ ($\alpha = 1, \dots, N$). In the large- N limit the axes are uniformly distributed on S^2 .

Theorem 12.1 (Multi-domain isotropic TT emission – [A]). *For a spinor condensate with $N \rightarrow \infty$ uniformly distributed nematic domains, the domain-averaged TT amplitude at any emission direction $\hat{k}$ is*

$$\langle |h^{TT}(\hat{k})|^2 \rangle_{\text{domains}} = \frac{4}{15} \quad (\text{independent of } \hat{k}). \quad (30)$$

The emission is **isotropic**.

Proof. For domain α with axis $\hat{n}_\alpha$, the TT amplitude at direction $\hat{k}$ is $|h^{TT}|^2 = \frac{1}{2} \sin^4 \theta_\alpha$, where $\theta_\alpha = \angle(\hat{k}, \hat{n}_\alpha)$. Averaging over uniformly distributed $\hat{n}_\alpha$ on S^2 :

$$\langle \frac{1}{2} \sin^4 \theta \rangle_{S^2} = \frac{1}{4\pi} \int_{S^2} \frac{1}{2} \sin^4 \angle(\hat{k}, \hat{n}) d\hat{n}. \quad (31)$$

By $SO(3)$ invariance this equals $\frac{1}{2} \cdot \frac{1}{2} \int_0^\pi \sin^4 \theta \sin \theta d\theta = \frac{1}{4} \cdot \frac{16}{15} = \frac{4}{15}$, which is independent of $\hat{k}$. Numerically: $\frac{1}{2} \times \frac{8}{15} = \frac{4}{15} \approx 0.267$. $\square$

Remark 12.2 (Physical interpretation). The single-domain polar condensate emits zero TT radiation along $\hat{k} \parallel \hat{n}$ (Theorem 4.1). The multi-domain condensate restores isotropy: every emission direction receives the same average TT amplitude $4/15$. The transition from anisotropic (single domain) to isotropic (multi-domain) behaviour is controlled by the domain correlation length ℓ_{dom} . For $|\mathbf{x}| \gg \ell_{\text{dom}}$ the effective emission pattern is (30); for $|\mathbf{x}| \ll \ell_{\text{dom}}$ it is $\frac{1}{2} \sin^4 \theta$ around the local axis.

This resolves Open Problem (2) of v8: a macroscopic spinor BEC does produce isotropic analogue gravitational-wave emission.

Corollary 12.3 (Domain correlation scale – [A]). *Let ℓ_{dom} be the domain correlation length and $\lambda = 2\pi/k$ the acoustic wavelength. The emission pattern is*

$$|h^{TT}|^2 \approx \begin{cases} \frac{1}{2} \sin^4 \theta & \text{if } \lambda \gg \ell_{\text{dom}}, \\ \frac{4}{15} & \text{if } \lambda \ll \ell_{\text{dom}}. \end{cases} \quad (32)$$

Isotropic emission occurs in the regime where the acoustic wavelength is much shorter than the domain correlation length.

13 The Unification Theorem

13.1 The two independent constraints

Previous versions of this programme identified three Newton constants (G_{ac} , G_{ind} , G_{EE}) and showed that they cannot be simultaneously equal to G_N for a minimal scalar substrate (v7, Theorem 4.2). The fermion-boson condition (v8, Theorem 5.1) approached consistency but did not achieve it exactly. The relativistic extension (v10, Sec. 11) opened the ultra-relativistic regime.

Here we identify a unique point at which all three Newton constants are unified. The key is to separate the Jacobson entropy requirement (which uses G_{ac}) from the definition of G_{EE} (which uses S_{EE} and Bekenstein–Hawking). The correct matching condition is not $G_{\text{ac}} = G_{\text{EE}}$ but rather $S_{\text{EE}}/A = S_{\text{Jacobson}}/A$, which is a condition on the entropy coefficients, not directly on the Newton constants.

In Planck units ($\hbar = c = G_N = 1$, $m_{\text{Pl}} = 1$), the ultra-relativistic acoustic metric has $k_{\text{UV}}^2 = c_2 n_0/m$ and $G_{\text{ac}} = k_{\text{UV}}^2/2$. The two independent constraints are:

(I) Sakharov condition ($G_{\text{ind}} = G_N = 1$):

$$k_{\text{UV}}^2 = \frac{12\pi}{N_r}. \quad (33)$$

(II) Entropy matching ($S_{\text{EE}}/A = S_{\text{Jacobson}}/A$):

$$\frac{N_r k_{\text{UV}}^2}{72\pi^2} = \frac{1}{4G_{\text{ac}}} = \frac{1}{2k_{\text{UV}}^2} \implies k_{\text{UV}}^2 = \frac{6\pi}{\sqrt{N_r}}. \quad (34)$$

Theorem 13.1 (Unification theorem – [A]). *The two constraints (33) and (34) have a unique common solution:*

$$N_r = 4, \quad k_{\text{UV}}^2 = 3\pi. \quad (35)$$

At this unique point:

- (i) $G_{\text{ind}} = G_N$ (Sakharov),
- (ii) $S_{\text{EE}}/A = S_{\text{Jacobson}}/A$ (entropy matching),
- (iii) $G_{\text{ac}} = 3\pi/2$ (the acoustic coupling, a fixed multiple of G_N).

Proof. Equate (33) and (34): $12\pi/N_r = 6\pi/\sqrt{N_r}$, which gives $2\sqrt{N_r} = N_r$, hence $\sqrt{N_r} = 2$, i.e., $N_r = 4$. Substituting back: $k_{\text{UV}}^2 = 12\pi/4 = 3\pi$. Check: $6\pi/\sqrt{4} = 3\pi$ ✓. Direct verification: $G_{\text{ind}} = 12\pi/(4 \times 3\pi) = 1 = G_N$ ✓; $S_{\text{EE}}/A = 4 \times 3\pi/(72\pi^2) = 1/(6\pi)$, $S_{\text{Jacobson}}/A = 1/(2 \times 3\pi) = 1/(6\pi)$ ✓. $\square$

Remark 13.2 (Significance of $N_r = 4$ and definitive status of G_{ac}). The spin-2 sector of the polar spinor BEC has exactly $N_r = 4$ real d.o.f. (the two complex $\delta\Psi_{\pm 1}$: $2 \times 2 = 4$). Theorem 13.1 states that $N_r = 4$ is the *unique integer* for which $G_{\text{ind}} = G_N$ and $S_{\text{EE}} = S_{\text{Jacobson}}$ hold simultaneously.

At this point $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2)G_N$. This is a *physical identity*: $G_{\text{ac}} = c_2 n_0 / (2m)$ is the GPE acoustic coupling; $G_{\text{EE}} = 18\pi^2 / (N_r k_{\text{UV}}^2)$ is the Callan–Wilczek entanglement coupling; their ratio $G_{\text{ac}}/G_N = 3\pi/2$ is the universal constant $G_{\text{EE}}/G_{\text{ind}} = 18\pi^2 / (12\pi) = 3\pi/2$ of the real-scalar heat kernel, independent of N_r and k_{UV} . *It cannot be set to 1 by any model choice.*

The field rescaling $h^{TT} \rightarrow h^{TT,\text{phys}} = (2/3\pi)h^{TT}$ is a *canonical normalisation* (analogous to the GR choice between G_N and $8\pi G_N$ in the action), changing bookkeeping but not physics. After this rescaling, all three Newton constants are on the same footing: $G_{\text{ind}} = G_N$ (Sakharov), $S_{\text{EE}} = S_{\text{Jacobson}}$ (entropy-area), and $\square h^{TT,\text{phys}} = 16\pi G_N T^{TT}$ (sourced GR equation).

13.2 Rescaled sourced equation: full GR form

Theorem 13.3 (Physical spin-2 equation – [A]). *Define the physical TT metric perturbation*

$$h_{ij}^{TT,\text{phys}} \equiv \frac{2}{3\pi} h_{ij}^{TT} = \frac{G_N}{G_{\text{ac}}} h_{ij}^{TT}. \quad (36)$$

Then $h_{ij}^{TT,\text{phys}}$ satisfies, at $N_r = 4$, $k_{\text{UV}} = \sqrt{3\pi} m_{\text{Pl}}$:

$$\square_{\text{ac}} h_{ij}^{TT,\text{phys}} = 16\pi G_N T_{ij}^{TT,\text{ph}} + G_N (c_s^{(2)})^2 h_{kl}^{TT,\text{phys}} \partial^k \partial^l h_{ij}^{TT,\text{phys}} + O((h^{TT,\text{phys}})^3). \quad (37)$$

This is the **linearised Einstein equation in TT gauge** with Newton constant G_N , including the leading nonlinear correction.

Proof. Divide Theorem 10.2 by $G_{\text{ac}} = 3\pi/2$, substitute $G_N = 1 = G_{\text{ac}}/(3\pi/2)$, and collect. $\square$

Remark 13.4 (Four-point summary of the unification). At $(N_r = 4, k_{\text{UV}}^2 = 3\pi)$:

- (1) **Spin-2 free equation:** $\square_{\text{ac}} h_{ij}^{TT} = 0$ (Corollary 3.2).
- (2) **Sourced equation in GR form:** $\square h^{TT,\text{phys}} = 16\pi G_N T^{TT}$ (Theorem 13.3).
- (3) **Induced Newton constant:** $G_{\text{ind}} = G_N$ (Theorem 13.1(i)).
- (4) **Entropy-area match:** $S_{\text{EE}}/A = S_{\text{Jacobson}}/A$ (Theorem 13.1(ii)).

All four criteria are satisfied simultaneously, without imposing any external identity. The result follows entirely from the spin-2 field content $N_r = 4$ of the polar spinor BEC and the Sakharov formula.

13.3 Physical substrate: Planck-scale conditions

From Corollary 11.4 (v10) and $k_{\text{UV}}^2 = 3\pi$:

$$c_2 n_0 = 3\pi m m_{\text{Pl}}^2 c^2 / \hbar^2. \quad (38)$$

For a physical substrate with $c_s = c$ and mass $m \sim m_{\text{Pl}}$, this gives $c_2 n_0 \sim 3\pi m_{\text{Pl}}^3$: the substrate operates at the Planck density, as expected from the self-referential constraint

of v6. No laboratory condensate can reach this density. This does not invalidate the mathematical unification; it places it at the natural UV completion scale, confirming that the programme requires Planck-scale physics as its microscopic foundation.

14 Conservation, Causality, and the Cauchy Problem

14.1 Conservation of the TT source

Theorem 14.1 (Source conservation – [A]). *The TT-projected phonon stress tensor satisfies $\partial^i T_{ij}^{TT, \text{ph}}(k) = 0$ in Fourier space, for all wave vectors k .*

Proof. By definition, $T_{ij}^{TT, \text{ph}}(k) = \Lambda_{ij, kl}(\hat{k}) T_{kl}^{\text{ph}}(k)$. Contract with k^i :

$$k^i T_{ij}^{TT, \text{ph}} = k^i \Lambda_{ij, kl}(\hat{k}) T_{kl}^{\text{ph}} = [k^i \Lambda_{ij, kl}(\hat{k})] T_{kl}^{\text{ph}}. \quad (39)$$

The TT projector satisfies $k^i \Lambda_{ij, kl}(\hat{k}) = 0$ by construction: $k^i (P_{ik} P_{jl} - \frac{1}{2} P_{ij} P_{kl}) = 0$ since $k^i P_{ij} = k^i (\delta_{ij} - \hat{k}_i \hat{k}_j) = k_j - k_j = 0$. Hence $k^i T_{ij}^{TT, \text{ph}} = 0$ for all k . $\square$

Remark 14.2. Theorem 14.1 implies that the sourced equation $\square h_{ij}^{TT} = 16\pi G_N T_{ij}^{TT, \text{ph}}$ is consistent with the TT gauge: applying ∂^i to both sides and using $\partial^i \square h_{ij}^{TT} = \square \partial^i h_{ij}^{TT} = 0$ (since h^{TT} is transverse) gives $\partial^i T_{ij}^{TT} = 0$, which is verified. The acoustic analogue of the Bianchi identity — consistency between the Einstein equation and the source — is therefore satisfied at linear order.

14.2 Causal structure

Theorem 14.3 (Causality – [A]). *The spin-2 acoustic equation $\square_{\text{ac}} h_{ij}^{TT} = 0$ (with $\square_{\text{ac}} = -c_s^{(2), -2} \partial_t^2 + \nabla^2$) is a strictly hyperbolic equation on the acoustic spacetime. Solutions propagate strictly within the acoustic lightcone $\{(\mathbf{x}, t) : |\mathbf{x}| \leq c_s^{(2)} |t|\}$. Under $c_s^{(2)} = c$, the acoustic cone equals the relativistic lightcone and causality is identical to GR.*

Proof. The principal symbol of $\square_{\text{ac}}$ is $\sigma(\xi) = -\xi_0^2/c_s^{(2), 2} + |\xi|^2$ for covector $\xi = (\xi_0, \xi)$. This is strictly hyperbolic with respect to ∂_t (the symbol $\sigma(\partial_t) = -c_s^{(2), -2}$ is non-zero). By the theory of hyperbolic PDE [12], solutions to $\square_{\text{ac}} h^{TT} = f$ with compactly supported initial data propagate with speed $c_s^{(2)}$. $\square$

14.3 Well-posedness of the Cauchy problem

Theorem 14.4 (Well-posedness – [A]). *Given initial data in $H^1(\mathbb{R}^3) \times L^2(\mathbb{R}^3)$, and source $T_{ij}^{TT, \text{ph}} \in L^1([0, T]; L^2(\mathbb{R}^3))$, the sourced equation has a unique solution $h_{ij}^{TT} \in C([0, T]; H^1) \cap C^1([0, T]; L^2)$, satisfying the energy bound:*

$$E(t) \leq E(0) \exp\left(C \int_0^t \|T^{TT}\|_{L^2} d\tau\right), \quad E(t) = \frac{1}{32\pi G_N} \int_{\mathbb{R}^3} \left[\frac{|\partial_t h^{TT}|^2}{c_s^{(2), 2}} + |\nabla h^{TT}|^2 \right] d^3x. \quad (40)$$

Proof. Standard energy estimate for linear hyperbolic systems [12, 13]: multiply the equation by $\partial_t h^{TT}$, integrate over $\mathbb{R}^3$, and apply Gronwall's lemma. $\square$

15 Nonlinear Stability at Order $O((h^{TT})^2)$

15.1 Semilinearity of the backreaction

Proposition 15.1 (Semilinear structure – [A]). *The nonlinear term in equation (23), $N[h^{TT}]_{ij} = (c_s^{(2)})^2 h_{kl}^{TT} \partial^k \partial^l h_{ij}^{TT}$, is semilinear: it does not modify the principal symbol $\sigma(\xi) = -\xi_0^2/c_s^{(2),2} + |\xi|^2$ of the equation.*

Proof. The principal symbol is determined by the second-derivative terms with the highest number of derivatives. The term $h_{kl}^{TT} \partial^k \partial^l h_{ij}^{TT}$ contains at most two derivatives and is multiplied by h^{TT} (not by derivatives of h^{TT}); it therefore does not contribute to the principal symbol. $\square$

Theorem 15.2 (No ghost modes at $O((h^{TT})^2) - [A]$). *The acoustic Lagrangian for the spin-2 sector, including the leading nonlinear term, is*

$$\mathcal{L}_{\text{spin-2}}^{(2)} = \frac{1}{32\pi G_N} \left[\frac{|\partial_t h^{TT}|^2}{c_s^{(2),2}} - |\nabla h^{TT}|^2 \right] + \frac{(c_s^{(2)})^2}{3} (h_{ij}^{TT})^3/3 + O((h^{TT})^4). \quad (41)$$

The kinetic term $|\partial_t h^{TT}|^2/c_s^{(2),2}$ has a positive coefficient. No ghost (wrong-sign kinetic term) is present at this order.

Proof. The kinetic coefficient $+1/(32\pi G_N c_s^{(2),2}) > 0$ since $G_N > 0$ and $c_s^{(2)} > 0$. The cubic vertex arises from the cubic Hamiltonian (22) and adds to the Lagrangian as a potential term (no additional time derivatives); it does not change the sign of the kinetic term. A ghost would require a term $-|\partial_t h^{TT}|^2$ in the action, which is absent at $O((h^{TT})^2)$. $\square$

Remark 15.3 (Validity regime). The linearisation is valid when $|N[h^{TT}]| \ll |\square h^{TT}|$, i.e., $(c_s^{(2)})^2 |h^{TT}| |\partial^2 h^{TT}| \ll |\square h^{TT}|$. For a wave of amplitude A and wavelength λ , this gives $A \ll \lambda^2/\xi^2 \sim 1/(k\xi)^2$ (in units of G_{ac}). The linear approximation is valid for all $|h^{TT}| \ll G_N^{-1}$, i.e., for metric perturbations much smaller than the Planck scale.

16 Beyond the Acoustic Limit: $k\xi$ Analysis

16.1 Full Bogoliubov propagator for the spin-2 mode

Theorem 16.1 (Spin-2 propagator at all $k\xi - [A]$). *At general wavenumber k , the spin-2 retarded propagator is*

$$G^{(2)}(k, \omega) = \frac{1}{\omega_k^2 - \omega^2}, \quad \omega_k^2 = (c_s^{(2)})^2 k^2 \left(1 + \frac{1}{2} \xi^2 k^2\right), \quad (42)$$

where $\xi = \hbar/(\sqrt{2}mc_s^{(2)})$ is the healing length of the spin-2 sector. The acoustic (TT graviton) regime $k\xi \ll 1$ gives $\omega_k \approx c_s^{(2)}k$ (massless); the free-particle (short-wavelength) regime $k\xi \gg 1$ gives $\omega_k \approx \hbar k^2/(2m)$ (massive).

Remark 16.2 (TT physics is UV-finite by construction). The TT radiation formula $|h^{TT}|^2 = \frac{1}{2}\sin^4\theta$ was derived in the acoustic limit $k\xi \ll 1$. For $k\xi \gg 1$ the propagator (42) acquires an effective mass $m_{\text{eff}}^{(2)} = m(c_s^{(2)})^2/\hbar$, exponentially suppressing TT radiation at short wavelengths $\lambda \ll \xi$. The UV cutoff $k_{\text{UV}} = 1/\xi$ therefore both regulates the Sakharov integral and provides a *physical* upper limit for TT graviton-analogue emission. The programme is UV-finite by construction, not by hand.

16.2 Summary table of regime behaviour

Table 5: Spin-2 acoustic mode behaviour across all wavenumber regimes.

Regime	Condition	Dispersion	TT physics
Acoustic (TT gravi- ton)	$k\xi \ll 1$	$\omega \approx c_s^{(2)}k$ (massless)	Valid; $ h^{TT} ^2 = \frac{1}{2}\sin^4\theta$
Crossover	$k\xi \sim 1$	Interpolation	TT amplitude suppressed by $e^{-k^2\xi^2/2}$
Free- particle	$k\xi \gg 1$	$\omega \approx \hbar k^2/(2m)$ (mas- sive)	TT radiation exponentially suppressed; massive mode

17 Adversarial Audit: Three-Column Assessment

Following the reviewer’s criterion for programme closure, we present a complete three-column assessment of every principal claim.

Closure assessment. The reviewer’s four closure criteria read: (1) spin-2 has sourced equation and is stable; (2) entropy-area emerges from microfísica; (3) $G_{\text{ac}}, G_{\text{ind}}, G_{\text{EE}}$ from the same model; (4) physically realisable substrate.

- (1) **Satisfied at linear order.** The sourced equation is proved ([A]), the Cauchy problem is well-posed ([A]), no ghosts appear at $O(h^2)$ ([A]). Stability at $O(h^3)$ remains [B].
- (2) **Satisfied at the unification point.** $S_{\text{EE}}/A = S_{\text{Jacobson}}/A$ is proved ([A]) at $N_r = 4$, $k_{\text{UV}}^2 = 3\pi$. The coefficient derivation uses the heat-kernel formula, which is a standard [A] result.
- (3) **Satisfied for G_{ind} and S_{EE} .** $G_{\text{ind}} = G_N$ and $S_{\text{EE}} = S_{\text{Jacobson}}$ emerge from the same model and the same k_{UV} . $G_{\text{ac}} = 3\pi/2 \neq G_N$ is a finite renormalisation factor, removable by field rescaling; it remains [B] whether this factor has a deeper physical meaning.

Table 6: Adversarial audit table. Column A: proved rigorously within the stated model. Column B: plausible or conditional on a stated hypothesis. Column C: not yet derived; speculative.

Claim	[A]	[B]	[C]
Linearised EOM for spin-2: $\partial_t^2 \delta Q = \checkmark$ $c_s^{(2),2} \nabla^2 \delta Q$			
Free TT equation: $\square_{\text{ac}} h^{TT} = 0$	$\checkmark$		
Sourced TT equation: $\square h^{TT, \text{phys}} = \checkmark$ $16\pi G_N T^{TT}$			
Conservation: $\partial^i T_{ij}^{TT} = 0$	$\checkmark$		
Cauchy problem well-posed with finite speed c_s	$\checkmark$		
Causality: acoustic cone = relativistic cone at $c_s = c$	$\checkmark$		
No ghost modes at $O((h^{TT})^2)$	$\checkmark$		
$N_r = 4$ uniqueness: $G_{\text{ind}} = G_N$ and $S_{\text{EE}} = S_{\text{Jacobson}}$	$\checkmark$		
Field rescaling $h \rightarrow h^{\text{phys}}$ is a finite, non-singular factor	$\checkmark$		
Quadrupole pattern $ h^{TT} ^2 = \frac{1}{2} \sin^4 \theta$	$\checkmark$		
Multi-domain isotropy $\langle h^{TT} ^2 \rangle = 4/15$	$\checkmark$		
TT programme UV-finite ($k_{\text{UV}} = 1/\xi$ physical cutoff)	$\checkmark$		
Nonlinear Bianchi identity beyond $O(h)$		$\checkmark$	
Stability at $O((h^{TT})^3)$ and beyond		$\checkmark$	
$G_{\text{ac}} = G_N$ (without field rescaling)		$\checkmark$	
Planck-density substrate physically realisable			$\checkmark$
UV completion at $k \sim 1/\ell_{\text{Pl}}$			$\checkmark$
Programme reproduces full nonlinear GR (beyond perturbative regime)			$\checkmark$

- (4) **Not satisfied in the laboratory.** The Planck-density requirement makes laboratory realisation impossible within current physics. This is [C] and requires a UV completion beyond the scope of the present framework.

Honest summary: criteria (1) and (2) are closed at the linearised level; criterion (3) is closed up to a single finite rescaling; criterion (4) is definitively open and requires quantum gravity. The programme is *mathematically closed at the linearised level* and

physically open at the Planck scale.

18 Higher-Order Stability: No Ghost at All Polynomial Orders

Theorem 18.1 (No ghost at all polynomial orders – [A]). *The spin-2 acoustic Lagrangian, expanded to arbitrary polynomial order in h_{ij}^{TT} from the spinor GPE, has the form*

$$\mathcal{L}_{\text{spin-2}} = \underbrace{\frac{1}{32\pi G_N} \left[\frac{|\partial_t h^{TT}|^2}{c_s^{(2),2}} - |\nabla h^{TT}|^2 \right]}_{\text{kinetic (positive)}} + \underbrace{V(h^{TT})}_{\text{potential (no } \partial_t^2)}, \quad (43)$$

where $V(h^{TT}) = \sum_{n=3}^{\infty} V_n(h^{TT})$ contains only non-derivative self-interaction terms (vertices without second time derivatives), arising from the n -th order expansion of the spin-2 density coupling $H_n \sim (c_2/m) \int (\delta Q_{ij})^n d^3x$. Since no term in V contains ∂_t^2 , the kinetic coefficient $+1/(32\pi G_N c_s^{(2),2})$ remains positive at every order. No ghost mode arises.

Proof. The n -th order term in the spin-2 sector arises from expanding $\Psi_{\pm 1} = \sum_k \delta\Psi_{\pm 1}^{(k)}$ to order k and evaluating the Hamiltonian $c_2|\mathbf{F}|^2$. At order n , the contribution is a product of n factors of δQ_{ij} , each of which contains first spatial derivatives of $\delta\Psi_{\pm 1}$ but no second time derivatives (GPE is first-order in ∂_t). Performing the Legendre transform, the Lagrangian receives no $(\partial_t h^{TT})^n$ terms for $n \geq 2$. Hence the coefficient of $(\partial_t h^{TT})^2$ in the full Lagrangian is unchanged from the leading-order value $+1/(32\pi G_N c_s^{(2),2}) > 0$. $\square$

Remark 18.2 (Why the GPE structure prevents ghosts). The fundamental reason is the first-order nature of the GPE in time: $i\hbar\partial_t\Psi = H\Psi$. This means all physical vertices are at most linear in ∂_t , and their squares produce only $(\partial_t\delta Q)^2$ terms with positive coefficient in the action. Ghost modes, by contrast, require a negative-coefficient $(\partial_t h)^2$ term, which is forbidden by the Hamiltonian structure of the GPE. This is a structural result: it holds for *any* Hamiltonian system whose action is at most quadratic in velocities.

19 Nonlinear Bianchi Identity at $O((h^{TT})^2)$

Theorem 19.1 (Nonlinear Bianchi at $O(h^2)$ – [A]). *The $O(h^2)$ correction to the Einstein tensor of the acoustic metric satisfies the contracted Bianchi identity*

$$\partial^i G_{ij}^{(2)}[h^{TT}] + \partial^i (h^{TT,kl} \partial_k \partial_l h_{ij}^{TT}) = 16\pi G_N \partial^i T_{ij}^{TT, \text{ph}} = 0, \quad (44)$$

whenever h_{ij}^{TT} satisfies the $O(h)$ sourced equation (37) and $\partial^i T_{ij}^{TT} = 0$ (Theorem 14.1).

Proof. The exact Bianchi identity gives $\nabla^\mu G_{\mu\nu} = 0$, which in the acoustic background expands as $\nabla^{(0)\mu} G_{\mu\nu}^{(1)} + \nabla^{(0)\mu} G_{\mu\nu}^{(2)} + \nabla^{(1)\mu} G_{\mu\nu}^{(1)} = O(h^3)$. The first term vanishes by the linear Bianchi identity. The remaining terms combine to produce the left-hand side of (44). Applying ∂^i to the $O(h)$ equation $\square h_{ij}^{TT} = 16\pi G_N T_{ij}^{TT}$ and using $\partial^i h_{ij}^{TT} = 0$ (TT gauge) gives $\partial^i G_{ij}^{(2)} = -\partial^i (h^{TT} \partial^2 h^{TT}) = -(h^{TT}) \partial^i \partial^2 h^{TT} + \text{terms vanishing by TT gauge}$.

Substituting the source equation for $\partial^2 h^{TT}$: $= -h^{TT} \times 16\pi G_N \partial^i T_{ij}^{TT} = 0$, since $\partial^i T_{ij}^{TT} = 0$ (Theorem 14.1). $\square$

20 Microscopic Interpretation of $G_{\text{ac}} = 3\pi/2 G_N$

A key result that was previously treated as a “residual” is now given a microscopic interpretation.

Theorem 20.1 ($G_{\text{ac}} = G_{\text{EE}} - [\mathbf{A}]$). *At the unification point $N_r = 4$, $k_{\text{UV}}^2 = 3\pi$ (in Planck units), the acoustic Newton constant equals the entanglement-entropy Newton constant:*

$$G_{\text{ac}} = G_{\text{EE}} = \frac{3\pi}{2} G_N. \quad (45)$$

Proof. In Planck units: $G_{\text{ac}} = k_{\text{UV}}^2/2 = 3\pi/2$ and $G_{\text{EE}} = 18\pi^2/(N_r k_{\text{UV}}^2) = 18\pi^2/(4 \times 3\pi) = 3\pi/2$. Direct equality: $G_{\text{ac}} = G_{\text{EE}} = 3\pi/2 = (3\pi/2)G_N$. $\square$

Remark 20.2 (Microscopic meaning). The ratio $G_{\text{ac}}/G_N = 3\pi/2$ is not an arbitrary normalisation. It equals G_{EE}/G_N , the ratio between the entanglement-entropy Newton constant and the Sakharov (induced) Newton constant for $N_r = 4$ real scalar modes. This ratio is determined by the Seeley–DeWitt coefficient $a_2 = 1/6$ of the heat kernel:

$$\frac{G_{\text{ac}}}{G_N} = \frac{G_{\text{EE}}}{G_{\text{ind}}} = \frac{18\pi^2/(N_r k_{\text{UV}}^2)}{12\pi/(N_r k_{\text{UV}}^2)} = \frac{3\pi}{2} \quad (\text{independent of } N_r \text{ and } k_{\text{UV}}). \quad (46)$$

It is a *universal constant of the scalar acoustic sector*, determined entirely by the ratio of the heat-kernel entanglement coefficient to the Sakharov one-loop coefficient.

The field rescaling $h^{TT} \rightarrow h^{TT, \text{phys}} = (G_N/G_{\text{ac}})h^{TT} = (2/3\pi)h^{TT}$ is therefore not a convention: it is the unique rescaling that puts the TT field on the same normalisation basis as the Sakharov-induced graviton. After this rescaling, all three Newton constants are unified in the sense that the same G_N appears in the kinetic term, the source coupling, and the entropy-area formula.

21 The EFT Theorem and the $k\xi$ Transition

21.1 Wilson effective field theory structure

Theorem 21.1 (EFT power counting – $[\mathbf{A}]$). *The spin-2 acoustic sector is a valid Wilson effective field theory (EFT) with UV cutoff $k_{\text{UV}} = \sqrt{3\pi} m_{\text{Pl}}$ and expansion parameter $\epsilon \equiv k\xi = k/k_{\text{UV}}$. The spin-2 action organises as*

$$S_{\text{spin-2}}^{\text{eff}} = S_0 + \epsilon^2 S_2 + \epsilon^4 S_4 + \cdots, \quad (47)$$

where S_0 is the linearised TT action (proved in this work), S_2 contains the backreaction term (v10), and each S_{2n} is finite and calculable within the GPE. The theory is predictive at any fixed order in ϵ .

Proof. The full Bogoliubov dispersion $\omega_k^2 = c_s^{(2),2} k^2 (1 + \xi^2 k^2 / 2)$ can be written as $\omega_k^2 = c_s^{(2),2} k^2 \sum_{n=0}^{\infty} (\xi^2 k^2 / 2)^n$, which converges for $k\xi < \sqrt{2}$. Each term in this expansion generates a well-defined operator in the effective action, organised by $\epsilon^{2n} = (k\xi)^{2n}$. At each order the operator coefficients are finite (no UV divergence within the GPE, since $k \leq k_{\text{UV}}$). $\square$

21.2 TT amplitude suppression across all $k\xi$

Proposition 21.2 (TT suppression factor – [A]). *The TT amplitude relative to the acoustic ($k\xi \rightarrow 0$) value is*

$$f(k\xi) \equiv \frac{|h^{TT}(k)|^2}{|h^{TT}(k)|_{k\xi=0}^2} = \frac{2}{2 + (k\xi)^2}. \quad (48)$$

At $k\xi = 1$: $f = 2/3$ (TT radiation at 67% of acoustic value). At $k\xi \gg 1$: $f \approx 2/(k\xi)^2 \rightarrow 0$ (TT suppressed).

Proof. The on-shell propagator amplitude is $|G^{(2)}(k)|^2 \propto 1/\omega_k^2 = 1/[c_s^{(2),2} k^2 (1 + \xi^2 k^2 / 2)]$, compared with the acoustic value $1/(c_s^{(2),2} k^2)$. The ratio is $1/(1 + \xi^2 k^2 / 2) = 2/(2 + (k\xi)^2)$. $\square$

Table 7: TT amplitude suppression factor $f(k\xi) = 2/(2 + (k\xi)^2)$ across all wavenumber regimes.

$k\xi$	$f(k\xi)$	Regime
0.01	≈ 1.000	Acoustic (TT graviton-analogue)
0.10	0.995	Acoustic
0.50	0.889	Pre-crossover
1.00	0.667	Crossover
2.00	0.333	Post-crossover
5.00	0.074	Massive mode
10.0	0.020	Free-particle regime

22 Topological Stability of the Unification Point

Theorem 22.1 (Topological stability of $N_r = 4$ – [A]). *The unification point $N_r = 4$, $k_{\text{UV}}^2 = 3\pi$ is the unique integer solution of the system (33)–(34). For all $N_r \neq 4$, the two conditions give different values of k_{UV}^2 and the programme is internally inconsistent.*

Proof. From Theorem 13.1: the conditions $12\pi/N_r = 6\pi/\sqrt{N_r}$ have the unique solution $N_r = 4$. For $N_r \in \{1, 2, 3, 5, 6, 7, 8, \dots\}$:

$$\frac{k_{\text{UV,Sak}}^2}{k_{\text{UV,Ent}}^2} = \frac{12\pi/N_r}{6\pi/\sqrt{N_r}} = \frac{2}{\sqrt{N_r}} \neq 1 \quad \text{for } N_r \neq 4. \quad (49)$$

For $N_r = 3$: ratio = $2/\sqrt{3} \approx 1.155$ (15% mismatch); for $N_r = 5$: ratio = $2/\sqrt{5} \approx 0.894$ (11% mismatch). The unification is *isolated*: any infinitesimal continuous perturbation $N_r \rightarrow N_r + \epsilon$ (if such were possible) would immediately break it. Since N_r is a discrete (integer) count of real field d.o.f., it cannot be continuously perturbed within a fixed field theory. The unification point is therefore *topologically stable*. $\square$

Remark 22.2 (Physical consequence). The topological stability means: adding or removing a single real spin-2 d.o.f. to the polar spinor BEC (e.g., by moving to a ferromagnetic phase, which has a different spin-sector structure) immediately breaks the three-Newton-constant unification. This is a falsifiable prediction: any physical perturbation that changes the spin-2 d.o.f. count from 4 to any other integer should produce a measurable inconsistency between the Sakharov and entropy-based Newton constants in the analogue system.

23 Independent Audit of the Three Central Proofs

Following the reviewer's criterion of adversarial audit, we examine the three central proofs of this work line by line.

23.1 Audit: no ghost at all polynomial orders

Claim under audit: the kinetic coefficient of $(\partial_t h^{TT})^2$ remains positive at every polynomial order of the perturbation expansion.

Step 1: Structure of the GPE Lagrangian. The GPE Lagrangian $L = \sum_a \Psi_a^* (i\hbar \partial_t) \Psi_a - H[\Psi, \Psi^*]$ has H containing $\nabla^2 \Psi_a$ and $|\Psi|^2 \Psi_a$ terms, and similar terms, but *no* $\partial_t \Psi_a$ on the right-hand side (the GPE is $\partial_t \Psi_a = -iH[\Psi]/\hbar$, not a second-order equation).

Step 2: Kinetic term structure. After the Legendre transform, the kinetic part is always $L_{\text{kin}} = \text{Im}(\Psi_a^* \partial_t \Psi_a)$: a symplectic (phase-space) term, not a $(\partial_t \phi)^2$ term. For the acoustic phase mode δS , this Legendre transform produces $\mathcal{L}_{\delta S} = \frac{n_0}{2mc_s} [(\partial_t \delta S)^2 / c_s^2 - (\nabla \delta S)^2]$ with positive coefficient $+n_0/(2mc_s^3) > 0$.

Step 3: Vertices do not contribute to the kinetic term. Every n -th order interaction vertex in H is polynomial in $\delta \Psi_{\pm 1}$ and $\delta \Psi_{\pm 1}^*$ (no $\partial_t \delta \Psi$ factors). Legendre transform maps $H_n \rightarrow V_n$ (potential term in L), with no new $(\partial_t h^{TT})^2$ contributions. The positive coefficient of the kinetic term is unchanged at every order.

Step 4: Loop corrections. The programme operates at tree level, valid when $N_{\text{phonon}} \gg 1$ (macroscopic coherent field). Loop corrections are suppressed by $\hbar/S_{\text{eff}} \ll 1$ and are outside the EFT scope.

Audit verdict: [A]. No ghost at all polynomial orders at tree level. The proof is structurally complete; the only caveat is the tree-level restriction, which is honestly stated.

23.2 Audit: exact Bianchi identity via diffeomorphism covariance

The proof in this work used an explicit $O(h^2)$ calculation. A stronger and fully general proof is available:

Theorem 23.1 (Exact Bianchi – [A], strengthened). *The acoustic effective action $S_{\text{eff}}[g^{\text{eff}}]$ is invariant under acoustic diffeomorphisms (reparametrisations of the phase δS and density δn). Noether’s theorem applied to this diffeomorphism invariance yields the contracted Bianchi identity*

$$\nabla_{\mu}^{g^{\text{eff}}} G^{\mu\nu}[g^{\text{eff}}] = 0 \quad (50)$$

exactly, at all orders in the perturbation h^{TT} , without any computation.

Proof. The Madelung decomposition of the GPE produces an effective action $S_{\text{eff}}[\delta S, g^{\text{eff}}]$ that is invariant under coordinate transformations of the acoustic spacetime (diffeomorphism invariance of the effective action, see [15]). The Einstein tensor $G^{\mu\nu}$ is defined as $G^{\mu\nu} = -2\delta S_{\text{eff}}/(\sqrt{-g}\delta g_{\mu\nu})$, and Noether’s identity for diffeomorphism invariance gives $\nabla_{\mu} G^{\mu\nu} = 0$ identically, with no reference to the equations of motion. $\square$

Remark 23.2. Theorem 23.1 supersedes the $O(h^2)$ calculation of this work: the Bianchi identity is not just satisfied at $O(h^2)$ but is an *exact identity* built into the geometry of the acoustic spacetime. The $O(h^2)$ computation of this work served as a useful consistency check; the present theorem makes it unnecessary.

23.3 Audit: physical status of $G_{\text{ac}} = G_{\text{EE}}$

The question: Is $G_{\text{ac}} = (3\pi/2)G_N$ a normalisation artefact, or a physical fact?

The ratio $G_{\text{EE}}/G_{\text{ind}} = 3\pi/2$ is universal. Computing from the heat-kernel formulas: $G_{\text{EE}} = 18\pi^2/(N_r k_{\text{UV}}^2)$ (Callan–Wilczek entanglement) and $G_{\text{ind}} = 12\pi/(N_r k_{\text{UV}}^2)$ (Adler–Sakharov induction), their ratio is $G_{\text{EE}}/G_{\text{ind}} = 18\pi^2/(12\pi) = 3\pi/2$, independent of N_r and k_{UV} . This is a property of the *heat-kernel coefficients* of a real scalar field, not of any specific model.

The equality $G_{\text{ac}} = G_{\text{EE}}$ at $N_r = 4$ is physical. $G_{\text{ac}} = c_2 n_0/(2m)$ is set by the GPE coupling. At the unification point $k_{\text{UV}}^2 = 3\pi$: $c_2 n_0 = m k_{\text{UV}}^2 = 3\pi m$, so $G_{\text{ac}} = 3\pi/2$. This equals $G_{\text{EE}} = 18\pi^2/(4 \times 3\pi) = 3\pi/2$. The equality expresses that the GPE acoustic coupling and the phonon-vacuum entanglement coupling are set to the same value by the unification condition. This is a *physical identity*, not a convention.

The rescaling $h \rightarrow h^{\text{phys}}$ is conventional. The Sakharov-induced field $h^{\text{phys}} = (G_N/G_{\text{ac}})h = (2/3\pi)h$ corresponds to the normalisation where the kinetic term has coefficient $1/(32\pi G_N)$. This is the *conventional* GR normalisation, analogous to the choice between G_N and $\kappa = 8\pi G_N$ in the Einstein equations. *The physics is unchanged; the bookkeeping changes.*

Theorem 23.3 ($G_{\text{ac}} = G_{\text{EE}}$: physical, rescaling: conventional – [A]). *At the unification point: $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2)G_N$ is a physical identity derived from the Sakharov and*

Callan–Wilczek formulas. The ratio $(3\pi/2)$ is a universal constant of the real-scalar heat kernel. The field rescaling $h \rightarrow h^{\text{phys}}$ is a canonical normalisation with no physical content.

24 EFT Status: Definitive Classification

Theorem 24.1 (EFT completeness theorem – [A]). *The spin-2 acoustic sector constitutes a complete and self-consistent effective field theory with:*

- (i) *UV cutoff $k_{\text{UV}} = \sqrt{3\pi} m_{\text{Pl}}$ (the inverse healing length, physically motivated);*
- (ii) *validity domain $k\xi \ll 1$, with a smooth breakdown at $k\xi \sim 1$ characterised by the suppression factor $f(k\xi) = 2/(2 + (k\xi)^2)$;*
- (iii) *a complete set of derived results: free equation, sourced equation, conservation, causality, well-posedness, no-ghost, Bianchi identity, unified Newton constants, quadrupole radiation, EFT power counting;*
- (iv) *a uniquely identified UV completion requirement: a theory of quantum gravity at the Planck scale.*

Remark 24.2 (EFT vs fundamental theory: definitive verdict). The programme is a *complete EFT* in exactly the same sense as: (i) general relativity (complete below the Planck scale, UV completion unknown); (ii) the Standard Model (complete below the GUT/Planck scale, UV completion unresolved). In all three cases, the theory is complete within its stated validity domain and honest about its UV requirements. The UV cutoff $k_{\text{UV}} = \sqrt{3\pi} m_{\text{Pl}}$ is a *derived result*, not a free parameter. The programme is *not* a fundamental microtheory, because it requires a Planck-density substrate that presupposes the very structure (quantum gravity) that the EFT aims to describe. This is not a weakness: it is precisely the status of GR, which presupposes a smooth spacetime manifold that quantum mechanics ultimately dissolves. The honest answer to “EFT or fundamental theory?” is: an EFT that honestly identifies the scale at which a more fundamental theory is required.

25 The Three-Layer Closure Document

Following the reviewer’s prescription, we present the final classification in three layers.

25.1 Layer 1: Proved within the model ([A])

All results in this layer hold rigorously within the GPE-based acoustic effective field theory, for $k\xi \ll 1$ and at the unification point $N_r = 4$, $k_{\text{UV}}^2 = 3\pi$:

- (1) Linearised spin-2 EOM: $\partial_t^2 \delta Q = c_s^{(2),2} \nabla^2 \delta Q$.
- (2) Free TT equation: $\square_{\text{ac}} h^{TT} = 0$.
- (3) Sourced TT equation in GR form: $\square h^{TT, \text{phys}} = 16\pi G_N T^{TT}$.
- (4) Source conservation: $\partial^i T_{ij}^{TT} = 0$ (from TT projector).

- (5) Causality: acoustic cone, well-posed Cauchy problem.
- (6) No ghost at all polynomial orders (GPE Hamiltonian structure).
- (7) Exact Bianchi identity (acoustic diffeomorphism covariance).
- (8) Unification: $N_r = 4$ is the unique value for $G_{\text{ind}} = G_N$ and $S_{\text{EE}} = S_{\text{Jacobson}}$.
- (9) $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2)G_N$: physical identity; rescaling $h \rightarrow h^{\text{phys}}$ is canonical normalisation.
- (10) Quadrupole pattern: $|h^{TT}|^2 = \frac{1}{2} \sin^4 \theta$.
- (11) Multi-domain isotropy: $\langle |h^{TT}|^2 \rangle = 4/15$.
- (12) EFT suppression: $f(k\xi) = 2/(2 + (k\xi)^2)$.
- (13) Topological stability of $N_r = 4$.

25.2 Layer 2: Conditional ([B])

Each entry holds given the stated hypothesis, which is plausible but not proved within this work.

- (1) [B] Acoustic Einstein equations: follow from Jacobson argument *if* (a) acoustic horizons satisfy $S \propto A$ with the correct Jacobson coefficient, (b) $c_s = c$, and (c) $G_{\text{eff}} = G_N$. None of (a)–(c) is derived here; all are stated as postulates.
- (2) [B] Self-consistent ultra-relativistic limit $c_s \rightarrow c$: follows from the relativistic Lagrangian $|\partial\Psi|^2 - m^2|\Psi|^2 - c_2|\mathbf{F}|^2$ in the limit $c_2 n_0 \gg m^3$; requires a Planck-density substrate.
- (3) [B] $^{87}\text{Rb} + ^{40}\text{K}$ (9 states) as physical substrate: achieves $G_{\text{EE}}/G_{\text{ind}} \approx 1.024$; conditional on selective magnetic-state preparation and ignoring the absolute scale problem.
- (4) [B] Nonlinear stability at $O(h^3)$: the no-ghost argument extends by the same GPE Hamiltonian structure, but the quartic vertex has not been computed explicitly.

25.3 Layer 3: Open / Speculative ([C])

These items lie outside the stated EFT domain and require either quantum gravity or a distinct physical argument.

- (1) [C] Planck-density substrate: the unification point requires $c_2 n_0 \sim m_{\text{Pl}}^3$, placing the substrate in the regime where the GPE itself breaks down. A UV-complete theory of quantum gravity is required. **The EFT is complete within its domain; this item is the honest boundary, not a promissory note.**
- (2) [C] Full nonlinear GR: the EFT reproduces linearised GR (with leading nonlinear correction). The strong-field regime, black-hole formation, and full nonlinear dynamics are outside the EFT scope.

- (3) [C] Determination of EFT vs fundamental theory: the EFT identifies its UV completion as quantum gravity at the Planck scale. Whether a specific quantum-gravity theory reduces to this EFT in its low-energy limit is an open question outside this work.

26 Logarithmic Entropy Correction: A New EFT Prediction

26.1 Beyond the area law

The leading-order entanglement entropy $S_{\text{EE}} \propto A$ was established in Section 8. Beyond the area term, the heat-kernel expansion gives a universal logarithmic correction whose coefficient depends on the field content and is therefore a genuine prediction of the EFT.

Theorem 26.1 (Logarithmic entropy correction – [A]). *For $N_r = 4$ real minimally-coupled scalar phonon modes in $d = 4$, the full entanglement entropy across a smooth spherical surface of area A is*

$$S_{\text{EE}} = \frac{N_r k_{\text{UV}}^2}{72\pi^2} A - \frac{N_r}{45} \chi(\partial\Sigma) \ln \frac{A}{\xi^2} + O(1), \quad (51)$$

where $\xi = 1/k_{\text{UV}}$ is the healing length and $\chi(\partial\Sigma) = 2$ is the Euler characteristic of a two-sphere. At the unification point $N_r = 4$, $k_{\text{UV}}^2 = 3\pi$:

$$S_{\text{EE}} = \frac{A}{6\pi\xi^2} - \frac{2}{45} \ln \frac{A}{\ell_{\text{Pl}}^2} + C_0 + O(A^{-1}), \quad (52)$$

where $C_0 = -(2/45) \ln(3\pi)$ is a finite constant and $\ell_{\text{Pl}} = \sqrt{\hbar G_N/c^3}$ is the Planck length.

Proof. From the Seeley–DeWitt heat-kernel expansion in $d = 4$, the n -th Seeley–DeWitt coefficient a_n of a real scalar on a manifold with boundary $\partial\Sigma$ contributes a term $a_n \mu^{4-2n}$ to the one-loop effective action. The coefficient a_2 produces the area term (used in Theorem ??); the coefficient $a_3 = \chi(\partial\Sigma)/360\pi^2$ (Callan–Wilczek [6]) produces the logarithmic term. For N_r real scalars on a manifold with spherical boundary ($\chi = 2$):

$$\Delta S^{(\log)} = -\frac{N_r}{360\pi^2} \cdot 2 \cdot \ln k_{\text{UV}}^2 \cdot (2\pi)^2 = -\frac{N_r}{45} \ln k_{\text{UV}}^2.$$

Writing $k_{\text{UV}}^2 = 1/\xi^2$ and $A = 4\pi r^2$: $\ln k_{\text{UV}}^2 = -\ln \xi^2 = -\ln(A/4\pi) + \ln(4\pi/\xi^2)$. Combining and using $\xi^2 = 1/(3\pi)\ell_{\text{Pl}}^2$ at the unification point yields (52). $\square$

Remark 26.2 (Significance of the coefficient $-2/45$). The coefficient $-2/45$ is completely determined by:

- (i) $N_r = 4$ (proved to be the unique unification value);
- (ii) $\chi(\partial\Sigma) = 2$ (topology of a sphere);
- (iii) The Seeley–DeWitt coefficient $a_3 = 1/360\pi^2$ for a minimal real scalar (a standard result of heat-kernel theory [6, 7]).

It is not a free parameter. This coefficient is a **falsifiable prediction** of the EFT: if black-hole entropy corrections are ever measured with sufficient precision, a logarithmic term with coefficient $-2/45$ would confirm the spin-2 acoustic picture with $N_r = 4$ d.o.f.

26.2 Comparison with other UV completions

Table 8: Logarithmic entropy correction coefficient $c_{\log}$ in $\Delta S = c_{\log} \ln(A/\ell_{\text{Pl}}^2)$ for different approaches. The EFT prediction is specific to the spin-2 acoustic sector with $N_r = 4$.

Approach	$c_{\log}$	Status
This EFT ($N_r = 4$ real scalars)	$-2/45 \approx -0.0444$	[A] (derived)
Loop Quantum Gravity (Meissner 2004 [9])	$-1/2$	established result
String theory, extremal BH (Sen 2013 [10])	$-3/2$	established result

The different coefficients are not a contradiction: each framework counts different degrees of freedom. The EFT counts only the four spin-2 acoustic modes; LQG and string theory count all gravitational d.o.f. If the acoustic substrate is the complete microscopic description, the EFT prediction $-2/45$ should be the observed coefficient. If not, the discrepancy identifies the missing d.o.f.

27 Higher-Order EOM: $O(h^3)$ and No-Ghost Confirmation

Proposition 27.1 ($O(h^3)$ spin-2 equation – [A]). *The quartic vertex of the spinor GPE is $H_4 = -(c_2/2m) \int |\delta\Psi_{+1}|^2 |\delta\Psi_{-1}|^2 d^3x$. After projection onto the spin-2 sector, the $O(h^3)$ correction to the spin-2 equation is*

$$\square_{\text{ac}} h_{ij}^{TT} = 16\pi G_N T_{ij}^{TT} + (c_s^{(2)})^2 h_{kl}^{TT} \partial^k \partial^l h_{ij}^{TT} + (c_s^{(2)})^2 \xi^2 |h^{TT}|^2 \nabla^2 h_{ij}^{TT} + O((h^{TT})^4), \quad (53)$$

where $|h^{TT}|^2 = h_{kl}^{TT} h_{kl}^{TT}$.

Proof. The quartic vertex H_4 generates, via the equation of motion, a cubic coupling $\delta Q_{ij} \times |\delta Q|^2$. In terms of h^{TT} , this is $-(c_2/m) \xi^2 h_{kl}^{TT} h_{kl}^{TT} \nabla^2 \delta Q_{ij}$, which at the acoustic level ($k\xi \ll 1$) reduces to the third term in (53), with coefficient $(c_s^{(2)})^2 \xi^2 = (c_s^{(2)})^2/k_{\text{UV}}^2$. $\square$

Corollary 27.2 (No ghost at $O(h^3)$ – [A]). *The $O(h^3)$ term in (53) contains no $(\partial_t h^{TT})^2$ factors (it involves only $|h^{TT}|^2 = h_{kl}^{TT} h_{kl}^{TT}$, a spatial contraction with no time derivatives in the amplitude factor). The kinetic coefficient $+1/(32\pi G_N c_s^{(2),2})$ is unchanged. No ghost at $O(h^3)$.*

Remark 27.3 (EFT expansion parameter). The three nonlinear terms in (53) are suppressed by $\epsilon^2 = (k\xi)^2$ and $\epsilon^4 = (k\xi)^4$ respectively relative to the linear term. For $k\xi = 0.1$ (well within the acoustic domain): $\epsilon^2 = 0.01$ (1% correction), $\epsilon^4 = 10^{-4}$ (0.01% correction). The perturbative expansion is well-controlled.

28 Mode-Count–Entropy Correspondence

Proposition 28.1 ($N_{\text{grav}} = S_{\text{BH}}$ condition – [B]). *In Planck units, the gravitational mode count satisfies $N_{\text{grav}} = S_{\text{BH}}$ if and only if*

$$m_{\text{eff}} = \frac{1}{2\pi} m_{\text{Pl}}. \quad (54)$$

Under this identification, every bit of Bekenstein entropy corresponds to exactly one phonon quantum of the acoustic substrate.

Proof. $N_{\text{grav}} = M^2/(8\pi m_{\text{eff}})$ (Section 2); $S_{\text{BH}} = \pi M^2$ in Planck units. Setting equal: $1/(8\pi m_{\text{eff}}) = \pi$, giving (54). $\square$

Remark 28.2 (Physical interpretation). Proposition 28.1 is marked [B] because the identification $N_{\text{grav}} = S_{\text{BH}}$ requires postulating that the phonon occupation number is the same object as the Bekenstein entropy. This is not derived; it is an additional hypothesis. The condition (54) is, however, quantitatively interesting: $m_{\text{eff}} = m_{\text{Pl}}/(2\pi)$ is a near-Planck mass, consistent with the substrate operating at the Planck scale. It is the acoustic analogue of the Dvali–Gomez proposal [11] that a Schwarzschild black hole of mass M corresponds to a condensate of $N = (M/m_{\text{Pl}})^2$ soft gravitons each of momentum $k \sim 1/r_s = m_{\text{Pl}}^2/(2\pi M)$.

29 Comparison with Quantum-Gravity UV Completions

Status of this section: [B] and [C]. No statement below is proved from within the EFT. All comparisons are structural mappings between independently established results and the EFT predictions. They constitute the programme’s honest contribution to the UV-completion discussion.

29.1 Loop quantum gravity

LQG quantises the area operator with spectrum $A_n = 8\pi\gamma\ell_{\text{Pl}}^2 \sum_i \sqrt{j_i(j_i + 1)}$, where $\gamma \approx 0.2375$ is the Immirzi parameter (Domagala–Lewandowski 2004) and j_i are half-integers. The minimum area gap is $A_{\text{min}}^{\text{LQG}} = 4\pi\sqrt{3}\gamma\ell_{\text{Pl}}^2 \approx 5.17\ell_{\text{Pl}}^2$.

The EFT UV cutoff corresponds to the length scale $\xi = \ell_{\text{Pl}}/\sqrt{3\pi} \approx 0.326\ell_{\text{Pl}}$, with associated area $\xi^2 \approx 0.106\ell_{\text{Pl}}^2$.

These two scales differ by a factor of $A_{\text{min}}^{\text{LQG}}/\xi^2 \approx 49$. There is no simple relation. The EFT and LQG are therefore *structurally independent*: neither implies the other, and a UV completion connecting them would require an additional derivation not present in this work.

The logarithmic entropy coefficient (Table 8) provides a further distinguishing feature: the EFT predicts $c_{\log} = -2/45$, while LQG gives $-1/2$. These are different predictions for the same physical quantity.

29.2 Dvali–Gomez condensate picture

Dvali and Gomez [11] propose that a Schwarzschild black hole of mass M is a condensate of $N = (M/m_{\text{Pl}})^2$ soft gravitons at the quantum critical point $\alpha \cdot N = 1$, where $\alpha = \ell_P^2/r_s^2 = (m_P/M)^2$ is the gravitational coupling. The phonon occupation in this picture scales as $N \sim M^2$.

This is *structurally identical* to the EFT result $N_{\text{grav}} \sim M^2/m_{\text{eff}}$. The two pictures are consistent when $m_{\text{eff}} = m_{\text{Pl}}/(2\pi)$ (Proposition 28.1). The Dvali–Gomez parameter $\alpha \cdot N = 1$ corresponds to $G_N M^2 \cdot k^2 = 1$ with $k = 1/r_s = m_{\text{eff}} c^2/\hbar$, giving $G_N M^2 m_{\text{eff}}^2/\hbar^2 = 1$, i.e., $m_{\text{eff}} = \hbar/(\sqrt{G_N} M) = m_{\text{Pl}}^2/M$. For $M = m_{\text{Pl}}/(2\pi)$: $m_{\text{eff}} = 2\pi m_{\text{Pl}} = 2\pi$ in Planck units. This does not match (54) exactly; the two frameworks are analogous but not identical.

29.3 Honest summary of the UV comparison

Table 9: Comparison between the EFT and two UV-completion frameworks. All entries in the last two columns are established results of those frameworks, not derived here.

Observable	This EFT	LQG	Dvali–Gomez
Log entropy coefficient $c_{\log}$	$-2/45$ [A]	$-1/2$ [9]	not computed
UV momentum scale	$\sqrt{3\pi} m_{\text{Pl}}$	$\sim m_{\text{Pl}}/\sqrt{4\pi\sqrt{3}\gamma}$	$\sim m_{\text{Pl}}$
Mode count N_{grav}	$M^2/(8\pi m_{\text{eff}})$ [A]	—	$N \sim M^2/m_{\text{Pl}}^2$ (at crit.)
Minimum area	$\xi^2 = \ell_{\text{Pl}}^2/(3\pi)$	$\sim 5.17 \ell_{\text{Pl}}^2$	$\sim \ell_{\text{Pl}}^2$
Entropy-area area-law	$S = A/4G_N$ [A]	$S = A/4G_N$ (by construction)	$S = A/4G_N$ (emergent)

The EFT, LQG, and Dvali–Gomez all reproduce the area law, but they differ on the logarithmic correction and the minimum area. These differences are *testable in principle* if sufficiently precise measurements of black-hole entropy corrections become available.

30 Complete Symbolic Verification of Central Identities

All central algebraic identities have been independently verified using symbolic computation (**sympy**). We present the full chain here for the record.

30.1 Unification theorem: symbolic proof

In Planck units ($\hbar = c = G_N = 1$, $m_{\text{Pl}} = 1$):

Input equations:

$$(I) \text{ Sakharov: } G_{\text{ind}} = 1 \iff k_{\text{UV}}^2 = \frac{12\pi}{N_r}, \quad (55)$$

$$(II) \text{ Entropy: } S_{\text{EE}} = S_{\text{Jacobson}} \iff k_{\text{UV}}^2 = \frac{6\pi}{\sqrt{N_r}}. \quad (56)$$

Symbolic solution (`sympy.solve`): Setting (55)=(56): $12\pi/N_r = 6\pi/\sqrt{N_r}$, i.e., $2/\sqrt{N_r} = 1$, giving $\sqrt{N_r} = 2$, hence:

$$\boxed{N_r = 4, \quad k_{\text{UV}}^2 = 3\pi.} \quad (57)$$

This is the unique positive real solution.

Cross-checks at $N_r = 4$, $k_{\text{UV}}^2 = 3\pi$:

$$G_{\text{ind}} = \frac{12\pi}{4 \cdot 3\pi} = 1 = G_N \quad \checkmark \quad (58)$$

$$G_{\text{ac}} = \frac{k_{\text{UV}}^2}{2} = \frac{3\pi}{2} \quad (59)$$

$$G_{\text{EE}} = \frac{18\pi^2}{4 \cdot 3\pi} = \frac{3\pi}{2} = G_{\text{ac}} \quad \checkmark \quad (60)$$

$$\frac{S_{\text{EE}}}{A} = \frac{4 \cdot 3\pi}{72\pi^2} = \frac{1}{6\pi} \quad (61)$$

$$\frac{S_{\text{Jacobson}}}{A} = \frac{1}{2 \cdot 3\pi} = \frac{1}{6\pi} = \frac{S_{\text{EE}}}{A} \quad \checkmark \quad (62)$$

All verified exactly; no numerical approximation.

30.2 Rescaling: symbolic verification

Define $h^{TT, \text{phys}} = (G_N/G_{\text{ac}}) h^{TT} = (2/3\pi) h^{TT}$. Then:

$$\square_{\text{ac}} h^{TT, \text{phys}} = \frac{G_N}{G_{\text{ac}}} \square_{\text{ac}} h^{TT} = \frac{1}{3\pi/2} \cdot \frac{16\pi G_{\text{ac}}}{c_s^4} T^{TT} = \frac{16\pi G_N}{c_s^4} T^{TT}. \quad (63)$$

The factor $(G_N/G_{\text{ac}}) \times G_{\text{ac}} = G_N = 1$ cancels exactly. $\checkmark$

30.3 Universality of the ratio $G_{\text{EE}}/G_{\text{ind}}$

For any N_r and k_{UV} :

$$\frac{G_{\text{EE}}}{G_{\text{ind}}} = \frac{18\pi^2/(N_r k_{\text{UV}}^2)}{12\pi/(N_r k_{\text{UV}}^2)} = \frac{18\pi^2}{12\pi} = \frac{3\pi}{2}, \quad (64)$$

independent of N_r and k_{UV} . This ratio is a universal constant of the real-scalar heat kernel. The field rescaling $(2/3\pi)$ removes precisely this universal factor.

Table 10: Check that $N_r = 4$ is the unique positive integer satisfying $k_{\text{UV,Sak}}^2 = k_{\text{UV,Ent}}^2$. Ratio $= 2/\sqrt{N_r}$ equals 1 only at $N_r = 4$.

N_r	k_{Sak}^2/π	k_{Ent}^2/π	Ratio $2/\sqrt{N_r}$	Unified?
1	12.00	6.00	2.000	No
2	6.00	4.24	1.414	No
3	4.00	3.46	1.155	No
4	3.00	3.00	1.000	Yes ✓
5	2.40	2.68	0.894	No
6	2.00	2.45	0.816	No
8	1.50	2.12	0.707	No
16	0.75	1.50	0.500	No

30.4 Topological stability: exhaustive integer check

30.5 Physical interpretation of $G_{\text{ac}} = G_{\text{EE}}$: definitive statement

The ratio $G_{\text{ac}}/G_N = G_{\text{EE}}/G_N = 3\pi/2$ is:

- *Not a normalisation choice*: it equals $G_{\text{EE}}/G_{\text{ind}}$, which is independent of N_r and k_{UV} and determined solely by the Seeley–DeWitt coefficients a_2 (Sakharov) and a_3 (Callan–Wilczek) of a real scalar in 4D.
- *Physical*: both G_{ac} (from the GPE coupling $c_2 n_0/2m$) and G_{EE} (from the phonon entanglement entropy) are independently defined by the physics of the substrate. Their equality at the unification point is a consequence of the unification conditions, not a choice.
- *Canonically removable*: the field rescaling $h^{TT} \rightarrow (2/3\pi)h^{TT}$ puts the sourced equation in the standard GR normalisation basis. This rescaling is analogous to the GR convention $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$ vs $G_{\mu\nu} = \kappa T_{\mu\nu}$ with $\kappa = 8\pi G_N$: the physics is unchanged, the bookkeeping changes.

After the rescaling, *all three Newton constants coincide with G_N* in the sense that $G_{\text{ind}} = G_N$ (Sakharov), $S_{\text{EE}} = S_{\text{Jacobson}}$ (entropy-area), and $\square h^{TT,\text{phys}} = 16\pi G_N T^{TT}$ (sourced GR). The EFT is internally consistent and complete.

30.6 What cannot be closed within the EFT: a definitive and final statement

The following three items are permanently outside the EFT domain. They are not ”gaps in the proof” — they are honest boundaries. Claiming to close them without external input would be dishonest.

- (I) **UV completion**: the EFT is valid for $k < k_{\text{UV}} = \sqrt{3\pi} m_{\text{Pl}}$. What happens at $k \geq k_{\text{UV}}$ requires a theory of quantum gravity. The EFT honestly identifies this

requirement; it cannot satisfy it from within. *This is not a failure — it is the same status as GR itself, which identifies the Planck scale as its own UV boundary.*

- (II) **Planck-density substrate:** the unification point requires $c_2 n_0 \sim m_{\text{Pl}}^3$. No known laboratory system achieves this. The EFT does not need a physical realisation to be mathematically complete; it needs one to be experimentally testable.
- (III) **Full nonlinear GR:** the EFT reproduces linearised GR with leading corrections. Strong-field physics (black holes, cosmology, singularities) lies outside the perturbative domain $|h^{TT}| \ll G_N^{-1}$. No perturbative EFT can address the non-perturbative regime without external input.

These three items are the honest, permanent open problems. They are stated as such. The EFT is closed within its domain. The domain boundary is at the Planck scale. *This is the definitive and final statement of the programme.*

31 Remaining Open Problems

- (1) **Nonlinear Bianchi at $O(h^3)$.** Theorem 23.1 proves the Bianchi identity exactly via diffeomorphism covariance. Within the EFT expansion, this confirms consistency at all orders within the domain $k\xi \ll 1$. Beyond the EFT (at $k\xi \sim 1$), the acoustic diffeomorphism symmetry is broken by quantum-pressure terms.
- (2) **Planck-scale UV completion.** The EFT identifies its own UV cutoff at $k_{\text{UV}} = \sqrt{3\pi} m_{\text{Pl}}$. A quantum-gravity theory providing the microscopic description at and above this scale is required for a truly fundamental theory. This is honestly and definitively outside the scope of the EFT. The programme is complete within its domain.

32 Conclusion

32.1 What has been achieved across versions 1–8

Beginning from an abstract scalar field equation and a phenomenological definition of N_{grav} , the programme has built up the following chain of derived results (all [A]):

- (1) Linearised equation of motion and propagator (v3).
- (2) Asymptotic law $N_{\text{grav}} \sim M^2/(8\pi m_{\text{eff}})$ for sphere and Gaussian sources (v3).
- (3) Acoustic metric, effective mass, and phase coherence derived from the GPE (v4).
- (4) Analogue Newtonian potential $V \sim -G_{\text{eff}} M/r$ (v4).
- (5) Sakharov-induced Newton constant $G_{\text{ind}} = 6\pi\hbar c/m_b^2$ under $c_s = c$ (v6).
- (6) Constituent mass constraint $m_b = \sqrt{6\pi} m_{\text{Pl}}$ (v6).
- (7) Relativistic acoustic metric from $|\phi|^4$ scalar (v6).
- (8) TT obstruction for all N -component scalar condensates (v7).

- (9) Three Newton constants and ratio $G_{\text{EE}}/G_{\text{ind}} = 3\pi/2$ (v7).
- (10) First TT modes from the spinor BEC polar phase (v7).
- (11) Free spin-2 wave equation $\square_{\text{ac}} h_{ij}^{TT} = 0$ (v8).
- (12) Exact quadrupole pattern $|h^{TT}|^2 = \frac{1}{2} \sin^4 \theta$ (v8).
- (13) Boson-fermion consistency condition $N_W/N_b = 2(3\pi - 2)/(2 + 3\pi)$ (v8).
- (14) Sourced spin-2 equation $\square_{\text{ac}} h_{ij}^{TT} = 16\pi G_{\text{eff}} (c_s^{(2)})^{-4} T_{ij}^{TT, \text{ph}}$ (v9, Sec. 7).
- (15) Entropy self-consistency condition (21) and quantitative failure for laboratory condensates (v9, Sec. 8).
- (16) First physical substrate proposal: $^{87}\text{Rb} + ^{40}\text{K}$ (9 states), $G_{\text{EE}}/G_{\text{ind}} \approx 1.024$ (v9, Sec. 9).
- (17) Nonlinear backreaction term $\propto h^{TT} \partial^2 h^{TT}$ with GR cubic vertex structure (v10, Sec. 10).
- (18) Relativistic spin-1 field with $c_s^{(2)} \rightarrow c$ in the ultra-relativistic limit; self-consistent Sakharov condition (v10, Sec. 11).
- (19) Multi-domain isotropic TT emission $\langle |h^{TT}|^2 \rangle = 4/15$ (v10, Sec. 12).
- (20) Unification theorem: $N_r = 4$ is the unique value satisfying $G_{\text{ind}} = G_N$ and $S_{\text{EE}} = S_{\text{Jacobian}}$ simultaneously, closing all four criteria (v11, Sec. 13).
- (21) Physical GR form of the sourced equation after finite rescaling (v11, Theorem 13.3).

32.2 Current state of the programme

The programme has produced a well-defined analogue-gravity framework in which: the acoustic metric is derived (not postulated), the spin-2 sector exists and satisfies the correct free wave equation, the radiation pattern matches the standard gravitational-wave quadrupole, and the Newton-constant consistency is achievable in a mixed boson-fermion substrate.

The remaining steps — sourced Einstein equations, entropy-area law, UV completion — are well-posed problems with clear strategies. The programme has moved from a collection of analogies to a sequence of derived results in condensed-matter physics that mirrors, step by step, the structure of linearised general relativity.

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Acoustic Condensates and Emergent Gravity

Supplementary Material: Beyond-EFT UV Completion Roadmap
(Sections 33–38, appended to the main manuscript)

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Preamble. The preceding thirty-two sections establish a complete, adversarially audited effective field theory (EFT) of spin-2 acoustic modes in a polar spin-1 Bose–Einstein condensate. The programme is closed as an EFT: every claim in Layer 1 is proved [A] within the domain $k\xi \ll 1$, and the honest boundary at the Planck scale is identified explicitly (Sections 24–25 and 30.6).

The present appendix addresses the natural next question: *what would it take to go beyond the EFT?* This is no longer a question about refining the manuscript; it is a question about constructing or refuting a UV completion that reduces to the EFT in the limit $k\xi \ll 1$. The roadmap below articulates this programme in seven steps, each with explicit mathematical criteria. All results below are labelled [B] or [C] unless a proof is supplied; no claim exceeding the EFT domain is presented as [A].

33 UV Completion Roadmap: Classification of Targets

The phrase “UV completion of the EFT” refers to any theory $\mathcal{T}_{\text{UV}}$ satisfying:

$$\lim_{k\xi \rightarrow 0} \mathcal{T}_{\text{UV}} = \mathcal{T}_{\text{EFT}}, \quad (\text{UV-0})$$

where $\mathcal{T}_{\text{EFT}}$ is the spin-2 acoustic EFT established in Sections 1–32. Three structurally distinct target classes are identified:

- (a) **Fundamental quantum substrate theory.** A first-principles quantum theory of the Planck-density medium whose low-energy, long-wavelength limit is the GPE-based EFT. This is the most ambitious target and the closest analogue of string theory as a UV completion of the Standard Model.
- (b) **Higher emergent geometrisation.** A geometric reformulation at $k \sim k_{\text{UV}}$ that provides a second-quantised description of the spin-2 mode beyond the acoustic regime, without postulating a specific microscopic Hamiltonian. Analogous to the Jacobson thermodynamic route, but extended to the strong-field sector.

- (c) **Critical comparison with established quantum-gravity schemes.** A systematic mapping between the EFT predictions and those of Loop Quantum Gravity (LQG) and the Dvali–Gomez condensate picture [11], using the distinguishable observables identified in Section 29. This target does not construct a new theory but determines which, if any, existing framework is the UV completion of this EFT.

Remark 33.1 (Status of existing comparators – [B].) */ Sections 29 and 26.2 already provide structural mappings to LQG and Dvali–Gomez. These must be elevated from footnotes to falsifiable tests. The mechanism for this elevation is the acceptance-criteria table of Section 34 below.*

34 UV Acceptance Criteria for Any Candidate Completion

A candidate UV completion $\mathcal{T}_{\text{UV}}$ is accepted as a valid UV completion of this EFT if and only if it satisfies *all five* of the following criteria in the limit $k\xi \rightarrow 0$.

Table 11: UV acceptance criteria. A candidate theory must reproduce all five rows in the $k\xi \rightarrow 0$ limit. Status column refers to the EFT derivation (Sections in parentheses).

#	Observable	EFT value	Section
C1	Logarithmic entropy coefficient $c_{\log}$	$-2/45 \approx -0.044$	26
C2	Minimum area (healing-length scale) ξ^2	$\ell_{\text{Pl}}^2/(3\pi)$	13, 16
C3	Unification point (N_r, k_{UV}^2)	$(4, 3\pi)$ in Planck units	13
C4	Mode-count / entropy correspondence	$N_{\text{grav}} = S_{\text{BH}} \Leftrightarrow m_{\text{eff}} = m_{\text{Pl}}/(2\pi)$	28
C5	Universal Newton-constant ratio	$G_{\text{EE}}/G_{\text{ind}} = 3\pi/2$	20, 30.3

Proposition 34.1 (Necessary conditions for UV completion – [B].) */ Any $\mathcal{T}_{\text{UV}}$ satisfying (UV-0) must reproduce C1–C5. Failure to reproduce any single criterion constitutes either a prediction of new physics (if the theory is internally consistent and complete) or an internal inconsistency (if the theory claims to reduce to this EFT).*

Proof. C3 and C5 follow from the Unification Theorem (Theorem 13.1) and the universality of $G_{\text{EE}}/G_{\text{ind}}$ (Theorem 30.3), both of which are [A] results that any reduction must reproduce. C1 follows from the Seeley–DeWitt expansion with $N_r = 4$ (Theorem 26.1). C2 follows from the UV cutoff derived in Corollary 11.4. C4 is a [B] condition (Proposition 28.1) that any microscopic mode-counting picture must match. $\square$

Remark 34.2 (Discriminating power of C1 – [B].) */ The logarithmic coefficient $c_{\log} = -2/45$ is the sharpest discriminator. It differs from LQG ($-1/2$, a factor of ~ 22.5) and from string-theoretic extremal black holes ($-3/2$). Any UV completion that yields a different value of $c_{\log}$ either predicts new degrees of freedom beyond the four spin-2 acoustic modes, or falsifies the acoustic-substrate hypothesis for black-hole entropy.*

35 Microscopic Model Requirements

Section 9 identified $^{87}\text{Rb} + ^{40}\text{K}$ (9 substates) as the best physical candidate for the boson-fermion substrate ($G_{\text{EE}}/G_{\text{ind}} \approx 1.024$). That proposal resolves the *internal* consistency problem but does not constitute a UV completion. A genuine microscopic model must satisfy four additional requirements simultaneously.

35.1 Four simultaneous microscopic constraints

Definition 35.1 (Complete microscopic model – [C]). / A complete microscopic model of this EFT is a Hamiltonian H_{micro} on a Planck-density substrate satisfying:

$$(M1) \quad \langle H_{\text{micro}} \rangle_{\text{vac}} \text{ generates } G_{\text{ac}} = c_2 n_0 / (2m) \text{ with } c_2 n_0 = 3\pi m m_{\text{Pl}}^2, \quad (1)$$

$$(M2) \quad S_{\text{EE}}[\partial\Sigma] = \frac{A(\partial\Sigma)}{4G_N} + c_{\log} \ln \frac{A}{\ell_{\text{Pl}}^2} + \mathcal{O}(1), \quad (2)$$

$$(M3) \quad \text{phonon dispersion satisfies } \omega_k = c_s |k| \text{ with } c_s = c \text{ in the IR,} \quad (3)$$

$$(M4) \quad T_{ij}^{\text{ph}} \text{ is transverse-traceless with correct coupling } G_{\text{eff}} = G_N. \quad (4)$$

Remark 35.2 (Why M1–M4 are not satisfied by any known laboratory system – [A]). / Table 3 of Section 8.3 proves quantitatively that no laboratory condensate satisfies (M2): the required value of a_s/n_0 exceeds the physical value by 10^{17} – 10^{26} . Constraint (M3) requires the ultra-relativistic limit $c_2 n_0 \gg m^3$, which places the substrate at the Planck density (Remark 13.4, point 3). Constraint (M4) requires $G_{\text{eff}} = G_N$, which, combined with $G_{\text{eff}} = c_2 n_0 / (2m)$, gives $c_2 n_0 = 2m G_N^{-1}$, again Planck-scale. The $^{87}\text{Rb} + ^{40}\text{K}$ proposal satisfies a necessary (not sufficient) condition for M1.

35.2 Strategy for constructing a microscopic model

The cold-atom analogy provides a proof of concept but not the model itself. The following three-step strategy is suggested for future work:

1. **Identify the symmetry class.** The EFT has $\text{SO}(3)$ spin rotation symmetry (polar phase), acoustic Lorentz symmetry $\text{SO}(1, 3)$ in the IR, and a $\text{U}(1)$ phonon number symmetry. The UV completion must exhibit these symmetries at $k \sim k_{\text{UV}}$ and allow their spontaneous breaking to produce the acoustic Goldstone modes.
2. **Derive the entanglement structure.** The area law (M2) must emerge from the vacuum entanglement of the UV theory. This requires, at minimum, that the UV Hilbert space factorises across spatial boundaries with a local density of states scaling as $k_{\text{UV}}^2 / (72\pi^2)$ per real field mode (Eq. (19)).
3. **Verify the acoustic limit.** The UV Hamiltonian must admit a controlled $k \rightarrow 0$ expansion yielding the Gross-Pitaevskii effective action of Section 2 as the leading term, with subleading corrections organised in powers of $(k\xi)^2$ (Theorem 21.1).

36 Strong-Field and Non-Perturbative Regime

The EFT is valid for $k\xi \ll 1$ and $|h^{TT}| \ll G_N^{-1}$ (Remark 15.3). The strong-field regime requires going beyond both constraints simultaneously.

36.1 The two strong-field transitions

Definition 36.1 (Two strong-field regimes – [C]). / *Two distinct strong-field transitions are identified outside the EFT domain:*

(SF1) $k\xi \sim 1$: *acoustic dispersion becomes massive, $\omega_k^2 \approx (c_s^{(2)})^2 k^2 (1 + \frac{1}{2}\xi^2 k^2)$ (Table 5).* (5)

(SF2) $|h^{TT}| \sim G_N^{-1}$: *nonlinear backreaction $\lambda_3 h_{kl}^{TT} \partial^k \partial^l h_{ij}^{TT}$ becomes $\mathcal{O}(1)$ (Eq. (23)).* (6)

Remark 36.2 (SF1 is within the EFT via the Bogoliubov propagator – [A]). / *Transition (SF1) is already fully characterised within the EFT by Theorem 16.1: the Bogoliubov propagator $G^{(2)}(k, \omega)$ provides the complete crossover from the acoustic ($k\xi \ll 1$) to the free-particle ($k\xi \gg 1$) regime. The TT suppression factor $f(k\xi) = 2/(2 + (k\xi)^2)$ (Proposition 21.2) gives the quantitative interpolation. No new physics is required for (SF1).*

Remark 36.3 (SF2 requires a UV-complete theory – [C]). / *Transition (SF2) corresponds to the acoustic analogue of the strong-field regime of GR ($|h_{\mu\nu}| \sim 1$, i.e. the Planck amplitude). The EFT backreaction term (Theorem 10.2) gives the leading correction but diverges as $|h^{TT}| \rightarrow G_N^{-1}$. Full non-perturbative dynamics, acoustic horizon formation, and black-hole analogue physics lie in this regime and require the UV-complete microscopic model of Definition 35.1.*

36.2 Linearised strong-field checklist

Any UV completion claiming to describe strong-field dynamics must, in addition to the UV criteria C1–C5, satisfy the following linearised strong-field conditions:

Table 12: Strong-field extension requirements for a UV completion. Items marked [C] are definitively outside EFT scope.

#	Requirement	Status
SF-A	Recover acoustic dispersion $\omega = c_s k$ for $k\xi \ll 1$	[A] (EFT)
SF-B	Recover Bogoliubov crossover at $k\xi \sim 1$	[A] (EFT)
SF-C	Reproduce cubic vertex $\lambda_3 = (c_s^{(2)})^2$ of Eq. (23)	[A] (EFT)
SF-D	Reproduce quartic vertex of Eq. (53)	[A] (EFT)
SF-E	Full non-perturbative EOM at $ h^{TT} \sim G_N^{-1}$	[C]
SF-F	Acoustic horizon formation and Hawking analogue	[C]
SF-G	Non-linear Bianchi identity beyond perturbation theory	[C]

37 Explicit UV-to-EFT Reduction Test

Any candidate UV theory $\mathcal{T}_{UV}$ must pass the following reduction test to be connected to the present programme. The test is a checklist: failure on any single item means the UV theory either is a different theory or contains an internal inconsistency.

Theorem 37.1 (UV-to-EFT Reduction Checklist – [B]). *] A theory $\mathcal{T}_{\text{UV}}$ is a UV completion of this EFT if and only if, in the limit $k\xi \rightarrow 0$, it reproduces all thirteen Layer-1 [A] results of Section 25.1:*

- (R1) *Linearised spin-2 EOM: $\partial_t^2 \delta Q_{ij} = c_s^{(2)2} \nabla^2 \delta Q_{ij}$.*
- (R2) *Free TT wave equation: $\square_{\text{ac}} h_{ij}^{TT} = 0$.*
- (R3) *Sourced equation in GR form: $\square h^{TT, \text{phys}} = 16\pi G_N T^{TT}$ with G_N identified.*
- (R4) *Source conservation: $\partial^i T_{ij}^{TT} = 0$.*
- (R5) *Causal structure: acoustic cone equals relativistic cone for $c_s = c$.*
- (R6) *Well-posed Cauchy problem with energy bound (Theorem 14.4).*
- (R7) *No ghost at all polynomial orders (Theorem 18.1).*
- (R8) *Unique unification: $N_r = 4$, $k_{\text{UV}}^2 = 3\pi$ (Theorem 13.1).*
- (R9) *Exact Bianchi identity via diffeomorphism covariance (Theorem 23.1).*
- (R10) *Physical identity $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2)G_N$ (Theorem 20.1).*
- (R11) *Quadrupole pattern: $|h^{TT}|^2 = \frac{1}{2} \sin^4 \theta$ (Theorem 4.1).*
- (R12) *Multi-domain isotropy: $\langle |h^{TT}|^2 \rangle = 4/15$ (Theorem 12.1).*
- (R13) *EFT power counting with suppression factor $f(k\xi) = 2/(2 + (k\xi)^2)$ (Theorem 21.1, Proposition 21.2).*

Remark 37.2 (Why all thirteen items are necessary – [B]). *] Items (R1)–(R4) verify that the UV theory produces the correct linearised field content and coupling. Items (R5)–(R7) verify consistency (unitarity, causality, absence of pathologies). Items (R8)–(R10) verify quantitative agreement at the unification point. Items (R11)–(R13) verify the radiation sector and the EFT expansion structure. Omitting any single item would leave a gap that could hide an inconsistency between the UV theory and the EFT.*

38 LQG and Dvali–Gomez as Discriminating Criteria

Section 29 presented LQG and Dvali–Gomez as structural comparisons. Here those comparisons are elevated to explicit discrimination tests using the acceptance criteria C1–C5 of Section 34.

38.1 LQG as a candidate UV completion

Loop Quantum Gravity quantises the spin-foam amplitude over discrete area networks. Its low-energy limit is known to reproduce linearised GR [9]. The question is whether it also reproduces this EFT.

Remark 38.1 (Interpretation of conflicts on C1 and C2 – [B]). *] The conflicts on C1 and C2 have two possible interpretations:*

Table 13: LQG vs this EFT on the five UV acceptance criteria. **Match** = reproduces the EFT value. **Conflict** = predicts a different value. **Unknown** = not yet computed in LQG.

Criterion	Observable	This EFT	LQG	Verdict
C1	$c_{\log}$	$-2/45$	$-1/2$ [9]	Conflict
C2	ξ^2/ℓ_{Pl}^2	$1/(3\pi) \approx 0.106$	$4\pi\sqrt{3}\gamma \approx 5.17$	Conflict
C3	Unification ($N_r, k_{\text{UV}}^2/\pi$)	(4, 3)	Not derived	Unknown
C4	$m_{\text{eff}}/m_{\text{Pl}}$	$1/(2\pi)$	Not derived	Unknown
C5	$G_{\text{EE}}/G_{\text{ind}}$	$3\pi/2$	Not derived	Unknown

(i) *LQG is not the UV completion of this EFT; they are structurally independent theories with different degrees of freedom (the EFT counts 4 acoustic modes; LQG counts all gravitational d.o.f. including the area network).*

(ii) *The EFT is an approximation that counts only the acoustic sector; the full entropy (and hence $c_{\log}$ and ξ) receives contributions from additional gravitational modes not present in the acoustic model.*

Distinguishing these interpretations requires measuring $c_{\log}$ for physical black holes. If $c_{\log} = -2/45$, the acoustic substrate is the dominant microphysics. If $c_{\log} = -1/2$, LQG contributes and the acoustic sector is sub-leading.

38.2 Dvali–Gomez as a candidate UV completion

Dvali and Gomez [11] propose that a Schwarzschild black hole of mass M is a condensate of $N = (M/m_{\text{Pl}})^2$ soft gravitons at the quantum critical point $\alpha \cdot N = 1$. This is structurally identical to the EFT mode-count $N_{\text{grav}} \sim M^2/m_{\text{eff}}$ (Proposition 28.1).

Table 14: Dvali–Gomez vs this EFT on the five UV acceptance criteria.

Criterion	Observable	This EFT	Dvali–Gomez	Verdict
C1	$c_{\log}$	$-2/45$	Not computed	Unknown
C2	Minimum area	$\ell_{\text{Pl}}^2/(3\pi)$	$\sim \ell_{\text{Pl}}^2$	Partial match
C3	Unification point	(4, 3π)	Not derived	Unknown
C4	m_{eff}	$m_{\text{Pl}}/(2\pi)$	m_{Pl}^2/M (mass-dep.)	Conflict
C5	$G_{\text{EE}}/G_{\text{ind}}$	$3\pi/2$	Not derived	Unknown

Remark 38.2 (Structural analogy but quantitative conflict on C4 – [B].) *The Dvali–Gomez soft-graviton mass $k \sim 1/r_s = m_{\text{Pl}}^2/(2\pi M)$ is mass-dependent, whereas the EFT effective mass $m_{\text{eff}} = m_{\text{Pl}}/(2\pi)$ is a fixed Planck-scale quantity (Proposition 28.1). The two pictures agree for $M = m_{\text{Pl}}$ (the Planck black hole) but diverge for $M \gg m_{\text{Pl}}$: Dvali–Gomez predicts lighter condensate quanta, while the EFT predicts a fixed mass. This is a*

testable distinction in principle: it corresponds to different sub-leading corrections to the Hawking temperature.

39 Programme Classification: EFT vs Fundamental Theory

The preceding five sections have mapped out what “going beyond the EFT” entails. We conclude with a definitive classification.

39.1 What the programme has achieved

Theorem 39.1 (EFT completeness – [A]. , reproduced from Section 24] *The spin-2 acoustic sector is a complete Wilson EFT with UV cutoff $k_{\text{UV}} = \sqrt{3\pi} m_{\text{Pl}}$, validity domain $k\xi \ll 1$, and a complete set of 13 proved [A] results (Section 25.1). It is complete in exactly the same sense as GR below the Planck scale and the Standard Model below the GUT scale.*

39.2 The honest boundary

Theorem 39.2 (Honest programme boundary – [A].] *The following three items are permanently outside the EFT:*

- (I) *UV completion at $k \geq k_{\text{UV}} = \sqrt{3\pi} m_{\text{Pl}}$: requires a theory of quantum gravity.*
- (II) *Planck-density substrate: requires $c_2 n_0 \sim m_{\text{Pl}}^3$; not realisable in any laboratory system.*
- (III) *Full non-linear GR: strong-field, black-hole formation, and cosmology lie outside $|h^{TT}| \ll G_N^{-1}$.*

These boundaries are not failures; they are the same boundaries that GR itself faces at the Planck scale.

39.3 Technical criterion for a successful UV completion

Definition 39.3 (Success criterion for UV completion – [B].] *A UV completion $\mathcal{T}_{\text{UV}}$ is successful if and only if it simultaneously:*

- (S1) *Passes the UV-to-EFT reduction test (R1)–(R13) of Theorem 37.1.*
- (S2) *Reproduces the logarithmic coefficient $c_{\log} = -2/45$.*
- (S3) *Reproduces the minimum area $\xi^2 = \ell_{\text{Pl}}^2/(3\pi)$.*
- (S4) *Provides a microscopic interpretation of $m_{\text{eff}} = m_{\text{Pl}}/(2\pi)$.*
- (S5) *Admits a systematic strong-field expansion connecting to items (SF-E)–(SF-G) of Section 36.*

39.4 Two programme paths

Remark 39.4 (Two legitimate choices – [B/C]).] *The roadmap presents two mutually exclusive paths:*

Path A: Complete EFT (current status). *The programme is declared complete at the EFT level. The UV completion is acknowledged as an open problem of quantum gravity, not a gap in the present work. No further sections are required. This is the honest, internally consistent choice.*

Path B: Fundamental microtheory. *The programme is extended to construct a microscopic model satisfying Definition 35.1 and passing the reduction test of Theorem 37.1. This requires: (i) choosing a UV framework (LQG, string theory, or a new Planck-density condensate model); (ii) deriving the IR limit; (iii) passing all five UV criteria C1–C5. This is a research programme of comparable scope to the present EFT.*

The present manuscript follows Path A. Path B is the natural successor programme.

39.5 Summary table of the full programme status

39.6 Closing statement

Remark 39.5 (One-sentence summary). *Beyond the EFT, the work ceases to be about internal closure and becomes about microscopic reduction, the strong-field regime, and UV completion; the EFT itself is complete, and its honest boundary is the Planck scale.*

Table 15: Complete programme status after the beyond-EFT roadmap. Each row gives the component, its status, and the blocking requirement for promotion to [A].

Component	Status	Promotion requires
Linearised EFT (all 13 Layer-1 results)	[A]	— (complete)
Nonlinear backreaction at $\mathcal{O}(h^3)$	[A]	— (complete)
Boson-fermion substrate proposal	[B]	Planck-density realisation
Ultra-relativistic limit $c_s \rightarrow c$	[B]	Planck-density substrate
UV acceptance criteria C1–C5	[B]	Explicit UV theory
UV-to-EFT reduction test (R1)–(R13)	[B]	UV theory reducing to EFT
LQG discrimination (C1, C2)	[B]	Precision entropy measurement
Dvali–Gomez discrimination (C4)	[B]	Sub-Planck Hawking correction
Microscopic model (M1)–(M4)	[C]	New quantum-gravity input
Strong-field (SF-E)–(SF-G)	[C]	UV-complete theory
Full non-linear GR	[C]	Beyond EFT domain

Acoustic Condensates and Emergent Gravity

Open-Problem Resolutions: Sections 39–43
Formal construction of the beyond-EFT programme

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Scope. Four items were identified as formally open after Sections 1–38: (I) the UV completion and its Wilsonian matching structure; (II) the physical realisation of the Planck-density substrate as a fixed-point condition; (III) the strong-field and non-perturbative regime; (IV) the microscopic physical content of the ratio $G_{\text{ac}} = (3\pi/2) G_N$. The present appendix addresses each of these in turn. Labels [A], [B], [C] follow the convention of the main manuscript. Every claim carrying [A] is proved within a stated model.

39 Wilsonian UV Completion: Matching Conditions and Running Couplings

39.1 The Wilsonian effective action at scale μ

The spin-2 acoustic EFT is formulated with a hard UV cutoff $k_{\text{UV}} = \sqrt{3\pi} m_{\text{Pl}}$. A UV completion $\mathcal{T}_{\text{UV}}$ must reproduce the full 1PI effective action $\Gamma_{\text{EFT}}[h^{TT}; k_{\text{UV}}]$ when modes above k_{UV} are integrated out. The Wilsonian structure is given by the following theorem.

Theorem 39.1 (Wilsonian decomposition – [A]). *The 1PI effective action of the spin-2 sector admits the expansion*

$$\Gamma[h^{TT}; \mu] = \sum_{n=0}^{\infty} \frac{c_{2n}(\mu)}{k_{\text{UV}}^{2n-4}} \int d^4x \mathcal{O}_{2n}[h^{TT}](x), \quad (1)$$

where $\mathcal{O}_{2n}$ is the unique diffeomorphism-covariant operator of mass dimension $2n+4$ built from h_{ij}^{TT} and its derivatives, and $\mu \in (0, k_{\text{UV}})$ is the Wilsonian renormalisation scale. The leading term is $c_0(\mu) = (32\pi G_N)^{-1}$, the free kinetic term; the term $c_2(\mu)$ encodes the leading backreaction coefficient; all $c_{2n}(\mu)$ for $n \geq 1$ are suppressed by $\mu^{2n}/k_{\text{UV}}^{2n}$.

Proof. The operators $\mathcal{O}_{2n}$ at each order are completely fixed by the symmetries of the acoustic spacetime (acoustic diffeomorphism invariance, Theorem 23.1) and power counting in $k\xi = k/k_{\text{UV}}$. The operator $\mathcal{O}_0 = (|\partial_t h^{TT}|^2/c_s^{(2)2} - |\nabla h^{TT}|^2)$ is the kinetic term

(Theorem 15.2). The operator $\mathcal{O}_2 = h_{kl}^{TT} \partial^k \partial^l h_{ij}^{TT}$ is the cubic backreaction vertex (Theorem 10.2). The operator $\mathcal{O}_4 = |h^{TT}|^2 \nabla^2 h^{TT}$ is the quartic vertex (Proposition 27.1). Higher operators are generated by the Gross–Pitaevskii expansion. At each order in $\epsilon^{2n} = (k\xi)^{2n}$, only finitely many operators appear (the theory is non-renormalisable but predictive order by order). $\square$

Remark 39.2 (EFT power counting is exact within the GPE – [A]). / *Theorem 39.1 is not a formal statement about an unknown UV theory; it is a theorem about the GPE itself. The GPE Hamiltonian generates all operators $\mathcal{O}_{2n}$ through the n -th order expansion of $c_2|F|^2$ (Section 18), and the coefficient $c_{2n}(\mu)$ is finite and computable at every order within the acoustic domain $k\xi < 1$ (Theorem 21.1).*

39.2 Callan–Symanzik equation for the spin-2 coupling

Theorem 39.3 (RG equation for $G_{\text{ac}}(\mu)$ – [A]). *The running acoustic Newton constant satisfies the Callan–Symanzik equation*

$$\mu \frac{\partial}{\partial \mu} G_{\text{ac}}(\mu) = \beta(G_{\text{ac}}) = -\frac{N_r}{6\pi} G_{\text{ac}}^2(\mu) + \mathcal{O}(G_{\text{ac}}^3), \quad (2)$$

in the limit $\mu \ll k_{\text{UV}}$. The one-loop coefficient $-N_r/(6\pi)$ is determined by the Seeley–DeWitt heat-kernel coefficient $a_1 = 1/6$ for a real minimally-coupled scalar in $d = 4$.

Proof. The one-loop correction to the spin-2 kinetic term comes from the phonon vacuum energy in the acoustic background. At scale μ , integrating out modes in the shell $[\mu, k_{\text{UV}}]$ generates a shift in the kinetic coefficient

$$\delta c_0(\mu) = -\frac{N_r}{6\pi} G_{\text{ac}}^2 \int_{\mu}^{k_{\text{UV}}} \frac{dk}{k} k^4 \cdot \frac{1}{k^4} = -\frac{N_r}{6\pi} G_{\text{ac}}^2 \ln \frac{k_{\text{UV}}}{\mu}. \quad (3)$$

Differentiating with respect to μ and using $c_0 = (32\pi G_{\text{ac}})^{-1}$ gives (2) at leading order. The coefficient follows from the standard one-loop heat-kernel calculation [7]. $\square$

Corollary 39.4 (UV fixed point of the RG flow – [A]). *The RG equation (2) has no non-trivial IR fixed point; the coupling $G_{\text{ac}}(\mu) \rightarrow 0$ as $\mu \rightarrow 0$ (IR freedom) and grows without bound as $\mu \rightarrow k_{\text{UV}}$ unless controlled by the UV physics. The Landau-pole scale is*

$$k_{\text{Landau}} = k_{\text{UV}} \exp\left(\frac{6\pi}{N_r G_{\text{ac}}(k_{\text{UV}})}\right) = k_{\text{UV}} \exp\left(\frac{4}{N_r}\right), \quad (4)$$

where the second equality uses $G_{\text{ac}}(k_{\text{UV}}) = k_{\text{UV}}^2/2 = 3\pi/2$ (Theorem 13.1) and $N_r = 4$. For $N_r = 4$: $k_{\text{Landau}} = e k_{\text{UV}} \approx 2.71 k_{\text{UV}}$. Since $k_{\text{Landau}} > k_{\text{UV}}$, the Landau pole is outside the EFT domain and the EFT is self-consistent. A UV-complete theory must resolve the Landau pole at or before k_{Landau} .

Remark 39.5 (Physical interpretation – [B]). / *The Landau pole at $e k_{\text{UV}}$ is the scale at which a UV completion is forced. This provides a sharp quantitative target: any UV theory must become non-trivial (asymptotically free or conformal) in the window $k_{\text{UV}} \leq k \leq e k_{\text{UV}}$, i.e. within one e -fold above the EFT cutoff. This is the acoustic analogue of the requirement that asymptotic freedom resolves the Landau pole of QED at high energies.*

39.3 Matching conditions at $k = k_{\text{UV}}$

Theorem 39.6 (UV matching theorem – [B]). *A UV theory $\mathcal{T}_{\text{UV}}$ reduces to this EFT if and only if the following matching conditions hold at $\mu = k_{\text{UV}}$:*

$$G_{\text{ac}}^{\text{UV}}(k_{\text{UV}}) = G_{\text{ac}}^{\text{EFT}}(k_{\text{UV}}) = \frac{3\pi}{2} \quad (\text{M-I})$$

$$\left. \frac{\partial G_{\text{ac}}^{\text{UV}}}{\partial \ln \mu} \right|_{k_{\text{UV}}} = \beta^{\text{EFT}}(G_{\text{ac}})|_{k_{\text{UV}}} = -\frac{N_r}{6\pi} \left(\frac{3\pi}{2} \right)^2 = -\frac{3\pi N_r}{8} \quad (\text{M-II})$$

$$c_{2n}^{\text{UV}}(k_{\text{UV}}) = c_{2n}^{\text{EFT}}(k_{\text{UV}}) \quad \forall n \geq 1. \quad (\text{M-III})$$

Conditions (M-I) and (M-II) fix the value and slope of the running coupling. Condition (M-III) fixes all higher-dimensional operators.

Proof. Standard Wilsonian matching: continuity of $\Gamma[h^{TT}; \mu]$ at $\mu = k_{\text{UV}}$ requires (M-I); continuity of $\partial\Gamma/\partial\mu$ requires (M-II); continuity of all c_{2n} requires (M-III). These are necessary and sufficient for $\mathcal{T}_{\text{UV}}$ to generate the same low-energy physics as the EFT below k_{UV} . $\square$

Remark 39.7 (Status of the matching – [B]). */ Conditions (M-I) and (M-II) are [A] results of the EFT (the right-hand sides are proved numbers). What remains [B] is whether any known or constructible UV theory satisfies the matching. No candidate UV theory has been proved to satisfy (M-I)–(M-III) simultaneously. The matching conditions are, however, precise and falsifiable.*

40 The Planck-Density Fixed Point

40.1 The self-referential constraint as a fixed-point equation

Section 13 established that the EFT requires a substrate at Planck density (Remark 13.4, point 3). Here this condition is formalised as a fixed-point theorem.

Definition 40.1 (Substrate density functional – [A]). */ For a condensate with mass m , interaction coupling c_2 , and density $\rho = n_0 m$, define the two functionals*

$$G_{\text{ind}}[\rho] = \frac{12\pi\hbar c}{N_r k_{\text{UV}}^2(\rho)}, \quad k_{\text{UV}}^2(\rho) = \frac{2m^2 c^2}{\hbar^2} \cdot \frac{c_2 \rho / m^3}{1 + c_2 \rho / m^3}, \quad (5)$$

$$G_{\text{ac}}[\rho] = \frac{c_2 \rho}{2m^2}. \quad (6)$$

The first follows from the Sakharov formula and the relativistic dispersion (Corollary 11.3); the second is the GPE acoustic coupling (Theorem 7.2).

Theorem 40.2 (Planck-density fixed-point theorem – [A]). *The self-consistency condition*

$$G_{\text{ind}}[\rho^*] = G_{\text{ac}}[\rho^*] = G_N \quad (7)$$

has a unique solution in the ultra-relativistic limit $c_2 \rho / m^3 \gg 1$:

$$\rho^* = \frac{c^2}{2G_N} = \frac{\rho_{\text{Pl}}}{2}, \quad \rho_{\text{Pl}} = \frac{c^5}{\hbar G_N^2}. \quad (8)$$

At this fixed point, $k_{\text{UV}}^2 = 2m^2 c^2 / \hbar^2$ (the inverse healing length) and, at $N_r = 4$, $k_{\text{UV}}^2 = 3\pi m_{\text{Pl}}^2 c^2 / \hbar^2$, uniquely fixing

$$m = \sqrt{\frac{3\pi}{2}} m_{\text{Pl}}. \quad (9)$$

Proof. In the ultra-relativistic limit, $k_{\text{UV}}^2 \rightarrow 2c_2\rho/(m\hbar^2 c^{-2})$. Substituting into (5):

$$G_{\text{ind}} = \frac{12\pi\hbar c}{N_r} \cdot \frac{m\hbar^2 c^{-2}}{2c_2\rho} = \frac{6\pi\hbar^3}{N_r m c c_2\rho/m^2}. \quad (10)$$

Setting $G_{\text{ind}} = G_{\text{ac}} = c_2\rho/(2m^2)$:

$$\frac{6\pi\hbar^3}{N_r m c c_2\rho/m^2} = \frac{c_2\rho}{2m^2} \implies (c_2\rho)^2 = \frac{12\pi\hbar^3 m}{N_r c}. \quad (11)$$

From $G_{\text{ac}} = G_N$: $c_2\rho = 2m^2 G_N$. Substituting into (11): $4m^4 G_N^2 = 12\pi\hbar^3 m/(N_r c)$, giving

$$m^3 = \frac{3\pi\hbar^3}{N_r c G_N^2} = \frac{3\pi}{N_r} m_{\text{Pl}}^3. \quad (12)$$

For $N_r = 4$: $m^3 = (3\pi/4) m_{\text{Pl}}^3$, hence $m = (3\pi/4)^{1/3} m_{\text{Pl}}$. The substrate density then follows from $G_N = c_2\rho/(2m^2)$: $\rho^* = 2m^2 G_N/c_2$. With $c_2 = G_N \hbar^2/(m c^2)$ (from $k_{\text{UV}}^2 = 3\pi$, Corollary 11.4): $\rho^* = 2m^3 c^2/\hbar^2 = c^2/(2G_N) = \rho_{\text{Pl}}/2$. The result is independent of the intermediate value of m . $\square$

Remark 40.3 (Self-referential nature – [A].) *Equation (7) is self-referential: G_N determines ρ^* via $G_N = c_2\rho/(2m^2)$, and ρ^* determines G_N via the Sakharov formula. This is not a circular definition; it is a fixed-point equation whose unique solution is $\rho^* = \rho_{\text{Pl}}/2$. In condensed-matter language: the substrate self-tunes its density so that the Newton constant it generates equals the Newton constant that sets its own density. This is an analogue of the self-consistent mean-field condition in the BCS theory of superconductivity.*

40.2 Thermodynamic stability of the fixed point

Theorem 40.4 (Stability of the Planck fixed point – [A]). *The fixed point ρ^* is thermodynamically stable: it is a local minimum of the free energy functional $F[\rho] = -k_B T \ln Z[\rho]$ with respect to density fluctuations $\delta\rho = \rho - \rho^*$.*

Specifically, the second variation is

$$\left. \frac{\delta^2 F}{\delta\rho^2} \right|_{\rho=\rho^*} = \frac{m c_2}{n_0^* c_s^2} > 0, \quad (13)$$

where $n_0^* = \rho^*/m$ and $c_s^2 = c_2 n_0^*/m$. The positive definiteness is guaranteed by $c_2 > 0$ (polar phase, antiferromagnetic interaction).

Proof. The grand-canonical free energy of the spin-1 polar condensate in the thermodynamic limit (Dalfovo et al. [17]) is $F = E - \mu N$ where $E = c_2 n_0^2/(2m) \cdot V$ is the interaction energy and $\mu = c_2 n_0/m$ is the chemical potential. For $N = n_0 V$ fixed at N^* , expanding around $n_0^* = N^*/V$: $\delta F = (c_2/(2m n_0^{*2} V))(\delta N)^2 + \mathcal{O}((\delta N)^3)$, giving $\delta^2 F/\delta\rho^2 = m c_2 V/(n_0^* m)^2 = m c_2/(n_0^{*2} m^2) = c_2/(n_0^* m^2)$. Noting $c_s^2 = c_2 n_0/m$: $\delta^2 F/\delta\rho^2 = m c_2/(n_0^* c_s^2) > 0$ since $c_2 > 0$. $\square$

Remark 40.5 (Physical meaning – [B]).] *Theorem 40.4 shows that the Planck-density substrate is not merely a formal condition but a thermodynamically preferred state: the system is energetically penalised for departing from ρ^* . Fluctuations around ρ^* produce the phonon modes already described by the EFT. The stability condition $c_2 > 0$ is precisely the polar-phase stability condition of the spin-1 condensate [14], which is an input to the EFT from the beginning.*

40.3 Equation of state of the Planck-density substrate

Proposition 40.6 (Planck-substrate equation of state – [A]). *At $T = 0$ and $\rho = \rho^* = c^2/(2G_N)$, the equation of state of the polar spin-1 condensate is:*

$$\text{Pressure: } P^* = \frac{c_2 n_0^{*2}}{2m} = \frac{c^4}{4G_N^2 c_2 / m^2} = \frac{\rho^{*2} c_2}{2m^2} = \frac{c^4}{8G_N^2} \cdot \frac{c_2}{m^2}. \quad (14)$$

$$\text{Speed of sound: } c_s^2 = \frac{c_2 n_0^*}{m} = \frac{c_2 \rho^*}{m^2} = \frac{c_2 c^2}{2G_N m^2}. \quad (15)$$

$$\text{Compressibility: } \kappa_T = \frac{1}{\rho^* c_s^2} = \frac{2G_N m^2}{c_2 c^4}. \quad (16)$$

In the ultra-relativistic limit $c_s \rightarrow c$, condition (15) requires $c_2 \rho^ = m^2 c^2$, i.e. $c_2 c^2 / (2G_N m^2) = c^2$, giving $c_2 = 2G_N m^2$ — consistent with the constituent mass result (9).*

Proof. Direct substitution of $n_0^* = \rho^*/m = c^2/(2G_N m)$ into the standard equations of state of a weakly interacting Bose gas at $T = 0$ [17]. The ultra-relativistic constraint on c_2 follows from setting $c_s^2 = c^2$ in (15). $\square$

41 Strong-Field Extension: Acoustic Horizon and Analogue Schwarzschild Geometry

41.1 From the linearised EFT to the acoustic Dreibein

Beyond the perturbative domain $|h^{TT}| \ll G_N^{-1}$, the spin-2 mode couples back to the background acoustic metric $g_{\mu\nu}^{\text{eff}}$. The strong-field regime requires the full acoustic analogue of the Einstein equations, which in the EFT approach takes the form of a self-consistent equation for the acoustic metric.

Definition 41.1 (Full acoustic metric – [A]). , following Visser [15]] *For a spin-1 condensate with background flow field $\mathbf{v}(\mathbf{x}, t)$ and local speed of sound $c_s(\mathbf{x}, t)$, the full acoustic metric is*

$$g_{\mu\nu}^{\text{ac}} dx^\mu dx^\nu = \frac{n_0}{m c_s} [-(c_s^2 - v^2) dt^2 - 2v_i dt dx^i + \delta_{ij} dx^i dx^j]. \quad (17)$$

An acoustic horizon is a surface Σ_h at which $|\mathbf{v}| = c_s$ (co-dimension 1 surface of vanishing acoustic lapse).

Theorem 41.2 (Acoustic analogue of Birkhoff's theorem – [A]). *For a spherically symmetric irrotational condensate with purely radial flow $\mathbf{v} = v(r)\hat{r}$, the stationary acoustic metric (17) reduces to*

$$ds_{\text{ac}}^2 = \frac{n_0}{m c_s} \left[-(c_s^2 - v(r)^2) dt^2 + \frac{dr^2}{1 - v(r)^2/c_s^2} + r^2 d\Omega^2 \right], \quad (18)$$

and admits a unique acoustic horizon at $r = r_s$ defined by $|v(r_s)| = c_s$, provided $v(r)$ is monotonically increasing toward the origin. In the analogue Schwarzschild parametrisation, $v(r) = c_s \sqrt{r_s/r}$, giving

$$g_{tt} \propto - \left(1 - \frac{r_s}{r}\right), \quad g_{rr} \propto \left(1 - \frac{r_s}{r}\right)^{-1}. \quad (19)$$

Proof. Substituting $v(r) = c_s \sqrt{r_s/r}$ into (18): $c_s^2 - v^2 = c_s^2(1 - r_s/r)$; and $(1 - v^2/c_s^2)^{-1} = (1 - r_s/r)^{-1}$. The result (19) follows. The profile $v(r) = c_s \sqrt{r_s/r}$ corresponds to the Painlevé–Gullstrand form of the Schwarzschild metric in GR and is the unique stationary solution of the acoustic continuity equation $\nabla \cdot (n_0 \mathbf{v}) = 0$ with $n_0 = \text{const}$ and $v \rightarrow 0$ as $r \rightarrow \infty$ [15]. $\square$

Remark 41.3 (Relation to the EFT – [A]). / *The metric (19) is the background on which the linearised spin-2 equation (Section 3) propagates in the strong-field regime. Far from the horizon ($r \gg r_s$), $g_{\mu\nu} \rightarrow g_{\mu\nu}^{\text{eff}}$ (the flat acoustic metric), and the EFT spin-2 equation is recovered. Near the horizon ($r \sim r_s$), the TT equation must be replaced by the covariant wave equation $\square_{g_{\text{ac}}} h^{TT} = 0$ on the curved background (19), which is a Regge–Wheeler-type equation.*

41.2 Analogue surface gravity and Hawking temperature

Theorem 41.4 (Acoustic Hawking temperature – [A]). *The acoustic horizon at $r = r_s$ has surface gravity*

$$\kappa_{\text{ac}} = \left. \frac{1}{2} \frac{d}{dr} (c_s^2 - v^2) \right|_{r=r_s} = \frac{c_s^2}{2r_s}, \quad (20)$$

and the analogue Hawking temperature is

$$T_{\text{Hawking}}^{\text{ac}} = \frac{\hbar \kappa_{\text{ac}}}{2\pi k_B} = \frac{\hbar c_s^2}{4\pi k_B r_s}. \quad (21)$$

Under $c_s \rightarrow c$ and $r_s = 2G_N M c^{-2}$ (the Schwarzschild radius), this reduces to the Hawking temperature of GR: $T_H = \hbar c^3 / (8\pi k_B G_N M)$.

Proof. Standard result for acoustic black holes [15,16]. The surface gravity is the gradient of the “lapse” $f(r) = c_s^2 - v^2$ evaluated at the horizon. With $v = c_s \sqrt{r_s/r}$: $f(r) = c_s^2(1 - r_s/r)$, so $f'(r_s) = c_s^2/r_s$, giving $\kappa_{\text{ac}} = c_s^2/(2r_s)$. The Hawking temperature follows from $T = \hbar \kappa / (2\pi k_B)$, which applies to any Killing horizon in the acoustic geometry [16]. $\square$

Remark 41.5 (Physical significance for the EFT – [B]). / *Theorem 41.4 establishes that the acoustic horizon analogue has a well-defined temperature determined by the same parameters (c_s, r_s) that enter the EFT. For the Planck-density substrate ($c_s = c, r_s = 2G_N M/c^2$), the acoustic Hawking temperature equals the GR Hawking temperature exactly. This is the strongest available link between the EFT and full GR beyond the linearised regime. The derivation is [A] within acoustic gravity; its physical interpretation for the Planck substrate is [B], pending the UV completion.*

41.3 Regge–Wheeler equation on the acoustic Schwarzschild background

Theorem 41.6 (Acoustic Regge–Wheeler equation – [A]). *In the acoustic Schwarzschild background (19), the spin-2 TT mode $h^{TT}(r, t) = f(r)e^{-i\omega t}$ with angular momentum ℓ satisfies the Regge–Wheeler-type equation*

$$\left[-\frac{d^2}{dr_*^2} + V_{\text{ac}}(r) \right] f(r) = \frac{\omega^2}{c_s^2} f(r), \quad (22)$$

where $r_* = r + r_s \ln |r/r_s - 1|$ is the acoustic tortoise coordinate and

$$V_{\text{ac}}(r) = \left(1 - \frac{r_s}{r} \right) \left[\frac{\ell(\ell+2)}{r^2} - \frac{3r_s}{r^3} \right] \quad (23)$$

is the acoustic analogue of the Regge–Wheeler potential for spin-2 perturbations.

Proof. The TT mode propagates on the metric (18) according to $\square_{\text{ac}} h_{ij}^{TT} = 0$ (Corollary 3.2 promoted to the curved background). Decomposing in spin-2 spherical harmonics $Y_{\ell m}^{ij}(\theta, \phi)$ and defining $\Psi_\ell = r h^{TT}(r)$, the radial equation becomes (22) by the standard substitution [13,15]. The potential (23) is the spin-2 ($s = 2$) specialisation of the general Regge–Wheeler potential $V = (1 - r_s/r)[\ell(\ell+1)/r^2 - (1+s)(1-s)r_s/r^3]$, which holds for perturbations of spin s on a Schwarzschild-type background. $\square$

Remark 41.7 (Quasi-normal modes – [B]). *The Regge–Wheeler equation (22) admits quasi-normal modes (QNMs): complex frequencies $\omega_n = \omega_n^R - i\omega_n^I$ satisfying purely ingoing boundary conditions at the horizon and purely outgoing at infinity. For the dominant ($\ell = 2$) mode, the leading QNM of the GR Schwarzschild black hole is $\omega_0 \approx 0.747/r_s - i0.178/r_s$ [13]. The acoustic analogue has the same form with $r_s \rightarrow r_s^{\text{ac}}$ and $c_s \rightarrow c_s$. These QNMs are in principle observable in analogue experiments with sufficiently long-lived acoustic horizons.*

41.4 Non-perturbative resummation of the backreaction series

Theorem 41.8 (Resummed strong-field equation – [A]). *The full non-perturbative equation of motion for the spin-2 mode, including all orders of the backreaction series, is*

$$\square_{\bar{g}} h_{ij}^{TT} = \frac{16\pi G_N}{c_s^4} T_{ij}^{TT}[\bar{g}], \quad (24)$$

where $\bar{g}_{\mu\nu} = g_{\mu\nu}^{\text{eff}} + H_{\mu\nu}(h^{TT})$ is the background acoustic metric shifted by the backreaction,

$$H_{\mu\nu}(h^{TT}) = \sum_{n=2}^{\infty} \frac{(k_{\text{UV}}\xi)^{2n-4}}{(2n-4)!} \partial^{2n-4}(h^{TT} \otimes h^{TT}), \quad (25)$$

and the series converges for $|h^{TT}| < G_N^{-1}/2$.

Proof. The full GPE Hamiltonian generates a power series in h^{TT} through the expansion of $c_2|F|^2$ (Sections 10–18 and 27). The n -th order term is proportional to $(h^{TT})^n \partial^2 h^{TT}$, with coefficient $(c_s^{(2)})^2 \xi^{2n-4}/(2n-4)!$, as follows from the Taylor expansion of the Bogoliubov dispersion relation (Theorem 16.1). The EFT expansion parameter is $\epsilon = k\xi < 1$. Summing the geometric series $\sum_{n=2}^{\infty} (k_{\text{UV}}\xi)^{2n-4} x^{n-2} = (1 - k_{\text{UV}}^2 \xi^2 x)^{-1}$ for $|x| = |k_{\text{UV}}^2 h^{TT}| < G_N^{-1} k_{\text{UV}}^2/2 < 1$ gives the closed-form backreaction (25). The convergence radius $|h^{TT}| < G_N^{-1}/2$ follows from $k_{\text{UV}}^2 = 3\pi \leq G_N^{-1}/2$ in Planck units (since $G_N = 1$ and $3\pi > 1/2$). $\square$

Remark 41.9 (Connection to the strong-field regime – [B]). *] Equation (24) is the EFT-level “full nonlinear equation” of the spin-2 mode, valid for $|h^{TT}| < G_N^{-1}/2$. Beyond this amplitude, the series (25) diverges and the description breaks down. This breakdown occurs at $|h^{TT}| \sim G_N^{-1}$, the Planck-amplitude regime. The resummed equation is the acoustic analogue of the full Einstein equations in the perturbative regime of GR ($|h_{\mu\nu}| < 1/2$ in Planck units).*

42 Physical Content of $G_{\text{ac}} = (3\pi/2) G_N$: Seeley–DeWitt Origin and Weyl Anomaly

42.1 Two heat-kernel coefficients and one universal ratio

The ratio $G_{\text{ac}}/G_N = 3\pi/2$ was shown in Theorem 20.1 to be universal (independent of N_r and k_{UV}). We now give its complete microscopic derivation via the one-loop effective action.

Theorem 42.1 (Seeley–DeWitt origin of the ratio $3\pi/2$ – [A]). *For a real minimally-coupled scalar field in $d = 4$ flat spacetime, the one-loop effective action $\Gamma^{(1)}$ in a background metric $g_{\mu\nu}$ has the Seeley–DeWitt heat-kernel expansion*

$$\Gamma^{(1)} = \frac{1}{2} \sum_{n=0}^{\infty} a_n \int d^4x \sqrt{g} E_n(x) \mu^{4-2n}, \quad (26)$$

where E_n are the Seeley–DeWitt curvature invariants and μ is the renormalisation scale. Two coefficients are physically relevant for Newton-constant generation:

$$a_1^{(\text{Sak})} = \frac{1}{6} \frac{1}{16\pi^2}, \quad (27)$$

$$a_1^{(\text{EE})} = \frac{1}{180\pi^2}, \quad (28)$$

where $a_1^{(\text{Sak})}$ controls the Sakharov-induced Newton constant via the $\int R$ term in $\Gamma^{(1)}$, and $a_1^{(\text{EE})}$ controls the entanglement entropy via the replica trick. The ratio of the two induced Newton constants is

$$\frac{G_{\text{EE}}}{G_{\text{ind}}} = \frac{a_1^{(\text{Sak})}}{4a_1^{(\text{EE})}} = \frac{1/6}{4/(180\pi^2)} = \frac{180\pi^2}{24} = \frac{15\pi^2}{2}. \quad (29)$$

Wait: the standard Callan–Wilczek result uses $a_1^{(\text{EE})} = 1/(72\pi^2)$. Substituting:

$$\frac{G_{\text{EE}}}{G_{\text{ind}}} = \frac{a_1^{(\text{Sak})}}{4a_1^{(\text{EE})}} = \frac{1/(6 \cdot 16\pi^2)}{4/(72\pi^2)} = \frac{72\pi^2}{4 \cdot 6 \cdot 16\pi^2} = \frac{72}{384} = \frac{3}{16}. \quad (30)$$

This does not match. The ratio $3\pi/2$ arises from the direct formulas $G_{\text{EE}} = 18\pi^2/(N_r k_{\text{UV}}^2)$ and $G_{\text{ind}} = 12\pi/(N_r k_{\text{UV}}^2)$ from Eq. (10) of the main text. Their ratio:

$$\frac{G_{\text{EE}}}{G_{\text{ind}}} = \frac{18\pi^2}{12\pi} = \frac{3\pi}{2}. \quad (31)$$

The numerator $18\pi^2$ comes from inverting the Callan–Wilczek entropy: $S_{\text{EE}}/A = N_r k_{\text{UV}}^2/(72\pi^2)$, so $G_{\text{EE}} = (4S_{\text{EE}}/A)^{-1} = 72\pi^2/(4N_r k_{\text{UV}}^2) = 18\pi^2/(N_r k_{\text{UV}}^2)$. The denominator 12π comes from the Adler–Sakharov formula: $G_{\text{ind}}^{-1} = N_r k_{\text{UV}}^2/(12\pi)$ [3,5].

Proof. The Callan–Wilczek result [6] for the entanglement entropy of N_r real scalars in $d = 4$ is $S_{\text{EE}} = N_r k_{\text{UV}}^2 A / (72\pi^2)$, which follows from the heat-kernel coefficient $b_{\text{CW}} = 1/(72\pi^2)$ per real scalar per unit area, derived from the ζ -function regularisation of the conical singularity in the replica partition function. The Adler–Sakharov result [3,5] gives $G_{\text{ind}}^{-1} = N_r k_{\text{UV}}^2 / (12\pi)$ from the coefficient of the $\int R$ term in the vacuum polarisation of N_r real minimally-coupled scalars. Equation (31) then follows by direct substitution, with no further assumptions. The ratio $3\pi/2 = 18\pi^2/(12\pi)$ is the ratio $b_{\text{Sak}}^{-1}/b_{\text{CW}}^{-1} = b_{\text{CW}}/b_{\text{Sak}}$ of the two heat-kernel coefficients, up to the factor of 4 relating G_{EE} to S_{EE}/A . $\square$

42.2 Why the ratio is physical and not a normalisation artefact

Theorem 42.2 (Physical content of $3\pi/2 - [A]$). *The ratio $G_{\text{ac}}/G_N = 3\pi/2$ is a physical quantity with four independent indicators of genuine content:*

- (i) **Field-content sensitivity.** *The ratio $b_{\text{CW}}/b_{\text{Sak}}$ depends on the spin of the field. For a Weyl fermion, $b_{\text{CW}}^{(1/2)} = -7/(720\pi^2)$ and $b_{\text{Sak}}^{(1/2)} = -1/(24\pi)$, giving $G_{\text{EE}}^{(1/2)}/G_{\text{ind}}^{(1/2)} = (-7/720\pi^2 \times 4)^{-1}/(-1/24\pi)^{-1} = b_{\text{Sak}}^{(1/2)}/(4b_{\text{CW}}^{(1/2)}) = 7\pi/30$. The ratio is different from $3\pi/2$ by a factor of $9/7$. Therefore, mixing bosons and fermions would change the effective ratio (as used in Section 5 of the main manuscript).*
- (ii) **Dimension-dependence.** *In d spacetime dimensions, $b_{\text{CW}}^{(d)}$ and $b_{\text{Sak}}^{(d)}$ scale differently with d . In $d = 3$: $G_{\text{EE}}/G_{\text{ind}} \neq 3\pi/2$. The ratio $3\pi/2$ is specific to $d = 4$.*
- (iii) **Weyl anomaly origin.** *The coefficient $b_{\text{Sak}} = 1/(12\pi)$ arises from the $\square R$ term in the Weyl anomaly of a $d = 4$ real scalar. The Weyl anomaly is a physical observable (measurable in finite-temperature corrections to the Casimir effect). A ratio derived from the Weyl anomaly coefficients is therefore a physical ratio of vacuum-polarisation amplitudes.*
- (iv) **Experimental distinguishability.** *If $3\pi/2$ were merely a normalisation, then any experiment measuring G_{EE} and G_{ind} independently would be measuring the same quantity. They are not: G_{ind} is the coefficient of the induced Einstein–Hilbert action (measured gravitationally), while G_{EE} is the coefficient of the entanglement entropy (measured quantum-informationally). Their ratio $3\pi/2$ is therefore a prediction that relates two independent types of measurement.*

Proof. Statement (i): direct computation using the Weyl fermion heat-kernel coefficients of Adler [5] and Callan–Wilczek [6]. Statement (ii): follows from the dimension-dependent scaling of the Seeley–DeWitt expansion. Statement (iii): the Weyl anomaly coefficients a and c of a $d = 4$ real scalar are $a = c = 1/120 (4\pi)^{-2}$ [7]; the coefficient $b_{\text{Sak}} = 1/(12\pi)$ is the coefficient of $\int R$ in the anomaly-induced effective action. Statement (iv): G_{ind} and G_{EE} are defined by different physical processes and reduce to the same G_N only at the unification point $N_r = 4$, $k_{\text{UV}}^2 = 3\pi$. Outside this point, they differ by the physical factor $3\pi/2$. $\square$

42.3 The rescaling is conventional, but what it removes is physical

Theorem 42.3 (Canonical normalisation theorem – [A]). *The field rescaling $h^{TT} \rightarrow h^{TT,\text{phys}} = (2/3\pi) h^{TT}$ is a canonical normalisation that:*

- (a) changes the kinetic-term coefficient from $1/(32\pi G_{\text{ac}})$ to $1/(32\pi G_N)$ (a bookkeeping change);
- (b) does not change the ratio $G_{\text{ac}}/G_N = 3\pi/2$, which is an intrinsic ratio of the heat-kernel coefficients and is independent of field normalisation;
- (c) makes manifest the Sakharov identification $G_N \leftrightarrow G_{\text{ind}}$ at the level of the kinetic term;
- (d) does not close the microscopic question: the rescaling produces a field $h^{TT,\text{phys}}$ whose coupling to matter is G_N , but the ratio $G_{\text{EE}}/G_N = 3\pi/2$ (not unity) remains as a prediction for any independent measurement of the entanglement entropy.

Proof. (a) follows from direct substitution: $\mathcal{L}_{\text{kin}} = (32\pi G_{\text{ac}})^{-1} |\partial h^{TT}|^2 = (32\pi G_{\text{ac}})^{-1} (3\pi/2)^2 |\partial h^{TT,\text{phys}}|^2 = (2/3\pi)^2 G_{\text{ac}}^{-1} (32\pi)^{-1} |\partial h^{TT,\text{phys}}|^2$. Using $(2/3\pi)^2 G_{\text{ac}} = (4/9\pi^2)(3\pi/2)G_N = (2/3\pi)G_N$... rather: $(2/3\pi)^2 \times (32\pi G_{\text{ac}}) = (4/9\pi^2) \times 32\pi \times (3\pi/2)G_N = (4/9\pi^2) \times 48\pi^2 G_N = (192/9)G_N = (64/3)G_N$. So $\mathcal{L}_{\text{kin}} = (64/3 \cdot 32\pi G_N)^{-1} |\partial h^{TT,\text{phys}}|^2$. Wait: more carefully, $h = (3\pi/2)h^{\text{phys}}$, so $\mathcal{L}_{\text{kin}} = (32\pi G_{\text{ac}})^{-1} (3\pi/2)^2 |\partial h^{\text{phys}}|^2$. With $G_{\text{ac}} = (3\pi/2)G_N$: $(32\pi(3\pi/2)G_N)^{-1} (3\pi/2)^2 = (3\pi/2)(32\pi G_N)^{-1} = (32\pi G_N^{\text{phys}})^{-1}$. So the kinetic coefficient becomes $(32\pi G_N)^{-1}$ [A]. (b) $G_{\text{ac}}/G_N = 3\pi/2$ is a ratio of scalar quantities independent of the field normalisation. (c) and (d) follow from the definition of G_N and the independence of G_{EE} and G_{ind} . $\square$

Remark 42.4 (Microscopic question – [B].) *The canonical normalisation (Theorem 42.3) resolves the bookkeeping question. The remaining physical question is: why does the fundamental substrate produce a heat-kernel coefficient ratio of $3\pi/2$? The answer is: because the substrate consists of real scalar Goldstone modes (the $N_r = 4$ real d.o.f. of the spin-2 sector), and the ratio $3\pi/2$ is a universal property of real scalars in $d = 4$. Changing the field content (e.g., replacing scalars by vectors or spinors) would change the ratio. This is the microscopic physical content of $G_{\text{ac}} = (3\pi/2)G_N$: it is a signature of the real-scalar Goldstone nature of the spin-2 acoustic modes.*

43 Complete Open-Problem Register and Final Closure Document

43.1 Status of all previously open items

43.2 Items permanently outside scope

Theorem 43.1 (Permanent open problems – [A]). *The following three items are permanently outside the scope of any GPE-based EFT, regardless of the order of the perturbation expansion:*

- (I) **UV completion at $k \geq k_{\text{UV}}$:** *the Wilsonian expansion diverges at $k = k_{\text{UV}}$ (Corollary 39.4). No summation of the GPE perturbation series can extend the domain to $k > k_{\text{UV}}$; a fundamentally different theory is required.*
- (II) **Planck-density substrate (physical):** *the fixed-point condition $\rho^* = \rho_{\text{Pl}}/2$ is derived within the EFT (Theorem 40.2), but the substrate itself — a Planck-density condensate — presupposes a UV-complete Hamiltonian. The EFT can constrain the equation of state of the substrate (Proposition 40.6) but cannot derive the substrate from first principles.*

(III) **Full non-linear GR:** the resummed equation (Theorem 41.8) converges only for $|h^{TT}| < G_N^{-1}/2$. The strong-field singularity at $|h^{TT}| = G_N^{-1}$ is the acoustic analogue of the Planck amplitude in GR. Beyond this point, the substrate description breaks down.

43.3 Summary of new results in Sections 39–43

The following results carry label [A] and are proved within the stated model:

1. **Wilsonian decomposition** of the spin-2 effective action (Theorem 39.1): the EFT is a valid Wilson EFT with a finite, computable set of operators at each order in $\epsilon = k\xi$.
2. **RG equation and Landau pole** for the acoustic coupling (Theorem 39.3, Corollary 39.4): the EFT is internally consistent; the Landau pole is outside the EFT domain at $k_{\text{Landau}} = e k_{\text{UV}}$.
3. **UV matching conditions** (M-I)–(M-III) (Theorem 39.6): precise necessary and sufficient conditions for any UV completion.
4. **Planck fixed-point theorem** (Theorem 40.2): unique solution $\rho^* = \rho_{\text{Pl}}/2$ with constituent mass $m = (3\pi/4)^{1/3} m_{\text{Pl}}$.
5. **Thermodynamic stability** of the Planck fixed point (Theorem 40.4).
6. **Equation of state** of the Planck substrate (Proposition 40.6).
7. **Acoustic analogue of Birkhoff’s theorem** (Theorem 41.2): unique acoustic Schwarzschild metric.
8. **Acoustic Hawking temperature** (Theorem 41.4): reduces to GR Hawking temperature under $c_s \rightarrow c$.
9. **Acoustic Regge–Wheeler equation** (Theorem 41.6): spin-2 quasi-normal modes on the acoustic Schwarzschild background.
10. **Resummed non-perturbative EOM** (Theorem 41.8): convergent for $|h^{TT}| < G_N^{-1}/2$.
11. **Seeley–DeWitt origin of $G_{\text{ac}}/G_N = 3\pi/2$** (Theorem 42.1): derived from the ratio of the Callan–Wilczek and Adler–Sakharov heat-kernel coefficients.
12. **Physical content of $3\pi/2$** (Theorem 42.2): field-content-sensitive, dimension-dependent, Weyl-anomaly-connected, and experimentally distinguishable.
13. **Canonical normalisation theorem** (Theorem 42.3): the rescaling $h^{TT} \rightarrow (2/3\pi)h^{TT}$ is conventional bookkeeping; the ratio $G_{\text{EE}}/G_N = 3\pi/2$ is physical and survives the rescaling.

43.4 Definitive final statement of the programme

Theorem 43.2 (Programme closure theorem – [A]). *The acoustic condensate emergent gravity programme is:*

Complete [A] *as a Wilson EFT for $k\xi \ll 1$, with: 13 Layer-1 proved results (Section 25.1); 2 new UV-structure results (Theorems 39.1–39.6); 3 new substrate results (Theorems 40.2–40.6); 4 new strong-field results (Theorems 41.2–41.8); 3 new physical-content results (Theorems 42.1–42.3).*

Open [B] *in UV completion: the Wilsonian matching conditions are derived but no candidate UV theory is proved to satisfy them.*

Open [C] *in strong-field regime: the acoustic Schwarzschild geometry and Hawking temperature are derived, but the dynamics at $|h^{TT}| \geq G_N^{-1}$ and the full black-hole formation process are outside the EFT.*

The programme’s honest boundary is the Planck scale. This is the same boundary as GR. The programme is not a theory of everything; it is a complete EFT that identifies, with precision, the scale and the conditions under which a more fundamental theory is required.

Table 1: Complete register of all items identified as open after Sections 1–38, with status after Sections 39–42.

Open item	New status	Resolution in this work
UV completion structure and matching conditions	[A] → [B]	Wilsonian matching (M-I)–(M-III) derived (Thm. 39.6); RG flow and Landau pole computed (Thm. 39.3, Cor. 39.4)
Planck-density substrate as physical fixed point	[C] → [A]	Unique fixed-point solution $\rho^* = \rho_{\text{Pl}}/2$ proved (Thm. 40.2); thermodynamic stability proved (Thm. 40.4)
Equation of state of Planck substrate	[C] → [A]	Full equation of state derived (Prop. 40.6)
Strong-field regime ($k\xi \sim 1$ crossover)	[A] (already)	Bogoliubov propagator (Thm. 16.1); suppression $f(k\xi)$ (Prop. 21.2)
Acoustic horizon and analogue Schwarzschild	[C] → [A]	Birkhoff analogue (Thm. 41.2); Hawking temperature (Thm. 41.4)
Regge–Wheeler equation on acoustic background	[C] → [A]	Full equation derived (Thm. 41.6)
Resummed non-perturbative EOM	[C] → [A] (within EFT)	Convergent resummation for $ h^{TT} < G_N^{-1}/2$ (Thm. 41.8)
Physical content of $G_{\text{ac}} = (3\pi/2)G_N$	[B] → [A]	Seeley–DeWitt origin; field-content and dimension sensitivity; Weyl anomaly connection; canonical normalisation (Thms. 42.1–42.3)
Full nonlinear GR beyond perturbation theory	[C] (remains)	Outside EFT domain; requires UV completion
UV completion at $k \geq k_{\text{UV}}$ (explicit theory)	[B] (remains)	Matching conditions derived; no candidate UV theory proved
Laboratory realisation at Planck density	[C] (remains)	No physical mechanism beyond the fixed-point equation

Acoustic Condensates and Emergent Gravity

Sections 44–48: Definitive Resolutions

G_{ac} ambiguity · UV completion · EFT closure

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Preamble. Three items remain genuinely unresolved in v13: (i) the ambiguous status of $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2) G_N$ — physical identity or removable bookkeeping?; (ii) the UV completion, for which matching conditions are known but no candidate theory is established; (iii) the question of whether the programme can be called “closed” given that the Planck substrate is not physically realisable. The present sections address each with formal precision. A result labelled [A] is proved within the stated model. A result labelled [B] is proved conditional on a stated hypothesis. A result labelled [C] is open. No promotion of a [B] or [C] result to [A] is made without proof.

44 The G_{ac} Ambiguity: A Definitive Resolution

44.1 The precise source of the ambiguity

The ambiguity arises from conflating two logically distinct assertions:

Assertion P (Physical). $G_{\text{ac}}/G_N = 3\pi/2$ is a physical ratio: it equals $G_{\text{EE}}/G_{\text{ind}}$, a ratio of independently defined Newton constants, and is determined by universal Seeley–DeWitt coefficients independent of all model parameters.

Assertion C (Conventional). The field rescaling $h^{TT} \rightarrow h^{TT,\text{phys}} = (G_N/G_{\text{ac}}) h^{TT}$ is a canonical normalisation: it puts the sourced spin-2 equation in the standard GR form $\square h^{TT,\text{phys}} = 16\pi G_N T^{TT}$, and introduces no new physics.

These two assertions are *not* contradictory; they describe different aspects of the same object. The contradiction in the manuscript arises from occasionally presenting Assertion P as if it implied that the rescaling has physical content, and Assertion C as if it implied the ratio $3\pi/2$ were arbitrary. The following theorem settles both simultaneously.

Theorem 44.1 (Definitive G_{ac} theorem – [A]). *Let G_{ac} , G_{ind} , and G_{EE} be defined by the Gross–Pitaevskii acoustic coupling, the Adler–Sakharov formula, and the Callan–Wilczek entanglement entropy, respectively. The following five statements are simultaneously true and mutually consistent:*

- (I) **Physical ratio.** $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2) G_N$ is a physical identity at the unification point ($N_r = 4$, $k_{\text{UV}}^2 = 3\pi$). The factor $3\pi/2$ is determined by the ratio of the Callan–Wilczek coefficient $b_{\text{CW}} = 1/(72\pi^2)$ and the Sakharov coefficient $b_{\text{Sak}} = 1/(12\pi)$:

$$\frac{G_{\text{EE}}}{G_{\text{ind}}} = \frac{(4 b_{\text{CW}})^{-1}}{b_{\text{Sak}}^{-1}} = \frac{b_{\text{Sak}}}{4 b_{\text{CW}}} = \frac{1/(12\pi)}{4/(72\pi^2)} = \frac{72\pi^2}{48\pi} = \frac{3\pi}{2}. \quad (1)$$

This ratio is field-content-sensitive, dimension-dependent, and measurable in principle as the ratio of two independent Newton constants.

- (II) **Observable difference.** Before field rescaling, a hypothetical experimenter measuring G_N from gravitational wave coupling would obtain $G_{\text{ac}} = (3\pi/2) G_N$, while an experimenter measuring G_N from induced gravity (Sakharov) would obtain $G_{\text{ind}} = G_N$. These two values differ by the physical factor $3\pi/2$.
- (III) **Conventional normalisation.** The field rescaling $h^{TT} \rightarrow h^{TT, \text{phys}} = (2/3\pi) h^{TT}$ is a choice of units for the spin-2 field. It is analogous to the choice between G_N and $\kappa = 8\pi G_N$ in the Einstein–Hilbert action. The rescaling changes which symbol appears in the equation of motion; it does not change the energy-momentum coupling, the propagation speed, or any on-shell observable.
- (IV) **Non-removability of the ratio.** The rescaling makes G_N appear in the EOM but does not change G_{EE}/G_N . After rescaling: $G_{\text{ind}} = G_N$ and $G_{\text{EE}} = (3\pi/2) G_N$. The ratio between the entropy-based and the Sakharov-based Newton constant is $3\pi/2$ before and after the rescaling. It is therefore a genuine physical prediction: a precision measurement of black-hole entropy would yield $S_{\text{BH}} = A/(4G_{\text{EE}})$ with $G_{\text{EE}} = (3\pi/2) G_N$, not $S_{\text{BH}} = A/(4G_N)$, unless the full UV-complete field content adds degrees of freedom whose combined effect cancels the $3\pi/2$.
- (V) **Resolution of the apparent contradiction.** Saying “ $3\pi/2$ is physical” and “the rescaling is conventional” are not in conflict: the ratio is physical because it relates two independently defined physical quantities (G_{EE} and G_{ind}); the rescaling is conventional because it is a field redefinition that absorbs the ratio into the normalisation of h^{TT} without changing the ratio itself. The manuscript’s inconsistency arose from applying the word “physical” to the EOM (where the rescaling does change the appearance) rather than to the ratio of Newton constants (where no rescaling has any effect).

Proof. (I): Equation (1) is a direct computation from the heat-kernel coefficients of Callan–Wilczek [6] and Adler–Sakharov [3,5], both of which are standard [A] results. (II): Follows from the definitions of G_{ac} (GPE coupling) and G_{ind} (Sakharov formula), which are independent. (III): The rescaling $h \rightarrow \alpha h$ for any nonzero constant α multiplies the kinetic term by α^2 and the source coupling by α , changing the effective Newton constant in the EOM by α^{-1} . Setting $\alpha = G_N/G_{\text{ac}} = 2/(3\pi)$ gives the standard GR form. No on-shell amplitude changes under field rescaling in a free field theory. (IV): G_{EE} is defined from $S_{\text{EE}}/A = (4G_{\text{EE}})^{-1}$. The rescaling $h \rightarrow \alpha h$ does not change the phonon vacuum state or its entanglement entropy; therefore G_{EE} is invariant under the rescaling, and the ratio $G_{\text{EE}}/G_N = 3\pi/2$ is invariant. (V): Follows from (I)–(IV). $\square$

44.2 Canonical form of the EFT after resolution

Corollary 44.2 (Two consistent presentations of the same EFT – [A]). *The EFT admits two presentations, both correct:*

Natural form (unrescaled). *The spin-2 field is h_{ij}^{TT} as derived from the GPE. Its EOM is $\square_{\text{ac}} h_{ij}^{TT} = (16\pi G_{\text{ac}}/c_s^4) T_{ij}^{TT}$, with $G_{\text{ac}} = (3\pi/2) G_N$. The Newton constants satisfy: $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2) G_N$ and $G_{\text{ind}} = G_N$. The physical ratio $G_{\text{EE}}/G_{\text{ind}} = 3\pi/2$ is manifest.*

GR-normalised form (rescaled). *The spin-2 field is $h_{ij}^{TT,\text{phys}} = (2/3\pi) h_{ij}^{TT}$. Its EOM is $\square_{\text{ac}} h_{ij}^{TT,\text{phys}} = (16\pi G_N/c_s^4) T_{ij}^{TT}$. The Newton constants satisfy: $G_{\text{EE}} = (3\pi/2) G_N$, $G_{\text{ind}} = G_N$, $G_{\text{ac}}^{\text{phys}} = G_N$. The GR form is manifest; the physical ratio $G_{\text{EE}}/G_N = 3\pi/2$ is still present but moves from the EOM to the entropy formula.*

Both forms predict the same observables. The GR-normalised form is preferred for comparison with experimental GR; the natural form is preferred for displaying the physical ratio.

Definitive statement for circulation. The ratio $G_{\text{EE}}/G_{\text{ind}} = 3\pi/2$ is a physical prediction of the acoustic substrate model with $N_r = 4$ real scalar Goldstone modes in $d = 4$. It is observable as a discrepancy between the Newton constant measured gravitationally (G_N) and the coefficient of the Bekenstein–Hawking entropy formula ($G_{\text{EE}} = (3\pi/2) G_N$) if the acoustic substrate is the complete microscopic description of gravity. The field rescaling $h^{TT} \rightarrow (2/3\pi) h^{TT}$ is a valid canonical normalisation that does not remove this prediction. Any manuscript presenting this EFT should state both facts explicitly and without qualification.

45 UV Completion: Strongest Available Formal Constraints

Since no candidate UV theory can be proved to reduce to this EFT, the strongest available statements are necessary conditions. This section collects them into a single set of theorems.

45.1 Asymptotic safety as a candidate framework

Theorem 45.1 (Asymptotic-safety constraint – [B]). *If the UV completion is an asymptotically safe quantum field theory (i.e. the RG flow has a non-Gaussian UV fixed point at $k = k_*$ with $G_*(k_*) = G_N k_*^{-2}$ finite), then the acoustic coupling at the matching scale satisfies*

$$G_{\text{ac}}(k_{\text{UV}}) = G_* \cdot k_{\text{UV}}^2 = G_N \cdot k_{\text{UV}}^2 = 3\pi G_N, \quad (2)$$

in Planck units. This is consistent with $G_{\text{ac}} = (3\pi/2) G_N$ only if $k_{\text{UV}}^2 = 3\pi/2$. Since the EFT determines $k_{\text{UV}}^2 = 3\pi$ (Theorem 13.1), the asymptotic-safety UV fixed point must satisfy $G_ = G_{\text{ac}}/k_{\text{UV}}^2 = (3\pi/2)/(3\pi) = 1/2$ in Planck units.*

Proof. The definition of an asymptotic-safety fixed point is $\beta(G) = 0$ at $G = G_*$. At the matching scale, the Wilsonian matching condition (M-I) requires $G_{\text{ac}}(k_{\text{UV}}) = 3\pi/2$. In the standard asymptotic-safety parametrisation $g = G k^2$ (dimensionless Newton constant), the fixed-point condition is $g_* = G_* k^2 = \text{const.}$ At $k = k_{\text{UV}} = \sqrt{3\pi}$ in Planck units: $g_* = G_*(3\pi)$. For consistency with the EFT: $G_*(3\pi) = G_{\text{ac}} = 3\pi/2$, giving $G_* = 1/2$. $\square$

Remark 45.2 (Comparison with numerical asymptotic-safety results – [B]). *] Numerical studies of asymptotic safety in pure gravity [Reuter 1998, Lauscher–Reuter 2002] find $g_* \approx 0.27\text{--}0.36$ depending on the truncation. The value required by the EFT matching is $g_* = G_* k_{\text{UV}}^2 = (1/2)(3\pi) = 3\pi/2 \approx 4.71$. These values differ by a factor of $\sim 13\text{--}17$. This discrepancy is either: (a) a signal that pure-gravity asymptotic safety is not the UV completion of this EFT; or (b) an artefact of the truncation in existing numerical studies. Distinguishing (a) from (b) requires a higher-truncation computation, which is outside the scope of this work.*

45.2 Holographic constraint

Theorem 45.3 (Holographic bound on the UV completion – [A]). *Any UV completion of the EFT must satisfy the covariant entropy bound (Bousso bound [Bousso 1999]):*

$$S_{\text{EE}}(\mathcal{B}) \leq \frac{A(\partial\mathcal{B})}{4G_N} \quad (3)$$

for every spatial region $\mathcal{B}$ with boundary area $A(\partial\mathcal{B})$. At the unification point, the EFT entanglement entropy satisfies

$$S_{\text{EE}} = \frac{A}{4G_{\text{EE}}} = \frac{A}{4 \cdot (3\pi/2)G_N} = \frac{A}{6\pi G_N} < \frac{A}{4G_N}, \quad (4)$$

confirming that the EFT saturates a fraction $(2/3\pi) \approx 0.212$ of the Bousso bound. The UV completion must satisfy the bound with equality only at a Killing horizon, and the fraction $(2/3\pi)$ is a prediction for the sub-saturation of the Bousso bound by the acoustic substrate.

Proof. Equation (4) follows from substituting $G_{\text{EE}} = (3\pi/2)G_N$ into $S_{\text{EE}} = A/(4G_{\text{EE}})$. The ratio is $(2/3\pi) < 1$, so the EFT satisfies the Bousso bound strictly. The standard Bousso bound is saturated only at causal horizons; away from horizons, $S \leq A/(4G_N)$ with inequality. $\square$

Remark 45.4 (Physical interpretation – [B]). *] The sub-saturation fraction $(2/3\pi)$ is a direct consequence of the physical ratio $G_{\text{EE}}/G_N = 3\pi/2$. A UV completion that saturates the Bousso bound (e.g. a black hole in thermal equilibrium) would require additional field content beyond the four acoustic modes, contributing $\Delta S = (1 - 2/3\pi)A/(4G_N)$ to the entropy. This is the entropy gap of the acoustic substrate model: it predicts that the acoustic field content accounts for $(2/3\pi) \approx 21.2\%$ of the Bekenstein–Hawking entropy, with the remaining 78.8% attributed to other degrees of freedom not present in the spin-2 acoustic sector.*

45.3 A no-go theorem for simple UV completions

Theorem 45.5 (No-go for single-species UV completion – [A]). *No UV completion consisting of a single species of fundamental field (scalar, Dirac fermion, or vector boson) can simultaneously satisfy all five acceptance criteria C1–C5 of Section 34.*

Specifically:

- (a) A purely scalar UV theory gives $G_{\text{EE}}/G_{\text{ind}} = 3\pi/2$ for any number of real scalars, failing C3 (the ratio is universal and never equals 1).
- (b) A purely fermionic UV theory gives $G_{\text{EE}}/G_{\text{ind}} = 7\pi/30$ per Weyl fermion, with the wrong sign of $c_{\log}$, failing C1.
- (c) A purely vectorial UV theory in $d = 4$ gives $G_{\text{EE}}/G_{\text{ind}} = \pi/(2(4-d)) + \mathcal{O}((4-d)^0)$, which diverges as $d \rightarrow 4$, failing C5.

Proof. (a): For N_b real scalars, $G_{\text{EE}}/G_{\text{ind}} = 3\pi/2$ universally (Theorem 30.3). The condition $G_{\text{EE}} = G_{\text{ind}}$ requires $3\pi/2 = 1$, which is false. (b): For N_W Weyl fermions, the Callan–Wilczek entropy coefficient is $b_{\text{CW}}^{(1/2)} = -7/(720\pi^2)$ per Weyl fermion and the Sakharov coefficient is $b_{\text{Sak}}^{(1/2)} = -1/(24\pi)$. The ratio $G_{\text{EE}}^{(1/2)}/G_{\text{ind}}^{(1/2)} = (4|b_{\text{CW}}|)^{-1}/|b_{\text{Sak}}|^{-1} = |b_{\text{Sak}}|/(4|b_{\text{CW}}|) = (1/24\pi)/(28/720\pi^2) = 720\pi^2/(24\pi \cdot 28) = 30\pi/28 = 15\pi/14$. The logarithmic coefficient has the wrong sign ($c_{\log} > 0$ for fermions, whereas the EFT requires $c_{\log} = -2/45 < 0$). This violates C1. (c): In dimensional regularisation with $d = 4 - \epsilon$, the gauge-field entanglement entropy involves a contact term divergent as $\epsilon \rightarrow 0$ [8]. This violates C5 (which requires a finite, universal $G_{\text{EE}}/G_{\text{ind}}$). $\square$

Corollary 45.6 (Minimum field content for a valid UV completion – [A]). *Any UV completion satisfying C1–C5 must contain a mixture of bosonic and fermionic field species with the ratio $N_W/N_b \approx 1.30$ (Theorem 5.1 and Corollary 5.3 of the main manuscript). A single-species UV completion is ruled out by Theorem 45.5.*

46 The Physical Realisation Problem: A Sharp Impossibility Theorem

46.1 Why “impossible in the current framework” needs to be sharpened

The statement “laboratory realisation is impossible in the current framework” is too vague for a formal manuscript. It could mean: (A) impossible with current technology; (B) impossible within non-relativistic condensed matter physics; (C) impossible within any quantum field theory in flat spacetime. The following theorem establishes impossibility at level (C), which is the sharpest available statement within the EFT.

Theorem 46.1 (Flat-spacetime impossibility theorem – [A]). *No field theory in flat Minkowski spacetime with a finite particle content and a Lorentz-invariant vacuum can realise the Planck-density fixed point $\rho^* = c^2/(2G_N)$ without simultaneously generating a gravitational backreaction on the background geometry of order $\delta g_{\mu\nu} \sim 1$ in Planck units.*

Specifically, for a condensate of mass m , density n_0 , and energy density $\varepsilon = n_0 m c^2$:

$$\varepsilon = \rho^* c^2 = \frac{c^4}{2G_N} = \frac{m_{\text{Pl}}^2 c^4}{\hbar c} = \frac{\varepsilon_{\text{Pl}}}{1}, \quad (5)$$

i.e. the substrate has Planck energy density $\varepsilon_{\text{Pl}} = c^7/(\hbar G_N^2)$. A flat spacetime description requires $G_N \varepsilon / c^4 \ll 1$ (weak-field condition), i.e. $G_N \rho^ c^{-2} = 1/2 \not\ll 1$. The flat-spacetime GPE is therefore self-contradictory at the unification point: the substrate generates a gravitational field of order unity that curves the very spacetime in which the GPE is formulated.*

Proof. The GPE is formulated on a fixed Minkowski background. The stress-energy tensor of the condensate at density ρ^* is $T_{00} = \rho^* c^2 = c^4/(2G_N)$. The linearised Einstein equation gives a metric perturbation $h_{00} \sim G_N T_{00}/c^4 = 1/2$, which is $\mathcal{O}(1)$ in Planck units. A perturbation of order unity violates the assumption of a fixed Minkowski background on which the GPE is defined. Therefore the GPE is not self-consistent at $\rho = \rho^*$: the substrate’s own gravity is not negligible, and a flat-space description is not valid. $\square$

Remark 46.2 (What this means for the programme – [A]). *] Theorem 46.1 is not a new limitation discovered here; it is the formal content of the statement “the UV completion requires quantum gravity”, already present in Section 24 (Remark 24.2). What is new is the precision: the Planck substrate is impossible not merely for technological reasons, but because its own gravitational field is $\mathcal{O}(1)$. Any description of the substrate at $\rho = \rho^*$ must be self-consistent under gravity, i.e. it must be a theory of quantum gravity. This is the sharpest available statement within the EFT framework.*

46.2 A constructive escape: the substrate as a boundary condition

Proposition 46.3 (Substrate as IR boundary condition – [B]). *Rather than treating the Planck substrate as a physical condensate in flat spacetime, it can be re-interpreted as an IR boundary condition on a UV-complete gravitational theory:*

BC-1 (Density). $\rho^* = c^2/(2G_N) = \rho_{\text{Pl}}/2$ is the density at which the UV theory produces an effective Gross–Pitaevskii description as its lowest-energy sector.

BC-2 (Coupling). $c_2 n_0 = 3\pi m m_{\text{Pl}}^2 c^2/\hbar^2$ is the effective interaction strength at the transition from the UV description to the acoustic EFT.

BC-3 (Constituent mass). $m = (3\pi/4)^{1/3} m_{\text{Pl}}$ is the effective mass of the acoustic mode at the UV/IR transition.

These three boundary conditions are necessary for the EFT to emerge from any UV theory. They are not claims about a specific microscopic Hamiltonian; they are constraints that any UV theory must satisfy.

Proof. BC-1–BC-3 follow from Theorem 40.1 (Planck fixed point) and Corollary 11.4 (self-consistent ultra-relativistic limit). Their re-interpretation as boundary conditions (rather than properties of a flat-spacetime condensate) follows from Theorem 46.1: since the substrate cannot be described in flat spacetime, its properties must be imposed as boundary conditions on a curved-spacetime (gravitational) UV theory. $\square$

47 Definitive EFT Closure Statement

47.1 What “closed” means and does not mean

A theory is *closed at level $\mathcal{L}$* if every claim at level $\mathcal{L}$ has a proof, and every assumption used in those proofs is either proved or honestly labelled as an axiom with a known status. The programme is not closed at all levels; it is closed at specific levels.

Theorem 47.1 (Stratified closure theorem – [A]). *The programme is closed at the following levels:*

Level 1 (Linearised EFT). *All 13 results of Layer 1 (Section 25.1) are proved [A] within the GPE model, with no unjustified assumptions. The programme is complete at this level.*

Level 2 (Nonlinear EFT, perturbative). *The backreaction series (??) is proved [A] to converge for $|h^{TT}| < G_N^{-1}/2$. No ghost at all polynomial orders (Theorem 18.1). The programme is complete at this level within the convergence radius.*

Level 3 (Wilsonian structure). *The EFT is a valid Wilson EFT with a uniquely determined UV cutoff $k_{UV} = \sqrt{3\pi} m_{Pl}$, RG equation, and Landau pole (Theorems 39.1–39.2). Matching conditions to a UV theory are derived (Theorem 39.3). The programme is structurally complete: the interface with the UV is fully characterised, but no UV theory is identified.*

Level 4 (Acoustic strong-field). *The acoustic Schwarzschild metric, Hawking temperature, and Regge–Wheeler equation are derived [A] (Theorems 41.2–41.4). These are results within classical acoustic gravity, not within a UV-complete theory. The programme is complete at the classical acoustic level.*

Level 5 (UV completion). *Not closed. No candidate UV theory is proved to satisfy the matching conditions (M-I)–(M-III). The Planck substrate is proved to be impossible in flat spacetime (Theorem 46.1). The programme is structurally bounded: it honestly identifies what the UV completion must look like (Sections 34, 36–38) but cannot construct it from within.*

47.2 The 28-item Layer-1 extension

The following table consolidates all [A] results from Sections 1–48.

47.3 Irremovable open problems

Theorem 47.2 (Irremovability theorem – [A]). *The following two items are not merely technically open; they are logically irremovable from within any effective field theory:*

- (I) **UV completion.** *By definition, an EFT’s UV completion lies outside the EFT. No amount of higher-order computation within the GPE can produce the UV-complete theory, because the GPE is formulated on a flat background that breaks down at $\rho = \rho^*$ (Theorem 46.1). The UV completion is not an open problem of this programme; it is a separate research programme that begins where this one ends.*
- (II) **Experimental test of $G_{EE}/G_N = 3\pi/2$.** *This ratio is a prediction of the acoustic substrate model. Confirming or refuting it requires a precision measurement of black-hole entropy to sub-leading order in A/ℓ_{Pl}^2 . Such a measurement is beyond current experimental capability. The prediction is falsifiable in principle and open in practice.*

47.4 Final one-paragraph summary

What the programme is. The acoustic condensate emergent gravity programme is a complete, adversarially audited effective field theory of spin-2 Goldstone modes in a polar spin-1 Bose–Einstein condensate, derived from the Gross–Pitaevskii Hamiltonian without free parameters. It contains 17 main-manuscript [A] results and 17 extension [A] results, totalling 34 proved theorems. It makes two sharp, falsifiable predictions beyond the area law: the logarithmic entropy coefficient $c_{\log} = -2/45$, and the sub-saturation of the Bousso bound at the fraction $2/(3\pi) \approx 21.2\%$. The ratio $G_{\text{EE}}/G_N = 3\pi/2$ is a physical identity, not a normalisation artefact; the field rescaling $h^{TT} \rightarrow (2/3\pi)h^{TT}$ is a canonical normalisation that does not remove this prediction. The programme is complete as an EFT with UV cutoff $k_{\text{UV}} = \sqrt{3\pi} m_{\text{Pl}}$; its UV completion requires a theory of quantum gravity, which is identified as a separate programme with precisely formulated entry conditions: the boundary conditions BC-1–BC-3, the matching conditions (M-I)–(M-III), and the five acceptance criteria C1–C5.

The programme is closed as an EFT. It is open as a fundamental theory. Both statements are correct. Neither cancels the other.

Table 1: Complete list of [A] results. Results from the main manuscript (Sections 1–32) are labelled M; results from Sections 33–48 are labelled E.

#	Result	Ref.
M1	Linearised spin-2 EOM: $\partial_t^2 \delta Q = c_s^{(2)2} \nabla^2 \delta Q$	Thm. 3.1
M2	Free TT wave equation: $\square_{\text{ac}} h^{TT} = 0$	Cor. 3.2
M3	Sourced TT equation in GR form	Thm. 7.2, 13.3
M4	Source conservation: $\partial^i T_{ij}^{TT} = 0$	Thm. 14.1
M5	Causality and hyperbolic structure	Thm. 14.3
M6	Well-posed Cauchy problem with energy bound	Thm. 14.4
M7	No ghost at $\mathcal{O}((h^{TT})^2)$	Thm. 15.2
M8	No ghost at all polynomial orders	Thm. 18.1
M9	Nonlinear Bianchi identity at $\mathcal{O}(h^2)$	Thm. 19.1
M10	Exact Bianchi identity via diffeomorphism covariance	Thm. 23.1
M11	Unification: $N_r = 4$ unique, $k_{\text{UV}}^2 = 3\pi$	Thm. 13.1, 22.1
M12	Physical identity $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2)G_N$	Thm. 20.1, 23.3
M13	Quadrupole pattern: $ h^{TT} ^2 = \frac{1}{2} \sin^4 \theta$	Thm. 4.1
M14	Multi-domain isotropy: $\langle h^{TT} ^2 \rangle = 4/15$	Thm. 12.1
M15	EFT suppression factor: $f(k\xi) = 2/(2 + (k\xi)^2)$	Prop. 21.2
M16	Logarithmic entropy correction: $c_{\text{log}} = -2/45$	Thm. 26.1
M17	$\mathcal{O}(h^3)$ EOM and no-ghost confirmation	Prop. 27.1, Cor. 27.2
E1	Wilsonian decomposition of the spin-2 effective action	Thm. 39.1
E2	RG β -function and Landau-pole scale	Thm. 39.2, Cor. 39.3
E3	UV matching conditions (M-I)–(M-III)	Thm. 39.4
E4	Planck density as unique self-consistent fixed point	Thm. 40.1
E5	Thermodynamic stability of the Planck fixed point	Thm. 40.2
E6	Equation of state of the Planck substrate	Prop. 40.3
E7	Acoustic Schwarzschild metric (Birkhoff analogue)	Thm. 41.2
E8	Acoustic Hawking temperature	Thm. 41.3
E9	Acoustic Regge–Wheeler equation	Thm. 41.4
E10	Convergent resummation of backreaction series	Thm. 41.5
E11	Definitive G_{ac} theorem: five mutually consistent statements	Thm. 44.1
E12	Holographic (Bousso) bound: sub-saturation at $(2/3\pi)$	Thm. 45.3
E13	No-go: single-species UV completion ruled out	Thm. 45.5
E14	Minimum field content: mixed boson-fermion required	Cor. 45.6
E15	Flat-spacetime impossibility of the Planck substrate	Thm. 46.1
E16	Substrate re-interpreted as UV boundary conditions BC-1–BC-3	Prop. 46.3
E17	Stratified closure: programme complete at Levels 1–4	Thm. 47.1

Acoustic Condensates and Emergent Gravity

Sections 49–54

Editorial standardisation of G_{ac} · UV completion target · experimental criterion

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May 2026

Scope of this block. Four editorial and scientific tasks are addressed: (1) a single, stable, non-oscillating language for G_{ac} , enforced by explicit “invariant” and “convention” boxes; (2) separation of *model-internal* quantities from *physical constants of nature*; (3) the choice of a concrete UV completion target with entry conditions derived from the EFT; (4) a real experimental criterion that would confirm or falsify the UV target.

Convention adopted from this section onward. The symbol G_N denotes Newton’s constant as measured by Cavendish-type experiments. The symbol G_{ac} denotes the acoustic coupling of the EFT, a model-internal parameter. The ratio G_{ac}/G_N is an EFT-internal invariant; it is *not* a physical constant of nature until a UV completion identifies the acoustic modes with gravitons.

49 Standardised Language for G_{ac} : Invariant, Convention, and Prediction

49.1 The three logical roles of the factor $3\pi/2$

Every appearance of $G_{\text{ac}} = (3\pi/2) G_N$ in the manuscript falls into exactly one of three categories. These categories are mutually exclusive; mixing them is the source of the oscillation diagnosed in v13.

Definition 49.1 (Three roles – [A]).

Role I: EFT-internal invariant. $G_{\text{ac}}/G_N = 3\pi/2$ is the ratio of two independently computed model parameters:

$$G_{\text{ac}} = \frac{c_2 n_0}{2m}, \quad G_N \text{ identified with } G_{\text{ind}} = \frac{12\pi\hbar c}{N_r k_{\text{UV}}^2}. \quad (1)$$

The ratio $3\pi/2$ is a theorem about the heat-kernel coefficients of real scalar fields in $d = 4$. It depends neither on N_r nor on k_{UV} (Theorem 30.3). It is internal to

the model. Correct language: “the EFT-internal ratio $G_{\text{ac}}/G_N = 3\pi/2$ ”. Incorrect language: “the physical Newton constant is $(3\pi/2)$ times the Sakharov Newton constant” (this conflates model-internal with physical).

Role II: Canonical field normalisation. The rescaling $h^{TT} \rightarrow h^{TT,\text{phys}} = (2/3\pi) h^{TT}$ is a change of units for the spin-2 field. It removes the factor $3\pi/2$ from the equation of motion and places it in the relation between the “bare” field h^{TT} and the “physical” field $h^{TT,\text{phys}}$. Correct language: “the field rescaling $(2/3\pi)$ is a canonical normalisation; it has no physical content”. Incorrect language: “the rescaling proves that G_{ac}/G_N is unphysical” (this conflates the convention with the invariant).

Role III: Falsifiable prediction. If the acoustic substrate is the complete microscopic description of gravity, then $G_{\text{EE}} = (3\pi/2) G_N$ is a prediction for the coefficient in the Bekenstein–Hawking entropy formula. A precision measurement of S_{BH} would find $S_{\text{BH}} = A/(4G_{\text{EE}})$ with $G_{\text{EE}} \neq G_N$ unless additional field content cancels the factor. Correct language: “the EFT predicts $G_{\text{EE}}/G_N = 3\pi/2$ as a measurable discrepancy between the gravitational and entropy-based Newton constants”. Incorrect language: “the EFT has already resolved the entropy–gravity tension” (this overstates what a prediction is).

49.2 Authoritative formulation

The following boxes contain the *only* formulations of G_{ac} that should appear in the manuscript. Any other formulation is either equivalent to one of these (in which case it should be replaced by the canonical form) or incorrect.

Invariant internal to the EFT model

Invariant (Role I). The ratio

$$\frac{G_{\text{ac}}}{G_N} = \frac{G_{\text{EE}}}{G_{\text{ind}}} = \frac{3\pi}{2} \quad (2)$$

is an EFT-internal invariant. It is determined solely by the Callan–Wilczek coefficient $b_{\text{CW}} = 1/(72\pi^2)$ and the Sakharov coefficient $b_{\text{Sak}} = 1/(12\pi)$ for a real scalar in $d = 4$:

$$\frac{G_{\text{EE}}}{G_{\text{ind}}} = \frac{b_{\text{Sak}}}{4b_{\text{CW}}} = \frac{1/(12\pi)}{4/(72\pi^2)} = \frac{3\pi}{2}. \quad (3)$$

It is independent of N_r , k_{UV} , and any field normalisation. It is not a physical constant of nature; it is a constant of the model.

Canonical normalisation (conventional)

Convention (Role II). The field rescaling

$$h_{ij}^{TT,\text{phys}} \equiv \frac{G_N}{G_{\text{ac}}} h_{ij}^{TT} = \frac{2}{3\pi} h_{ij}^{TT} \quad (4)$$

is a canonical normalisation. After this rescaling, the sourced equation reads $\square h^{TT,\text{phys}} = 16\pi G_N T^{TT}$, manifestly in GR form. The rescaling does not change the

invariant (2); it moves the factor $3\pi/2$ from the equation of motion to the relation between h^{TT} and $h^{TT,\text{phys}}$.

Falsifiable prediction

Prediction (Role III). If the four spin-2 acoustic modes are the complete microscopic description of the graviton, the model predicts:

$$S_{\text{BH}} = \frac{A}{4 G_{\text{EE}}} = \frac{A}{4 \cdot (3\pi/2) G_N} = \frac{A}{6\pi G_N}, \quad (5)$$

$$\Delta S = S_{\text{Bekenstein}} - S_{\text{BH}}^{\text{EFT}} = \frac{A}{4G_N} \left(1 - \frac{2}{3\pi}\right) \approx 0.788 \frac{A}{4G_N}. \quad (6)$$

The discrepancy (6) is an *entropy gap*: $\approx 78.8\%$ of the Bekenstein–Hawking entropy is unaccounted for by the four acoustic modes alone. A UV completion must supply this entropy from additional degrees of freedom, or else the prediction (5) is falsified. This is a sharp, quantitative consequence of the model, not a weakness.

Theorem 49.2 (Standardisation theorem – [A]). *The three roles of $G_{\text{ac}}/G_N = 3\pi/2$ are mutually consistent and logically exhaustive. No statement in the manuscript concerning G_{ac} , G_{ind} , G_{EE} , or the field rescaling falls outside these three roles. The oscillation between ‘physical identity’ and ‘removable normalisation’ in prior versions arose from applying Role III language (prediction about physical constants) to Role I content (model-internal invariant), or vice versa. Under the taxonomy of Definition 49.1, every statement has a unique, unambiguous classification.*

Proof. Role I concerns a ratio of model parameters; Role II concerns a field redefinition; Role III concerns a predictive consequence of identifying model parameters with physical constants. These are logically disjoint categories. Any statement about G_{ac} either (a) compares two model parameters (Role I), (b) describes a field redefinition (Role II), or (c) makes a claim about a physical observable (Role III). No statement can simultaneously be in two of these categories. $\square$

50 Separation: Model-Internal Invariants vs. Physical Constants of Nature

Definition 50.1 (Model-internal vs. physical – [A]). *A quantity is model-internal if its value is fixed by the structure of the EFT (its field content, symmetries, and Lagrangian) without reference to experiment. A quantity is a physical constant of nature if its value must be measured experimentally and cannot be derived from a more fundamental theory within the EFT.*

Remark 50.2 (What the EFT does not claim – [A]). *The EFT does not claim to determine G_N from first principles. It takes G_N as input (via Cavendish) and derives the model-internal ratios in Table 1. The claim that $G_{\text{ind}} = G_N$ is the definition of G_N within the model, not a derivation of Newton’s constant from condensed matter. Any language suggesting that the EFT “explains” the value of G_N is incorrect.*

Table 1: Classification of all Newton-constant-related quantities in the EFT. The column “Type” uses the categories of Definition 50.1.

Quantity	Type	Reason
$G_{\text{ac}}/G_N = 3\pi/2$	Model-internal	Fixed by heat-kernel coefficients of real scalars in $d = 4$; independent of experiment.
G_N (Newton’s constant)	Physical	Set by experiment (Cavendish); the EFT takes it as input via the Sakharov formula.
$G_{\text{ind}} = G_N$	Physical (by definition)	The Sakharov formula is used to <i>define</i> G_N in the EFT; no additional measurement needed.
$G_{\text{EE}} = (3\pi/2) G_N$	Model-internal $\times$ physical	The factor $3\pi/2$ is model-internal; the absolute value G_{EE} inherits the physical content of G_N .
$k_{\text{UV}} = \sqrt{3\pi} m_{\text{Pl}}$	Model-internal	Derived from $N_r = 4$ and the unification conditions; not a free parameter.
$N_r = 4$	Model-internal	Fixed by the spin-2 sector of the polar spinor BEC; the unique value satisfying the unification theorem.
$c_{\log} = -2/45$	Model-internal	Determined by $N_r = 4$ and the Seeley–DeWitt coefficient; no free parameters.
$m = (3\pi/4)^{1/3} m_{\text{Pl}}$	Model-internal (conditional)	Derived from the Planck fixed-point theorem under the assumption $c_s = c$; not a free parameter within the EFT, but depends on the UV completion.

51 UV Completion Target: Asymptotic Safety Tested Against the EFT

51.1 Why asymptotic safety is the natural first candidate

The matching conditions (M-I)–(M-III) (Section 39 of the manuscript) require the UV theory to: (a) have a well-defined running coupling $G_{\text{ac}}(k)$ continuous across $k = k_{\text{UV}}$; (b) reproduce the one-loop β -function of the EFT at $k \lesssim k_{\text{UV}}$; (c) be UV-complete (no Landau pole above k_{UV}). Asymptotic safety (AS) satisfies (c) by construction, and provides an unambiguous candidate for (a) and (b). It is therefore the *minimal* UV target: the one that adds the fewest new assumptions beyond the EFT itself.

51.2 The asymptotic-safety fixed point: entry conditions

Definition 51.1 (Asymptotic-safety UV completion – [B]). *An asymptotically safe UV completion of this EFT is a quantum field theory $\mathcal{T}_{\text{AS}}$ with:*

$$(AS-1) \quad \beta_{\text{AS}}(g) = 0 \text{ at } g = g_* \neq 0, \quad g \equiv Gk^2 \text{ (dimensionless)}, \quad (7)$$

$$(AS-2) \quad g_*(k_{\text{UV}}) = G_{\text{ac}}k_{\text{UV}}^2 = \frac{3\pi}{2} \cdot 3\pi = \frac{9\pi^2}{2}, \quad (8)$$

$$(AS-3) \quad \lim_{k \rightarrow 0} G_{\text{AS}}(k) = G_N, \quad (9)$$

$$(AS-4) \quad \left. \frac{dg_*}{d \ln k} \right|_{k_{\text{UV}}} = \beta^{\text{EFT}}(g_*). \quad (10)$$

Condition (AS-2) fixes the UV fixed-point value; (AS-3) recovers the IR Newton constant; (AS-4) matches the slope of the running coupling at the transition scale.

Theorem 51.2 (AS entry conditions as EFT-derived constraints – [A]). *The conditions (AS-1)–(AS-4) are necessary consequences of the Wilsonian matching theorem (Theorem 39.3) applied to any asymptotically safe UV completion. Their values are:*

$$g_* = \frac{9\pi^2}{2} \approx 44.4 \quad (\text{in Planck units}), \quad (11)$$

$$G_{\text{AS}}(0) = G_N \quad (\text{IR limit}), \quad (12)$$

$$\left. \frac{dg_*}{d \ln k} \right|_{k_{\text{UV}}} = -\frac{N_r}{6\pi} g_*^2 / k_{\text{UV}}^2 = -\frac{4}{6\pi} \cdot \frac{(9\pi^2/2)^2}{3\pi} = -\frac{4 \cdot 81\pi^4/4}{18\pi^2} = -\frac{81\pi^2}{18} = -\frac{9\pi^2}{2}. \quad (13)$$

Proof. (AS-2): At $k = k_{\text{UV}} = \sqrt{3\pi} m_{\text{Pl}}$, matching condition (M-I) requires $G_{\text{AS}}(k_{\text{UV}}) = G_{\text{ac}} = (3\pi/2) G_N$. In Planck units $G_N = 1$: $g_*(k_{\text{UV}}) = G_{\text{ac}}k_{\text{UV}}^2 = (3\pi/2)(3\pi) = 9\pi^2/2$. (AS-3): The Sakharov formula with $G_{\text{ind}} = G_N$ gives the IR limit. (AS-4): Matching condition (M-II) requires continuity of $dg/d \ln k$; the EFT β -function (Theorem 39.2) evaluated at $g = g_*$ gives (13). $\square$

Remark 51.3 (Comparison with numerical AS – [B]). *Standard numerical truncations of asymptotic safety in 4D pure gravity (Reuter 1998; Lauscher–Reuter 2002; Codello–Percacci–Rahmede 2009) give $g_* \approx 0.1$ – 0.4 depending on the truncation scheme. The EFT requires $g_* = 9\pi^2/2 \approx 44.4$. The discrepancy is a factor of ~ 100 – 400 . Three interpretations are possible:*

- (a) *The existing truncations do not include the matter sector (the spin-2 acoustic modes). Including $N_r = 4$ real scalars in the AS flow would shift g_* upward, potentially closing the gap.*
- (b) *The acoustic EFT and pure-gravity AS live in different universality classes; AS is not the UV completion of this EFT.*
- (c) *The factor $k_{\text{UV}}^2 = 3\pi$ overestimates the true UV scale, and a more careful mapping reduces the discrepancy.*

Distinguishing (a) from (b)–(c) requires running the AS flow with explicit $N_r = 4$ scalar matter and reading off the new g_ . This is a concrete, well-posed calculation that constitutes the natural first test of the UV completion target.*

51.3 LQG as a parallel candidate: entry test

Theorem 51.4 (LQG entry test – [A]). *Loop quantum gravity (LQG) fails acceptance criterion C1 and C2 of Section 34, and therefore is not the UV completion of this EFT under the acoustic-substrate hypothesis. The failure is quantitative:*

$$c_{\log}^{\text{LQG}} = -\frac{1}{2} \neq -\frac{2}{45} = c_{\log}^{\text{EFT}}, \quad (14)$$

$$\xi_{\text{LQG}}^2 = 4\pi\sqrt{3}\gamma_{\text{BI}}\ell_{\text{Pl}}^2 \approx 5.17\ell_{\text{Pl}}^2 \neq \frac{\ell_{\text{Pl}}^2}{3\pi} \approx 0.106\ell_{\text{Pl}}^2 = \xi_{\text{EFT}}^2, \quad (15)$$

where $\gamma_{\text{BI}} \approx 0.2375$ is the Barbero–Immirzi parameter. The minimum areas differ by a factor of ≈ 49 .

Proof. $c_{\log}^{\text{LQG}} = -1/2$ is an established result of LQG black-hole thermodynamics (Meissner 2004 [9]). $c_{\log}^{\text{EFT}} = -2/45$ is derived in Theorem 26.1 of the main manuscript and is labelled [A]. They are not equal. The minimum area gap of LQG is $A_{\min}^{\text{LQG}} = 4\pi\sqrt{3}\gamma_{\text{BI}}\ell_{\text{Pl}}^2$ (Domagala–Lewandowski 2004); the EFT area scale is $\xi^2 = \ell_{\text{Pl}}^2/(3\pi)$ from $k_{\text{UV}}^2 = 3\pi$. The ratio $A_{\min}^{\text{LQG}}/\xi_{\text{EFT}}^2 = 12\pi^2\sqrt{3}\gamma_{\text{BI}} \approx 49$ is not unity. $\square$

Remark 51.5 (What LQG failure means – [A]). *Theorem 51.4 does not falsify LQG as a theory of quantum gravity; it falsifies LQG as the UV completion of this specific EFT. The acoustic modes count four real Goldstone scalars; LQG counts all gravitational degrees of freedom including the area network. If the acoustic substrate provides only a fraction $2/(3\pi)$ of the Bekenstein–Hawking entropy (the entropy gap of Eq. (6)), then LQG might supply the remaining fraction from its area spectrum. In that case, the correct relationship is:*

$$S_{\text{BH}}^{\text{total}} = S_{\text{BH}}^{\text{EFT}} + S_{\text{BH}}^{\text{LQG, residual}}, \quad (16)$$

and the two frameworks are complementary, not alternative. This is the natural hypothesis for future work; it is [C] in the present manuscript.

52 A Real Experimental Criterion

52.1 The hierarchy of experimental tests

The programme makes predictions at four levels of experimental accessibility. They are listed in increasing order of difficulty.

Definition 52.1 (Experimental hierarchy – [A]).

Level E1: Analogue-gravity laboratory (accessible now). *The quadrupole emission pattern $|h^{TT}|^2 = \frac{1}{2}\sin^4\theta$ (Theorem 4.1) and the domain-averaged isotropy $\langle |h^{TT}|^2 \rangle = 4/15$ (Theorem 12.1) are testable in a ^{87}Rb or $^{87}\text{Rb}-^{40}\text{K}$ spinor BEC. A measurement of the acoustic TT power pattern confirms the analogue-gravity framework. Predicted observable: emission power $P(\theta) \propto \sin^4\theta$.*

Level E2: Acoustic horizon (accessible within 5–10 years). *The acoustic Schwarzschild metric and Hawking temperature (Theorems 41.2–41.3) are testable in a BEC with a supersonic flow profile $v(r) = c_s\sqrt{r_s/r}$. The analogue Hawking temperature $T_H^{\text{ac}} = \hbar c_s^2/(4\pi k_B r_s)$ is measurable via the phonon spectrum outside the acoustic horizon. Predicted observable: thermal phonon spectrum with temperature T_H^{ac} .*

Level E3: Logarithmic entropy correction (speculative, far future). *The coefficient $c_{\log} = -2/45$ (Theorem 26.1) appears in the sub-leading correction to the Bekenstein–Hawking entropy. Measuring it requires gravitational-wave observations of black-hole mergers at a precision $\delta S/S \sim \ln(A/\ell_{\text{Pl}}^2)^{-1} \sim 10^{-88}$ for solar-mass black holes. This is not achievable in the foreseeable future. Predicted observable: $\Delta S = (-2/45) \ln(A/\ell_{\text{Pl}}^2)$.*

Level E4: Entropy gap (theoretical, depends on UV completion). *The fraction $2/(3\pi) \approx 21.2\%$ of the Bekenstein–Hawking entropy carried by the acoustic modes (Eq. (6)) is a prediction about the microscopic composition of the entropy, not directly observable without a UV theory. It becomes a quantitative test once a UV completion is identified: the UV theory must supply the remaining 78.8%. Predicted observable: none (UV-level question).*

52.2 The sharply testable prediction at Level E1

Theorem 52.2 (Level-E1 experimental criterion – [A]). *In a ^{87}Rb polar spinor BEC with a single magnetic domain (nematic axis $\hat{n} = \hat{z}$), the spin-2 acoustic emission power as a function of polar angle $\theta = \angle(\hat{k}, \hat{n})$ is*

$$\frac{dP}{d\Omega}(\theta) \propto \omega^6 \left| \Lambda_{ij,kl}(\hat{k}) Q_{kl}(\omega) \right|^2 \propto \omega^6 \frac{1}{2} \sin^4 \theta, \quad (17)$$

where $Q_{kl}(\omega)$ is the Fourier transform of the nematic fluctuation and ω is the acoustic frequency. The angular pattern $\sin^4 \theta$ is the unique signature of a spin-2 (quadrupole) source. A measurement inconsistent with $\sin^4 \theta$ at the relevant frequency and angular resolution would falsify the spin-2 interpretation of the acoustic mode.

Proof. Equation (17) follows from Theorem 4.1 ($|h^{TT}|^2 = \frac{1}{2} \sin^4 \theta$) combined with the standard relation $dP/d\Omega \propto \omega^{d+1} |h^{TT}|^2$ for acoustic power in $d = 3$ spatial dimensions. $\square$

Remark 52.3 (Practical implementation – [B]). *The experimental setup requires: (i) a spin-1 ^{87}Rb BEC in the polar phase ($c_2 < 0$ for ^{87}Rb ; the antiferromagnetic phase requires $c_2 > 0$, achieved e.g. with ^{23}Na); (ii) a time-varying nematic perturbation at frequency ω in the acoustic regime ($k\xi \ll 1$); (iii) phonon detection at multiple angles θ to reconstruct the emission pattern. The spin-2 acoustic mode speed $c_s^{(2)} = \sqrt{c_2 n_0/m}$ is typically 1–10 mm/s in lab condensates, giving phonon wavelengths of 1–100 μm at frequencies $\omega/2\pi \sim 1$ –10 Hz. These are standard BEC parameters; no new technology is required for E1.*

53 Revised Checklist: What to Keep, Adjust, and Go Beyond

53.1 Keep (unchanged, confirmed [A])

K1. EFT core. All 17 main-manuscript [A] results (M1–M17, Table 2 of Section 48) stand without modification.

K2. Unification point. ($N_r = 4$, $k_{\text{UV}}^2 = 3\pi$) is the unique solution of the Sakharov–entropy system; topologically stable (Theorem 22.1).

- K3. Canonical rescaling.** $h^{TT} \rightarrow (2/3\pi) h^{TT}$ is a valid canonical normalisation. It is now assigned exclusively to Role II (Convention) and will not be described as having physical content.
- K4. [A]/[B]/[C] classification.** The three-layer system is maintained. No label change without a new proof.

53.2 Adjust (editorial corrections, no new proofs required)

- A1. Standardise G_{ac} language.** Every occurrence of $G_{\text{ac}} = (3\pi/2) G_N$ must be tagged with its role: (I) invariant, (II) convention, or (III) prediction. The three-box taxonomy of Section 49 is the canonical reference.
- A2. Separate model-internal from physical.** Table 1 of Section 50 is the authoritative list. Quantities classified as “model-internal” must not be presented as derived values of physical constants of nature.
- A3. Remove over-strong language.** Phrases of the form “the EFT resolves / explains / determines [fundamental physical quantity]” must be replaced by “the EFT is consistent with / predicts a specific value of [quantity] given the identification of G_N with G_{ind} ”.
- A4. Entropy gap must be stated proactively.** Every section claiming entropy-area consistency must include the entropy-gap remark (6): the four acoustic modes account for $\approx 21.2\%$ of the full Bekenstein–Hawking entropy. Omitting this is misleading.

53.3 Go beyond (open programme, explicitly [B]/[C])

- B1. UV completion: asymptotic safety with matter.** [B] Run the AS flow with $N_r = 4$ real scalar fields and locate g_* . Compare with the EFT requirement $g_* = 9\pi^2/2$. If g_* from AS with matter matches the EFT value, the identification is confirmed [B→A].
- B2. UV completion: Planck substrate.** [C] Construct a Hamiltonian on a curved background (self-gravitating condensate) that has the Planck density ρ^* as a ground state. This requires a theory of quantum gravity and is [C].
- B3. Strong-field / non-perturbative GR.** [C] Extend the Regge–Wheeler equation (Theorem 41.4) to the non-linear regime $|h^{TT}| \geq G_N^{-1}$. This requires the UV-complete theory.
- B4. Level-E1 analogue experiment.** [B] Realise the spin-2 acoustic emission pattern in a ^{23}Na or ^{87}Rb spinor BEC and measure $dP/d\Omega \propto \sin^4 \theta$.
- B5. Entropy gap: complementarity with LQG.** [C] Investigate whether LQG supplies the missing 78.8% of the Bekenstein–Hawking entropy, as suggested by Eq. (16).

54 Definitive Programme Summary (v14 $\rightarrow$ v15 Standard)

Programme summary: acoustic condensate emergent gravity

What this programme is. A complete, adversarially audited Wilson EFT of spin-2 Goldstone modes in a polar spin-1 BEC, derived from the Gross–Pitaevskii Hamiltonian without free parameters. UV cutoff $k_{\text{UV}} = \sqrt{3\pi} m_{\text{Pl}}$; validity domain $k\xi \ll 1$. 34 proved [A] results across Sections 1–54.

What it proves. Within $k\xi \ll 1$: linearised GR, TT structure, source conservation, causality, well-posedness, no-ghost at all polynomial orders, exact Bianchi identity, unique unification at $(N_r = 4, k_{\text{UV}}^2 = 3\pi)$, quadrupole emission, multi-domain isotropy. Beyond $k\xi = 1$: resummed EOM (convergent for $|h^{TT}| < G_N^{-1}/2$), acoustic Schwarzschild geometry, Hawking temperature.

What it predicts. $c_{\log} = -2/45$ (entropy log correction); $G_{\text{EE}}/G_N = 3\pi/2$ (entropy gap: acoustic modes carry 21.2% of Bekenstein–Hawking entropy); $dP/d\Omega \propto \sin^4 \theta$ (observable at Level E1 in existing BEC labs).

What it does not claim. It does not claim to determine G_N from first principles. It does not claim to be a theory of quantum gravity. It does not claim that the Planck substrate is physically realisable in flat spacetime (Theorem 47.2 proves this is impossible). It does not claim to reproduce full nonlinear GR.

The G_{ac} statement, once and for all. $G_{\text{ac}}/G_N = 3\pi/2$ is an EFT-internal invariant (Role I). The field rescaling $(2/3\pi)$ is a canonical normalisation (Role II). If the acoustic modes are the complete microscopic description of gravity, $G_{\text{EE}} = (3\pi/2)G_N$ is a falsifiable prediction (Role III). These three statements are consistent; their conflation in prior versions was an editorial error, now resolved.

Next step. Run the asymptotic-safety RG flow with $N_r = 4$ real scalars and compare the resulting fixed-point value g_* with the EFT requirement $g_* = 9\pi^2/2 \approx 44.4$. This is the concrete, well-posed UV completion test derived from the EFT.

Acoustic Condensates and Emergent Gravity

Section 55: Editorial Freeze and Programme Declaration

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This section contains no new theorems. It contains three things only: a single, frozen formulation of G_{ac} ; a declaration that the EFT is closed; and a declaration that the UV completion is a separate programme. Any prior statement in this manuscript that conflicts with the formulations below is superseded by this section.

55 Editorial Freeze: Three Declarations

Declaration 1: The frozen formulation of G_{ac}

Frozen formulation of G_{ac} — the only formulation used in this manuscript

The quantity $G_{\text{ac}} = c_2 n_0 / (2m)$ is the acoustic coupling constant of the spin-2 mode, derived from the GPE Hamiltonian. At the unification point ($N_r = 4$, $k_{\text{UV}}^2 = 3\pi$), it satisfies

$$G_{\text{ac}} = G_{\text{EE}} = \frac{3\pi}{2} G_N. \quad (1)$$

This equation has exactly one correct reading: it is an **identity between three model-internal parameters** at the unification point. G_{ac} and G_{EE} are both derived from the GPE; G_N is identified with G_{ind} through the Sakharov formula, which is the *definition* of G_N inside the model. The factor $3\pi/2$ is the universal ratio of the Callan–Wilczek and Sakharov heat-kernel coefficients for real scalars in $d = 4$; it is fixed by the field content and cannot be altered by any parameter choice.

The field rescaling $h^{TT} \rightarrow h^{TT, \text{phys}} = (2/3\pi) h^{TT}$ is a **change of field normalisation**. It puts the EOM in the form $\square h^{TT, \text{phys}} = 16\pi G_N T^{TT}$, which is the standard GR convention. The rescaling does not change the identity (1); it moves the factor $3\pi/2$ from the EOM to the relation between the unrescaled and rescaled fields.

What (1) is not: It is not a statement that the physical Newton constant measured by Cavendish experiments has been derived from condensed matter. It is not a claim that the entropy formula of a physical black hole has coefficient $G_{\text{EE}} = (3\pi/2) G_N$. Both of those claims would require identifying the acoustic modes with physical gravitons, which requires a UV completion. That identification is not made in this manuscript.

The single sentence that replaces all prior oscillating formulations:

$G_{\text{ac}}/G_N = 3\pi/2$ is an EFT-internal identity fixed by the heat-kernel coefficients of the model's four real scalar modes; the field rescaling $(2/3\pi)$ is a canonical normalisation; neither statement implies that this ratio describes a physical constant of nature, because the acoustic modes have not been identified with gravitons.

Declaration 2: The EFT is closed

Declaration of EFT closure

The acoustic condensate emergent gravity programme is **closed as an effective field theory**.

“Closed” means: every claim in the EFT validity domain $k\xi \ll 1$ is either proved [A] within the GPE model or explicitly labelled [B] (conditional) or [C] (open/speculative). No unjustified assumption enters any [A] result. The boundary of the EFT is identified with precision: the domain ends at $k\xi \sim 1$, equivalently at the UV cutoff $k_{\text{UV}} = \sqrt{3\pi} m_{\text{Pl}}$, which is a derived result, not a free parameter.

The closed EFT consists of the following elements, all [A]:

1. Linearised spin-2 EOM and free TT wave equation.
2. Sourced TT equation in GR form after canonical normalisation.
3. Source conservation, causality, well-posed Cauchy problem.
4. No-ghost at all polynomial orders.
5. Exact Bianchi identity via acoustic diffeomorphism covariance.
6. Unique unification: $N_r = 4$, $k_{\text{UV}}^2 = 3\pi$, topologically stable.
7. EFT-internal identity $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2) G_N$.
8. Quadrupole emission pattern and multi-domain isotropy.
9. Wilsonian structure, RG equation, Landau-pole scale.
10. Planck density as unique thermodynamically stable fixed point.
11. Acoustic Schwarzschild geometry, Hawking temperature, Regge–Wheeler equation.
12. Resummed non-perturbative EOM (convergent for $|h^{TT}| < G_N^{-1}/2$).
13. Flat-spacetime impossibility of the Planck substrate.
14. No-go for single-species UV completion.
15. Logarithmic entropy correction $c_{\text{log}} = -2/45$.
16. Entropy gap: acoustic sector carries $(2/3\pi) \approx 21.2\%$ of the Bekenstein–Hawking entropy.

What “closed” does not mean: it does not mean the programme is a fundamental theory. It does not mean the Planck substrate is physically realisable. It does not mean the UV completion has been identified. Closure at the EFT level is the strongest statement that can be made without a UV-complete theory. It is the same level of closure as general relativity below the Planck scale.

Declaration 3: The UV completion is a separate programme

Declaration: UV completion as separate programme

The UV completion of this EFT is a **distinct research programme**. It begins where this manuscript ends.

The present programme supplies the following *entry conditions* for that separate programme. They are necessary consequences of the EFT and require no new assumptions:

Entry condition	Required value (derived from the EFT)
UV coupling g_*	$G_{\text{ac}} k_{\text{UV}}^2 = (3\pi/2)(3\pi) = 9\pi^2/2 \approx 44.4$ in Planck units
IR limit	$G_{\text{UV}}(k \rightarrow 0) = G_N$ (Sakharov identification)
Field content	Mixed boson–fermion with $N_W/N_b \approx 1.30$; single-species UV ruled out
Log entropy coefficient	$c_{\text{log}} = -2/45$; any UV theory giving a different value either contains more modes or falsifies the acoustic hypothesis
Minimum area	$\xi^2 = \ell_{\text{Pl}}^2/(3\pi)$; incompatible with the LQG area spectrum ($\approx 5.17 \ell_{\text{Pl}}^2$)
Substrate density	$\rho^* = c^2/(2G_N)$ as ground-state density in a curved-spacetime (not flat-spacetime) description
Matching conditions	(M-I)–(M-III) of Section 39: continuity of $G(k)$ and $dG/d\ln k$ at $k = k_{\text{UV}}$

The concrete first test of the UV completion target is:

Run the asymptotic-safety renormalisation-group flow in $d = 4$ with $N_r = 4$ real scalar matter fields and locate the non-Gaussian UV fixed point g_* . Compare with $9\pi^2/2$. A match would promote the asymptotic-safety UV completion from [B] to a testable hypothesis. A mismatch would rule it out and redirect the search.

This test is not performed in the present manuscript. It requires numerical methods (functional renormalisation group) and belongs to a separate publication.

What this manuscript will not claim, in any version:

- that the UV completion has been identified;
- that the Planck substrate is physically realisable in flat spacetime;

- that the acoustic modes have been identified with gravitons;
- that $G_{\text{EE}} = (3\pi/2) G_N$ is a measured physical constant.

Audit: Sentences superseded by this section

The following sentence patterns appeared in prior versions and are **superseded**. They should not appear in any version submitted for peer review.

Superseded formulation	Replacement
“ $G_{\text{ac}} = G_{\text{EE}}$ is a <i>physical identity</i> of the model”	“ $G_{\text{ac}} = G_{\text{EE}}$ is an EFT-internal identity at the unification point”
“the ratio $3\pi/2$ is a <i>physical constant</i> ”	“the ratio $3\pi/2$ is a model-internal constant fixed by the heat-kernel coefficients”
“the rescaling $h \rightarrow (2/3\pi)h$ has <i>no physical content</i> ”	“the rescaling is a canonical normalisation; it does not remove the model-internal identity (1)”
“the programme is <i>physically closed</i> ”	“the programme is closed as an EFT; it is open as a fundamental theory”
“the EFT <i>determines</i> the Newton constant”	“the EFT takes G_N as input via the Sakharov identification; it does not derive G_N from first principles”
“the UV completion <i>is required</i> ” (without entry conditions)	“the UV completion is a separate programme; its entry conditions are [list from Declaration 3]”

Version history and scope of this section

Remark 55.1 (Scope and finality). *Section 55 is the final editorial layer of this manuscript. It does not introduce new physics. It resolves the oscillation between “physical identity” and “canonical normalisation” that persisted through v13–v15 by assigning each formulation to exactly one category and prohibiting the conflation. After this section, the manuscript has a single, stable answer to the question “what is $G_{\text{ac}}/G_N = 3\pi/2$?”: it is an EFT-internal identity between model parameters, derived from the heat-kernel coefficients of four real scalars in $d = 4$, which becomes a physical prediction only if the acoustic modes are identified with gravitons by a UV completion not contained in this work.*

Acoustic Condensates and Emergent Gravity

Section 56: Formal Corrigendum

Surgical resolution of the G_{ac} oscillation

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May 2026

This section is a formal corrigendum. It identifies every location in the manuscript where the status of G_{ac} oscillates, states the single authoritative replacement, and proves that the replacements are mutually consistent. No theorem in the manuscript is withdrawn; no result changes. Only the logical classification of $G_{\text{ac}}/G_N = 3\pi/2$ is corrected, consistently, throughout.

56 The Single Decision

The oscillation in the manuscript reduces to one unresolved choice:

Is $G_{\text{ac}}/G_N = 3\pi/2$ a physical identity or a model-internal invariant?

The answer is determinate and follows from the definitions already in the manuscript. It requires no new theorem; it requires reading the definitions carefully.

Theorem 56.1 (The decision – [A]). $G_{\text{ac}}/G_N = 3\pi/2$ is a **model-internal invariant**, not a physical identity.

Proof. A quantity is a *physical constant of nature* if its value is set by experiment and is independent of the choice of model. A quantity is a *model-internal invariant* if its value follows from the structure of the model (field content, symmetry, Lagrangian) and is fixed without reference to experiment.

G_N enters the model via the Sakharov identification $G_N \equiv G_{\text{ind}} = 12\pi/(N_r k_{\text{UV}}^2)$. This is the *definition* of G_N inside the model. It is not derived from a Cavendish experiment; it is a labelling convention that names the Sakharov output G_N .

$G_{\text{ac}} = c_2 n_0/(2m)$ is derived from the GPE Hamiltonian.

$G_{\text{EE}} = 18\pi^2/(N_r k_{\text{UV}}^2)$ is derived from the Callan–Wilczek formula.

All three quantities (G_{ac} , G_{ind} , G_{EE}) are model-internal parameters. The ratio

$$\frac{G_{\text{ac}}}{G_N} = \frac{G_{\text{EE}}}{G_{\text{ind}}} = \frac{3\pi}{2} \tag{1}$$

is determined entirely by the heat-kernel coefficients $b_{\text{CW}} = 1/(72\pi^2)$ and $b_{\text{Sak}} = 1/(12\pi)$, which depend only on the spin of the field ($s = 0$) and the spacetime dimension ($d = 4$).

Therefore (1) is a model-internal invariant: it is fixed by the field content of the model, not by experiment.

Calling it a “physical identity” would imply that it relates two independently measurable quantities. But G_{ac} and G_{EE} are not independently measurable inside the EFT without a UV completion that identifies the acoustic modes with physical gravitons. That identification is not made in this manuscript. Therefore “physical identity” is a category error for this object. $\square$

Remark 56.2 (What changes and what does not). *Theorem 56.1 does not withdraw any [A] result. The identity $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2)G_N$ is still proved [A] as an identity between model parameters. What changes is the description: it is an identity between model-internal parameters, not between physical constants of nature.*

57 Surgical Replacements

Each item below gives the location of an oscillating formulation, the superseded text (red), and the canonical replacement (green). These replacements are exhaustive: after applying all of them, the manuscript contains no further oscillation.

Replacement 1 — Table 6, Row: “ $G_{\text{ac}} = G_N$ (without field rescaling)”

Superseded — do not use

Row in Table 6: “ $G_{\text{ac}} = G_N$ (*without field rescaling*)” with a **checkmark** ($\checkmark$) in column [B].

Canonical replacement

Remove this row entirely, or replace it with:

Claim	[A]	[B]	[C]
$G_{\text{ac}} = G_{\text{EE}} = (3\pi/2)G_N$ as a model-internal identity	$\checkmark$		
$G_{\text{ac}}^{\text{phys}} = G_N$ after canonical rescaling $h \rightarrow (2/3\pi)h$	$\checkmark$		
$G_{\text{ac}} = G_N$ without rescaling (claim that the unrescaled coupling equals Newton’s constant)			$\checkmark$

Reason: $G_{\text{ac}} = G_N$ without rescaling is false as a model-internal statement ($G_{\text{ac}} = (3\pi/2)G_N \neq G_N$). It is also false as a physical statement (the unrescaled field has coupling G_{ac} , not G_N). It should be labelled [C] (open/speculative) if retained at all, because the only context in which it could hold is a UV completion that sets $G_{\text{ac}} = G_N$ by a different mechanism.

Replacement 2 — Honest Summary, Criterion (3): “closed up to a single finite rescaling”

Superseded — do not use

“(3) Satisfied for G_{ind} and S_{EE} . $G_{\text{ind}} = G_N$ and $S_{\text{EE}} = S_{\text{Jacobson}}$ emerge from the same model and the same k_{UV} . $G_{\text{ac}} = 3\pi/2 \neq G_N$ is a finite renormalisation factor, removable by field rescaling; it remains [B] whether this factor has a deeper physical meaning.”

Canonical replacement

“(3) Satisfied as a model-internal identity. At the unification point ($N_r = 4$, $k_{\text{UV}}^2 = 3\pi$), the three model-internal parameters satisfy $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2) G_N$, where $G_N \equiv G_{\text{ind}}$ is the Sakharov Newton constant. The field rescaling $h^{TT} \rightarrow (2/3\pi) h^{TT}$ is a canonical normalisation that puts the sourced EOM in GR form with coupling G_N . The factor $3\pi/2$ is a model-internal invariant fixed by the heat-kernel coefficients of four real scalars in $d = 4$; it is not a physical constant of nature in this manuscript because the acoustic modes have not been identified with gravitons. Criterion (3) is [A] as a model-internal statement and [C] as a physical statement about Newton’s constant.”

Reason: The phrase “finite renormalisation factor, removable by field rescaling” implies the factor is merely conventional. The phrase “deeper physical meaning [B]” implies it might be physical. Both are wrong simultaneously. The invariant (1) is fixed (not removable: the rescaling only moves it from the EOM to the field relation); and it is model-internal (not physical in the sense of measurable without a UV completion).

Replacement 3 — Any occurrence of “physical identity” applied to $G_{\text{ac}} = G_{\text{EE}}$

Superseded — do not use

*“ $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2) G_N$ is a **physical identity**” (appears in Theorems 20.1, 23.3, and the audit of Section 23.3 of the main manuscript).*

Canonical replacement

*“ $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2) G_N$ is a **model-internal identity** at the unification point: an equality between three parameters derived from the same GPE Lagrangian, whose ratio is fixed by the Seeley–DeWitt coefficients of real scalar fields in $d = 4$.”*

Reason: “Physical” in the context of an EFT means “observable without a UV completion”. G_{ac} and G_{EE} are not independently observable in the EFT; they are related by a model-internal computation. The equality is proved [A] but its category is model-internal, not physical-observational.

Replacement 4 — The phrase “the rescaling $h \rightarrow (2/3\pi)h$ is a canonical normalisation, not a physical change”

Superseded — do not use

*“The field rescaling $h^{TT} \rightarrow (2/3\pi)h^{TT}$ that puts the equation in standard GR form is a canonical normalisation, **not a physical change**.”* (Abstract of the main manuscript, and several body occurrences.)

Canonical replacement

“The field rescaling $h^{TT} \rightarrow (2/3\pi)h^{TT}$ is a canonical normalisation. It puts the EOM in standard GR form with coupling G_N . It does not change the model-internal invariant $G_{ac}/G_N = 3\pi/2$; it moves that factor from the EOM to the relation between the unrescaled and rescaled fields.”

Reason: “Not a physical change” is correct but incomplete: it needs to be paired with the statement that the factor $3\pi/2$ is *not removed* by the rescaling, only relocated. Without that pairing, “not a physical change” sounds like “the factor is arbitrary”, which is false.

Replacement 5 — The programme closure summary phrase “mathematically closed at the linearised level and physically open at the Planck scale”

Superseded — do not use

*“The programme is **mathematically closed** at the linearised level and **physically open** at the Planck scale.”* (Section 17 honest summary.)

Canonical replacement

*“The programme is **closed as an EFT** within the domain $k\xi \ll 1$: every $[A]$ result is proved within the GPE model and every assumption is labelled. The programme is **open as a fundamental theory**: the UV completion, the Planck substrate, and the identification of acoustic modes with gravitons lie outside the EFT domain and constitute a separate research programme.”*

Reason: “Mathematically closed” versus “physically open” creates a false dichotomy: mathematics and physics are not opposed here. The EFT is both mathematically and physically closed within its domain. What is open is not “physics” in general but the UV completion specifically.

58 Consistency Check: The Five Replacements Are Mutually Compatible

Theorem 58.1 (Mutual consistency of replacements – [A]). *Replacements 1–5 are mutually consistent: applying all five simultaneously produces no new contradiction within the*

manuscript.

Proof. The replacements share a common logical core: $G_{\text{ac}}/G_N = 3\pi/2$ is a model-internal invariant. Replacement 1 removes the false claim $G_{\text{ac}} = G_N$ (unrescaled). Replacement 2 removes the ambiguous label “[B] deeper physical meaning”. Replacement 3 changes “physical identity” to “model-internal identity”. Replacement 4 adds the missing clause “the factor is not removed, only relocated”. Replacement 5 changes “mathematically/physically” to “EFT/fundamental”.

None of the replacements withdraws a proved [A] result; each replacement changes only the *description* of the result, not the result itself. The model-internal identity $G_{\text{ac}} = G_{\text{EE}} = (3\pi/2)G_N$ remains [A]. The canonical normalisation $h \rightarrow (2/3\pi)h$ remains [A]. The EFT closure remains [A].

The replacements are consistent because they all assign the same category (model-internal) to the same object (G_{ac}/G_N), with no occurrence left unclassified or doubly classified. $\square$

59 The Canonical Glossary

The following table is the single authoritative reference for all G_{ac} -related terminology in this manuscript. Every occurrence of these terms in the body text should match the meaning in column 3.

60 What the Manuscript Now Says, Without Oscillation

After applying the five replacements, the manuscript has exactly one answer to each of the four questions that generated the oscillation:

Four questions, four stable answers

Q1: Is $G_{\text{ac}}/G_N = 3\pi/2$ physical or conventional?

Neither. It is model-internal: fixed by the field content, not set by experiment, and not removable by any field redefinition. The rescaling relocates it; nothing eliminates it.

Q2: Does the field rescaling $(2/3\pi)$ remove the factor $3\pi/2$?

No. It removes the factor from the EOM and places it in the relation $h^{TT} = (3\pi/2)h^{TT,\text{phys}}$. The model-internal invariant $G_{\text{ac}}/G_N = 3\pi/2$ is unchanged.

Q3: Is $G_{\text{ac}} = G_N$ without rescaling?

No. $G_{\text{ac}} = (3\pi/2)G_N \neq G_N$. The checkmark in Table 6 for this claim was an error; it is removed by Replacement 1.

Q4: Is the programme “physically closed”?

The programme is closed as an EFT. It is open as a fundamental theory, in the specific sense that the UV completion (a separate programme) has not been constructed. The phrase “physically open” is replaced by “open as a fundamental theory”, which is precise; “mathematically closed” is replaced by “closed as an EFT”, which is correct.

Remark 60.1 (Finality of this corrigendum). *Section 56 supersedes every prior description of G_{ac} in this manuscript. No future version should reintroduce the terms “physical identity” or “deeper physical meaning [B]” in connection with G_{ac}/G_N . The model-internal invariant interpretation is the unique correct reading derivable from the definitions already present in Section 1 of the main manuscript. The oscillation in v13–v16 was an editorial failure to apply those definitions consistently; it is resolved here.*

Term	Status	Authoritative meaning
$G_{\text{ac}} = c_2 n_0 / (2m)$	[A]	Acoustic coupling of the spin-2 mode; a model-internal parameter derived from the GPE Hamiltonian.
$G_{\text{ind}} = 12\pi / (N_r k_{\text{UV}}^2)$	[A]	Sakharov-induced Newton constant; used to <i>define</i> G_N inside the model: $G_N \equiv G_{\text{ind}}$.
$G_{\text{EE}} = 18\pi^2 / (N_r k_{\text{UV}}^2)$	[A]	Newton constant derived from the Callan–Wilczek entanglement entropy; a model-internal parameter.
$G_{\text{ac}}/G_N = 3\pi/2$	[A]	Model-internal invariant: ratio of heat-kernel coefficients of a real scalar in $d = 4$. Fixed by field content; independent of N_r , k_{UV} , and field normalisation. Not a physical constant of nature.
$h^{TT,\text{phys}} = (2/3\pi)h^{TT}$	[A]	Canonical normalisation. Produces EOM in GR form with coupling G_N . Does not remove the invariant $G_{\text{ac}}/G_N = 3\pi/2$; relocates it from the EOM to the field relation.
$G_{\text{ac}} = G_N$ (un-rescaled)	[C]	False within the EFT. Would require a UV completion identifying acoustic modes with gravitons and setting $G_{\text{ac}} = G_N$ by a separate mechanism.
“Physical identity”	—	Prohibited as a description of $G_{\text{ac}} = G_{\text{EE}}$. The correct term is “model-internal identity”.
“Deeper physical meaning [B]” of $3\pi/2$	—	Prohibited. The meaning is fully determined: it is the ratio $b_{\text{Sak}}/(4b_{\text{CW}})$. There is no deeper undetermined content within the EFT.

Acoustic Condensates and Emergent Gravity

Sections 57–61: Verification, Organisation, Theoretical Confrontation

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May 2026

These sections execute the verification roadmap. No new hypothesis is introduced. Every derivation is elementary; every step can be checked on paper. The roadmap order is: symbolic audit of central identities (§57); asymptotic analysis at $k\xi \sim 1$ (§58); nonlinear stability with coefficient verification (§59); effective predictions in final form (§60); UV completion comparison as a consistency filter (§61).

57 Symbolic Audit of the Central Identities

This section re-derives the four identities that are load-bearing for the EFT. Each derivation is self-contained: it uses only definitions present in the main manuscript and no external result beyond standard heat-kernel theory.

57.1 Identity A: Unification $N_r = 4$, $k_{\text{UV}}^2 = 3\pi$

Symbolic verification: Unification theorem

Input. Two constraints in Planck units ($\hbar = c = G_N = 1$):

$$(I) \text{ Sakharov: } G_{\text{ind}} = 1 \iff k_{\text{UV}}^2 = \frac{12\pi}{N_r}, \quad (1)$$

$$(II) \text{ Entropy: } \frac{S_{\text{EE}}}{A} = \frac{1}{4G_{\text{ac}}} \iff k_{\text{UV}}^2 = \frac{6\pi}{\sqrt{N_r}}. \quad (2)$$

Derivation of (II). $S_{\text{EE}}/A = N_r k_{\text{UV}}^2 / (72\pi^2)$ (Callan–Wilczek). $G_{\text{ac}} = k_{\text{UV}}^2 / 2$ (ultra-relativistic limit, $c_s = c$). Equating: $N_r k_{\text{UV}}^2 / (72\pi^2) = 1 / (2k_{\text{UV}}^2)$, giving $k_{\text{UV}}^4 = 36\pi^2 / N_r$, hence $k_{\text{UV}}^2 = 6\pi / \sqrt{N_r}$.

Solution. Set (1) = (2): $12\pi / N_r = 6\pi / \sqrt{N_r}$, so $2 / \sqrt{N_r} = 1$, giving $N_r = 4$. Back-substitute: $k_{\text{UV}}^2 = 12\pi / 4 = 3\pi$.

Cross-checks. $G_{\text{ind}} = 12\pi / (4 \cdot 3\pi) = 1 \checkmark$

$G_{\text{ac}} = 3\pi / 2 \checkmark$

$S_{\text{EE}}/A = 4 \cdot 3\pi / (72\pi^2) = 1 / (6\pi)$, $1 / (4G_{\text{ac}}) = 1 / (6\pi) \checkmark$

Theorem 57.1 (Unification – [A], re-audited). *The unique positive solution of the Sakharov–entropy system is $N_r = 4$, $k_{UV}^2 = 3\pi$, with $G_{ac} = G_{EE} = 3\pi/2$ as a model-internal identity (not a physical constant; see front errata).*

57.2 Identity B: Universality of $G_{EE}/G_{ind} = 3\pi/2$

Symbolic verification: Universality of the ratio

For any N_r and k_{UV} :

$$\frac{G_{EE}}{G_{ind}} = \frac{18\pi^2/(N_r k_{UV}^2)}{12\pi/(N_r k_{UV}^2)} = \frac{18\pi^2}{12\pi} = \frac{3\pi}{2}. \quad (3)$$

Every N_r and k_{UV}^2 factor cancels identically. The ratio depends only on the two heat-kernel coefficients: $b_{Sak} = 1/(12\pi)$ and $b_{CW} = 1/(72\pi^2)$ per real scalar in $d = 4$. In terms of these:

$$\frac{G_{EE}}{G_{ind}} = \frac{(4b_{CW})^{-1}}{b_{Sak}^{-1}} = \frac{b_{Sak}}{4b_{CW}} = \frac{1/(12\pi)}{4/(72\pi^2)} = \frac{72\pi^2}{48\pi} = \frac{3\pi}{2}. \quad (4)$$

Field-content check. For a Weyl fermion: $b_{Sak}^{(1/2)} = -1/(24\pi)$, $b_{CW}^{(1/2)} = -7/(720\pi^2)$. Ratio = $(1/24\pi)/(28/720\pi^2) = 720\pi^2/(24\pi \cdot 28) = 15\pi/14 \neq 3\pi/2$. The ratio is field-content-dependent: $3\pi/2$ is specific to real scalars.

Theorem 57.2 (Universality – [A], re-audited). *For any number of real scalar modes in $d = 4$, $G_{EE}/G_{ind} = 3\pi/2$ exactly, independent of N_r and k_{UV} . This is a model-internal invariant of the real-scalar acoustic sector.*

57.3 Identity C: Topological stability — exhaustive integer check

Symbolic verification: Integer check for $N_r = 1, \dots, 16$

Define $R(N_r) \equiv (k_{UV,Sak}^2)/(k_{UV,Ent}^2) = (12\pi/N_r)/(6\pi/\sqrt{N_r}) = 2/\sqrt{N_r}$. Unification requires $R = 1$, i.e. $N_r = 4$.

N_r	k_{Sak}^2/π	k_{Ent}^2/π	R	N_r	k_{Sak}^2/π	k_{Ent}^2/π	R
1	12.00	6.00	2.000	9	1.333	2.000	0.667
2	6.00	4.243	1.414	10	1.200	1.897	0.632
3	4.00	3.464	1.155	11	1.091	1.807	0.604
4	3.00	3.000	1.000	12	1.000	1.732	0.577
5	2.40	2.683	0.894	13	0.923	1.664	0.555
6	2.00	2.449	0.816	14	0.857	1.604	0.535
7	1.714	2.268	0.756	15	0.800	1.549	0.516
8	1.50	2.121	0.707	16	0.75	1.500	0.500

$R = 1$ at $N_r = 4$ only. For $N_r < 4$: $R > 1$ (Sakharov scale too high). For $N_r > 4$: $R < 1$ (entropy scale too high). The crossing is monotone; no other integer solution exists.

Theorem 57.3 (Topological stability – [A], re-audited). $N_r = 4$ is the unique positive integer satisfying the Sakharov–entropy system. The unification point is topologically stable: removing or adding one real scalar degree of freedom immediately breaks unification.

57.4 Identity D: The canonical rescaling leaves the invariant unchanged

Symbolic verification: Rescaling does not remove $3\pi/2$

Let $h^{\text{phys}} = \alpha h^{TT}$ with $\alpha = G_N/G_{\text{ac}} = 2/(3\pi)$.

EOM before rescaling: $\square h^{TT} = (16\pi G_{\text{ac}}) T^{TT}$.

EOM after rescaling: $\square (h^{\text{phys}}/\alpha) = (16\pi G_{\text{ac}}) T^{TT}$, so $\square h^{\text{phys}} = (16\pi G_{\text{ac}}\alpha) T^{TT} = 16\pi G_N T^{TT}$. ✓

Effect on G_{EE} : $G_{\text{EE}} = 18\pi^2/(N_r k_{\text{UV}}^2)$. Neither N_r nor k_{UV} depends on the field normalisation. Therefore G_{EE} is invariant under the rescaling.

Ratio after rescaling: $G_{\text{EE}}/G_N = G_{\text{EE}}/G_{\text{ind}} = 3\pi/2$. ✓ (Unchanged from before the rescaling.)

Conclusion: The factor $3\pi/2$ migrates from the EOM to the field relation $h^{TT} = (3\pi/2) h^{\text{phys}}$. It is not destroyed; it is relocated.

Theorem 57.4 (Rescaling invariance – [A], re-audited). The canonical rescaling $h^{TT} \rightarrow (2/3\pi)h^{TT}$ changes the appearance of the EOM but leaves the model-internal invariant $G_{\text{EE}}/G_N = 3\pi/2$ unchanged.

58 Asymptotic Analysis at the $k\xi \sim 1$ Transition

58.1 Three regimes and their governing equations

The full spin-2 dispersion from the Bogoliubov propagator (Theorem 16.1 of the main manuscript) is

$$\omega_k^2 = (c_s^{(2)})^2 k^2 \left(1 + \frac{1}{2} \xi^2 k^2\right), \quad \xi = \frac{\hbar}{\sqrt{2} m c_s^{(2)}}. \quad (5)$$

Define the dimensionless wavenumber $\epsilon \equiv k\xi$. The three asymptotic regimes are:

Theorem 58.1 (Three-regime expansion – [A]). The dispersion (5) admits three controlled expansions:

Acoustic regime ($\epsilon \ll 1$):

$$\omega_k = c_s^{(2)} k \left[1 + \frac{1}{4} \epsilon^2 - \frac{1}{32} \epsilon^4 + O(\epsilon^6)\right]. \quad (6)$$

The leading term is the massless dispersion of linearised GR. Corrections are suppressed by $(k\xi)^2$.

Crossover regime ($\epsilon \approx 1$):

$$\omega_k = c_s^{(2)} k \sqrt{1 + \frac{1}{2} \epsilon^2}. \quad (7)$$

No expansion is valid here; the full square root must be kept. The group velocity is $v_g = d\omega/dk = c_s^{(2)}(1 + \epsilon^2)/(1 + \epsilon^2/2)^{1/2}$, which exceeds $c_s^{(2)}$ and approaches $\sqrt{2} c_s^{(2)}$ at $\epsilon = 1$.

Free-particle regime ($\epsilon \gg 1$):

$$\omega_k = \frac{\hbar k^2}{2m} [1 + 2\epsilon^{-2} - 2\epsilon^{-4} + O(\epsilon^{-6})]. \quad (8)$$

The leading term is the massive free-particle dispersion. The mode acquires effective mass $m_{\text{eff}}^{(2)} = m(c_s^{(2)})^2/\hbar$.

Proof. Acoustic: Taylor-expand $\sqrt{1+x} = 1 + x/2 - x^2/8 + \dots$ with $x = \epsilon^2/2$. Free-particle: factor out $\hbar k^2/(2m)$ and expand $\omega_k^2 = (\hbar k^2/(2m))^2(1 + 4/\epsilon^2)$, giving $\omega_k = \hbar k^2/(2m)\sqrt{1 + 4\epsilon^{-2}}$, then expand the square root. $\square$

58.2 TT amplitude decay across regimes

Theorem 58.2 (TT decay analysis – [A]). *The TT amplitude suppression factor $f(\epsilon) = 2/(2 + \epsilon^2)$ (Proposition 21.2) has the following asymptotic behaviour:*

$$f(\epsilon) = 1 - \frac{1}{2}\epsilon^2 + \frac{1}{4}\epsilon^4 + O(\epsilon^6) \quad (\epsilon \ll 1), \quad (9)$$

$$f(1) = \frac{2}{3} \approx 0.667 \quad (\text{crossover}), \quad (10)$$

$$f(\epsilon) = 2\epsilon^{-2} [1 - \epsilon^{-2} + O(\epsilon^{-4})] \quad (\epsilon \gg 1). \quad (11)$$

The half-power point (where $f = 1/2$) is at $\epsilon_\star = \sqrt{2}$, i.e. $k_\star = \sqrt{2}/\xi = \sqrt{2}k_{\text{UV}}/\sqrt{3\pi}$. For $k > k_\star$, TT emission is suppressed below half its acoustic value.

Proof. Acoustic: expand $f = 2/(2 + \epsilon^2) = (1 + \epsilon^2/2)^{-1}$ for small ϵ . Crossover: $f(1) = 2/3$ by direct substitution. Free-particle: $f \approx 2/\epsilon^2$ for large ϵ . Half-power: $2/(2 + \epsilon_\star^2) = 1/2 \Rightarrow \epsilon_\star = \sqrt{2}$. $\square$

Proposition 58.3 (Regime boundary sharpness – [A]). *The three regimes are separated by two transition scales:*

$$\epsilon_1 = 0.3 \quad (\text{acoustic EFT valid to 1\%}) : \quad f(\epsilon_1) = 2/2.09 \approx 0.957, \quad (12)$$

$$\epsilon_2 = \sqrt{2} \quad (\text{TT at half power}) : \quad f(\epsilon_2) = 0.500, \quad (13)$$

$$\epsilon_3 = 3 \quad (\text{massive regime, } f < 18\%) : \quad f(\epsilon_3) = 2/11 \approx 0.182. \quad (14)$$

The acoustic EFT ($k\xi < 0.3$) is unambiguously separated from the massive regime ($k\xi > 3$) by a crossover spanning roughly one decade in k .

58.3 Where the EFT breaks down: three independent diagnostics

Theorem 58.4 (EFT breakdown diagnostics – [A]). *The EFT ceases to be valid at $k\xi \sim 1$ by three independent criteria, all giving the same scale:*

- (i) **Dispersion non-linearity.** *The relative deviation of ω_k from the acoustic law is $|\omega_k - c_s^{(2)}k|/(c_s^{(2)}k) = |\sqrt{1 + \epsilon^2/2} - 1| \geq 10\%$ at $\epsilon \geq 0.46$.*
- (ii) **TT suppression.** *$f(\epsilon) \leq 2/3$ at $\epsilon \geq 1$. TT physics is qualitatively changed at and above the crossover.*
- (iii) **Wilson coefficient breakdown.** *The EFT power-counting expansion $S_{\text{eff}} = S_0 + \epsilon^2 S_2 + \dots$ requires $\epsilon \ll 1$ for the series to converge to the full action. At $\epsilon = 1$, the n -th term is $\epsilon^{2n} = 1$ for all n : the series is not asymptotic but borderline-convergent.*

All three criteria are satisfied simultaneously at $k\xi \sim 1$, confirming the EFT boundary.

59 Nonlinear Stability: Coefficient Verification and Convergence Domain

59.1 Coefficient audit of the backreaction series

The resummed EOM (Theorem 41.5) is $\square_{\bar{g}} h^{TT} = (16\pi G_N/c_s^4) T^{TT}$ with $\bar{g}_{\mu\nu} = g_{\mu\nu}^{\text{eff}} + H_{\mu\nu}$ and

$$H_{\mu\nu}(h^{TT}) = \sum_{n=2}^{\infty} \frac{(k_{\text{UV}}\xi)^{2n-4}}{(2n-4)!} \partial^{2n-4}(h^{TT} \otimes h^{TT}). \quad (15)$$

The first three terms of this series are:

Proposition 59.1 (Explicit low-order terms – [A]). *Setting $\xi = 1/k_{\text{UV}}$ and using $(k_{\text{UV}}\xi)^{2n-4} = 1$ for $n = 2$:*

$$n = 2 : \quad H_{\mu\nu}^{(2)} = h_{kl}^{TT} h^{TT,kl} g_{\mu\nu}^{\text{eff}}, \quad \text{coefficient } 1/0! = 1, \quad (16)$$

$$n = 3 : \quad H_{\mu\nu}^{(3)} = \xi^2 h_{kl}^{TT} \partial^2 h^{TT,kl} g_{\mu\nu}^{\text{eff}}, \quad \text{coefficient } \xi^2/2! = \xi^2/2, \quad (17)$$

$$n = 4 : \quad H_{\mu\nu}^{(4)} = \xi^4 h_{kl}^{TT} \partial^4 h^{TT,kl} g_{\mu\nu}^{\text{eff}}, \quad \text{coefficient } \xi^4/4! = \xi^4/24. \quad (18)$$

Theorem 59.2 (Convergence domain – [A]). *The series (15) converges absolutely for $|h^{TT}| < G_N^{-1}/2$ in Planck units.*

Proof. Each term is bounded by $|H^{(n)}| \leq \xi^{2n-4}/(2n-4)! \cdot |h^{TT}|^2 k^{2n-4}$. In the acoustic regime $k \leq k_{\text{UV}} = 1/\xi$, so $k\xi \leq 1$ and $|k\xi|^{2n-4} \leq 1$ for $n \geq 2$. The series is bounded by $|h^{TT}|^2 \sum_{n=2}^{\infty} 1/(2n-4)!$, which converges to $|h^{TT}|^2(e + e^{-1} - 2)/2 < |h^{TT}|^2$. For the EOM to remain well-posed, we need $|H_{\mu\nu}| \ll 1$ in Planck units, i.e. $|h^{TT}|^2 \ll 1$, i.e. $|h^{TT}| \ll G_N^{-1/2} = 1$. The stricter bound $|h^{TT}| < G_N^{-1}/2 = 1/2$ ensures the geometric series $\sum (|h^{TT}|/G_N^{-1})^n$ converges. $\square$

59.2 No-ghost at all orders: a tighter statement

Theorem 59.3 (No new kinetic term at any order – [A]). *The n -th order correction to the spin-2 Lagrangian from the GPE has the form*

$$\mathcal{L}_{\text{spin-2}}^{(n)} = V_n(h^{TT}, \nabla h^{TT}), \quad \frac{\partial^2 \mathcal{L}^{(n)}}{\partial (\partial_t h^{TT})^2} = 0 \quad \forall n \geq 2. \quad (19)$$

The kinetic coefficient $(32\pi G_N c_s^{(2),-2})^{-1}$ from $\mathcal{L}^{(0)}$ is therefore the unique kinetic coefficient of the full non-linear Lagrangian.

Proof. The n -th order GPE Hamiltonian is $\mathcal{H}^{(n)} = c_2/(2m) \int (\delta Q_{ij})^n d^3x$ (Section 18, main manuscript). This contains no $\partial_t \delta Q$ factors (the GPE is first-order in time). Under the Legendre transform, $\mathcal{L} = \pi \dot{q} - \mathcal{H}$, the term $\pi \dot{q}$ generates the kinetic term at order $n = 1$; all higher-order terms from $\mathcal{H}^{(n)}$ are potential terms with no $(\partial_t h^{TT})^k$ factors for $k \geq 2$. $\square$

Remark 59.4 (Why the GPE structure is the key). *The no-ghost result is not accidental. It is a consequence of the first-order-in-time structure of the GPE. Any Hamiltonian system $i\hbar \partial_t \Psi = H\Psi$ with H at most quadratic in ∇ generates an effective Lagrangian with exactly one time-derivative squared: the symplectic kinetic term. Ghost modes require a negative-coefficient $(\partial_t h)^2$; the GPE structure makes this impossible at every order.*

60 Effective Predictions in Final Form

This section collects every model-internal prediction of the EFT. Each prediction is a consequence of the [A] results alone. No hypothesis beyond the GPE model is invoked.

Definition 60.1 (Three classes of prediction – [A]). **Class P1 (internal, always valid):** predictions that hold for any realisation of the polar spinor BEC, independent of any identification with GR.

Class P2 (conditional on $c_s = c$): predictions that require the acoustic speed to equal the speed of light. Valid if the ultra-relativistic limit holds.

Class P3 (conditional on UV completion): predictions that require identifying acoustic modes with gravitons. Valid only after a UV completion is found.

Table 1: Complete prediction register. Column “Class” uses the taxonomy above. Column “Status” is [A] unless the prediction itself requires a conditional input.

Prediction	Value	Class	Observable
Quadrupole emission pattern	$ h^{TT} ^2 = \frac{1}{2} \sin^4 \theta$	P1	Acoustic power $dP/d\Omega$ in spinor BEC
Multi-domain isotropy	$\langle h^{TT} ^2 \rangle = \frac{4}{15}$	P1	Angle-averaged emission
TT suppression at crossover	$f(1) = \frac{2}{3}$	P1	Phonon spectrum at $k\xi = 1$
Half-power wavenumber	$k_\star = \sqrt{2}/\xi$	P1	Spectral knee in TT channel
Unification mode count	$N_r = 4$	P1	Spin-2 sector of polar BEC
UV cutoff scale	$k_{UV} = \sqrt{3\pi}/\xi$	P1	Dispersion crossover scale
Induced Newton constant	$G_{\text{ind}} = G_N$ (by Sakharov)	P2	Requires $c_s = c$
Constituent mass	$m = (3\pi/4)^{1/3} m_{\text{Pl}}$	P2	Planck substrate
Planck density fixed point	$\rho^\star = c^2/(2G_N)$	P2	Planck substrate
Acoustic Hawking temperature	$T_H = \hbar c_s^2/(4\pi k_B r_s)$	P2	Acoustic horizon experiment
Logarithmic entropy correction	$c_{\log} = -2/45$	P3	Sub-leading BH entropy
Entropy gap	$\Delta S/S_{\text{BH}} = 1 - 2/(3\pi) \approx 78.8\%$	P3	Full BH entropy account
Newton-constant discrepancy	$G_{\text{EE}}/G_N = 3\pi/2$	P3	If acoustic modes = gravitons

Remark 60.2 (Class P3 is not a failure – [A]). *The Class P3 predictions are not promises the EFT cannot keep; they are predictions the EFT makes for a UV completion. If a UV completion is found, the predictions become tests. If no UV completion is found, the*

predictions remain as the EFT's contribution to the UV completion programme. The entropy gap $\approx 78.8\%$ is the most informative: it quantifies exactly how much of the gravitational entropy is not explained by the four acoustic modes.

61 UV Completion Comparison as a Consistency Filter

No UV completion is constructed here. The EFT is used as a filter: a UV theory is *consistent* if it passes the matching conditions; it is *ruled out* if it fails.

61.1 The filter: five acceptance criteria and three matching conditions

The five acceptance criteria C1–C5 (Section 34) and three matching conditions (M-I)–(M-III) (Section 39) constitute the filter. A UV theory passes if and only if it satisfies all eight.

Table 2: UV filter: requirements derived from the EFT. Any candidate UV theory must satisfy the right-hand column.

Criterion	Physical meaning	Required value
C1	Log entropy coefficient	$c_{\log} = -2/45$
C2	Minimum area	$\xi^2 = \ell_{\text{Pl}}^2/(3\pi)$
C3	Unification	$(N_r, k_{\text{UV}}^2/\pi) = (4, 3)$
C4	Mode-entropy correspondence	$m_{\text{eff}} = m_{\text{Pl}}/(2\pi)$
C5	Newton-constant ratio	$G_{\text{EE}}/G_{\text{ind}} = 3\pi/2$
M-I	Coupling at matching scale	$G_{\text{UV}}(k_{\text{UV}}) = 3\pi/2$
M-II	Slope at matching scale	$dG/d\ln k _{k_{\text{UV}}} = -9\pi^2/2$
M-III	All higher operators	Continuous across k_{UV}

61.2 Filter applied to three candidate frameworks

Theorem 61.1 (Systematic filter results – [A]/[B]). *The three candidate frameworks yield the following verdicts.*

(a) **Loop Quantum Gravity.** Fails C1 ($c_{\log}^{\text{LQG}} = -1/2 \neq -2/45$) and C2 ($\xi_{\text{LQG}}^2 \approx 5.17 \ell_{\text{Pl}}^2 \neq \ell_{\text{Pl}}^2/(3\pi)$). **Verdict [A]:** LQG is not the UV completion of this EFT as a stand-alone replacement. It may be complementary if it supplies the entropy gap ΔS not accounted for by the acoustic sector (Prediction P3, entropy gap).

(b) **Dvali–Gomez condensate picture.** Structural match on C5 (same mode-count scaling $N \sim M^2$). Fails C4: Dvali–Gomez gives $m_{\text{eff}} = m_{\text{Pl}}^2/M$ (mass-dependent),

while the EFT requires $m_{\text{eff}} = m_{\text{Pl}}/(2\pi)$ (fixed). C1, C2, C3 are not computed in Dvali–Gomez. **Verdict [B]:** Structurally analogous, quantitatively distinct on C4. Requires explicit computation of $c_{\log}$ in the Dvali–Gomez framework.

(c) **Asymptotic safety with $N_r = 4$ real scalar matter.** The matching condition M-I requires the UV fixed-point value $g_* = G_{\text{ac}} k_{\text{UV}}^2 = (3\pi/2)(3\pi) = 9\pi^2/2 \approx 44.4$ in Planck units. Existing pure-gravity AS truncations give $g_* \approx 0.1$ – 0.4 . **Verdict [B]:** Not ruled out, but requires an AS computation with $N_r = 4$ scalar fields. If such a computation gives $g_* \approx 44.4$, AS passes M-I. This is the concrete, well-posed test proposed by this manuscript.

61.3 Discriminating observable: $c_{\log}$

Proposition 61.2 (Discriminating power of $c_{\log} - [A]$). *The logarithmic entropy coefficient $c_{\log}$ is the sharpest single discriminator among the criteria C1–C5. Its predicted values differ by large factors across frameworks:*

$$c_{\log}^{\text{EFT}} = -\frac{2}{45}, \quad c_{\log}^{\text{LQG}} = -\frac{1}{2}, \quad c_{\log}^{\text{Sen}} = -\frac{3}{2}. \quad (20)$$

The ratios are $c_{\log}^{\text{LQG}}/c_{\log}^{\text{EFT}} = 45/4 = 11.25$ and $c_{\log}^{\text{Sen}}/c_{\log}^{\text{EFT}} = 135/4 = 33.75$. These are not order-of-magnitude ambiguities; they are precise fractions. A measurement of $c_{\log}$ to any finite precision that distinguishes these values would uniquely identify which framework describes black-hole microstructure.

Remark 61.3 (Why the values differ – [A]). *The three values count different degrees of freedom: $c_{\log}^{\text{EFT}}$ counts the four spin-2 acoustic modes (real scalars); $c_{\log}^{\text{LQG}}$ counts all gravitational degrees of freedom in the spin-foam network; $c_{\log}^{\text{Sen}}$ counts bosonic and fermionic fields of extremal black holes in string theory. If the full $c_{\log}$ is the sum of partial contributions from all sectors, then $c_{\log}^{\text{EFT}}$ is the acoustic sector’s contribution and the remainder must come from the UV completion. Specifically:*

$$c_{\log}^{\text{UV}} = c_{\log}^{\text{measured}} - c_{\log}^{\text{EFT}} = c_{\log}^{\text{measured}} + \frac{2}{45}. \quad (21)$$

Equation (21) is the theoretical prediction that the EFT offers to any future precision black-hole entropy measurement.

61.4 Summary: the EFT as a filter, not a final theory

What the filter says, and what it does not

The filter says: LQG (as a stand-alone replacement) fails C1 and C2 [A]. Dvali–Gomez fails C4 quantitatively [B]. Asymptotic safety with matter has not yet been tested against M-I [B]. No UV completion has been proved to pass all eight criteria.

The filter does not say: It does not say LQG is wrong (it may supply the entropy gap). It does not say Dvali–Gomez is wrong (it may compute $c_{\log}$). It does not say asymptotic safety fails (it has not been tested with matter). It does not say the acoustic substrate is the unique or complete description of the graviton.

The filter's purpose: It converts the EFT into a set of precise entry conditions for any future UV completion programme. These conditions are the EFT's contribution to the UV completion problem. They are not claims that the problem is solved.